

Firm Heterogeneity and Aggregate Fluctuations

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Abstract

We propose a bottom-up approach combining an accounting-based aggregation framework with the Generalized Random Forest (GRF) to study how firm heterogeneity shapes the transmission of aggregate shocks. The framework decomposes aggregate responses into the average firm-level sensitivity and the covariance between firms' sensitivities and aggregate weights. The GRF estimates sensitivities as nonparametric functions of a high-dimensional set of characteristics. Applying it to U.S. public companies, we document nonlinear heterogeneity in responses to business-cycle fluctuations, driven by non-financial characteristics. Firms with larger aggregate weights are less cyclical, so heterogeneity dampens aggregate responses. Results extend to monetary policy and oil price shocks.

JEL Codes: D22, E32, C14

Keywords: Firm Heterogeneity, Firm Sensitivity, Aggregate Fluctuations, Machine Learning, Generalized Random Forest.

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1 Introduction

A substantial body of literature in corporate finance and macroeconomics documents pronounced cross-sectional heterogeneity in how firms respond to aggregate shocks and business cycle fluctuations (Crouzet and Mehrotra, 2020). Some firms adjust sales, investment, employment, or financing sharply as economic conditions change, while others remain largely unaffected. This heterogeneity appears to reflect differences in observable characteristics—such as size, leverage, and liquidity (Gertler and Gilchrist, 1994; Ottonello and Winberry, 2020; Jeenas, 2023). Far less is known, however, about its macroeconomic importance—whether it amplifies or dampens aggregate fluctuations, and how large these effects are. The question is central for policy design, since aggregate outcomes are shaped not only by the average firm’s response but also by how responses are distributed across firms (Chang et al., 2024b; Lenza and Savoia, 2026).

We propose a bottom-up approach to study how heterogeneity in firms’ sensitivities to aggregate shocks shapes macroeconomic outcomes. We combine a transparent accounting-based aggregation framework with nonparametric, machine-learning estimates of firm-level sensitivities. The framework maps micro-level responses to aggregate outcomes by weighting each firm’s sensitivity according to its contribution to aggregate economic activity. This yields a decomposition of the aggregate response into two components: the average firm-level sensitivity and the covariance between firms’ sensitivities and their weights. The first measures how the average firm responds to a shock, while the second captures whether the firms that account for most economic activity are more or less responsive than that average. A covariance term opposite in sign to the average sensitivity implies that firms with larger aggregate weights are less responsive than the average firm, so that heterogeneity dampens the aggregate response; a covariance with the same sign amplifies it.

We estimate firms’ sensitivities to aggregate shocks using the Generalized Random Forest (GRF) algorithm of Athey et al. (2019). While firms’ aggregate weights are directly observable, their sensitivities are not, so estimating them is essential for quantifying the aggregate role of cross-sectional heterogeneity. The GRF estimates each firm’s sensitivity as a flexible function of a high-dimensional vector of firm characteristics, revealing which features drive the heterogeneity in responsiveness across firms. Relative to a standard linear panel model, the GRF captures nonlinearities and interactions among covariates without requiring the econometrician to specify their functional form ex ante. Unlike machine-learning methods designed primarily for prediction, the GRF is specifically designed to estimate heterogeneous treatment effects. These features make it well suited to characterizing the distribution of firm-level sensitivities and quantifying its implications for aggregate outcomes. To our knowledge, this is the first use of the GRF to estimate firm-level responses to aggregate shocks.

We illustrate our approach in two applications using Compustat data on U.S. publicly listed

firms from 1990 to 2019. The first examines firms’ sensitivities to business-cycle fluctuations, while the second studies their dynamic responses to identified monetary policy and oil price shocks. In both applications, we estimate the sensitivities of sales, employment, investment, and debt issuance as flexible functions of several financial and non-financial firm characteristics widely studied in the literature, including leverage, liquidity, distance to default, firm size, and industry scope.

In our first application, we find that business-cycle fluctuations, which we measure by the one-year growth rate of real GDP, generate widely dispersed responses across firms. A 1 percentage point increase in real GDP growth is associated with a 2.1 percent increase in firms’ sales, on average, while investment, employment, and debt issuance increase by 0.9, 1.2, and 1.2 percent, respectively. These averages, however, mask substantial cross-sectional heterogeneity: a large share of firms are more than twice as sensitive as the median firm, and some firms’ investment and debt issuance respond in the opposite direction to the average.

Non-financial characteristics—above all industry scope and firm size—drive most of the heterogeneity over the cycle, accounting for roughly 70 percent of the total, on average across outcomes. Industry scope is the dominant driver for sales, accounting for about 40 percent, while firm size accounts for nearly 50 percent in investment and about 40 percent in debt issuance. Employment sensitivity is instead more evenly distributed across sales volatility, profitability, and industry scope, with firm size playing a limited role. Financial characteristics such as leverage, liquidity, and distance to default account for relatively little of this dispersion, suggesting that differences in firms’ cyclical sensitivities reflect differences in their non-financial characteristics more than in their financial conditions ([Crouzet and Mehrotra, 2020](#)). These relationships are also strongly nonlinear: while the average GRF sensitivity is statistically indistinguishable from that of a standard linear panel model, the two distributions differ markedly in their higher moments.

This heterogeneity matters at the macro level, as it systematically dampens the aggregate response to business-cycle fluctuations. The covariance term of our decomposition is negative for every outcome: the firms accounting for the largest shares of economic activity are systematically less cyclical than the average firm, so heterogeneity dampens the aggregate response. The dampening reduces the aggregate responses of sales and employment by about 5 and 18 percent, respectively, and that of investment by 44 percent, while nearly offsetting the response of debt issuance. Heterogeneity operates both across and within sectors: assigning each firm the median sensitivity of its sector eliminates only about half of the covariance term, so within-sector variation matters as much as cross-sector differences. Finally, nonlinearities matter for aggregation: relative to a linear specification, the GRF changes the estimated aggregate response by up to 25 basis points, with the difference arising almost entirely from differences in the covariance term.

In the second application, we combine the GRF with a local projection approach ([Jordà, 2005](#)) to study firms’ responses to exogenously identified monetary policy and oil price shocks over time.

Monetary policy shocks are measured using interest-rate surprises around Federal Reserve announcements, identified as in [Bauer and Swanson \(2023\)](#); oil price shocks are structural oil supply news shocks identified using high-frequency OPEC announcement surprises following [Känzig \(2021\)](#) in an augmented VAR that incorporates financial variables, as proposed by [Mori and Peersman \(2024\)](#). This extension complements the contemporaneous business cycle analysis by accounting for the delayed propagation of aggregate shocks and isolating the effects of specific structural disturbances from broader cyclical movements in GDP.

The GRF uncovers substantial and increasingly pronounced heterogeneity in firm-level responses, with dispersion in sensitivities roughly doubling at longer horizons. Despite this widening dispersion, firms’ relative sensitivity remains remarkably persistent: those most responsive on impact remain the most responsive over time, reflecting the stability of the firm characteristics underlying this heterogeneity. In contrast to business cycle fluctuations, the heterogeneity in responses to monetary policy is dominated by financial characteristics—default risk, liquidity, and debt maturity—providing data-driven support for the conditioning variables emphasized in the monetary policy literature ([Ottonello and Winberry, 2020](#); [Jeenas, 2023](#)). For oil price shocks, heterogeneity reflects a more balanced mix of financial and non-financial characteristics, with no single characteristic accounting for more than 20 percent of the heterogeneity. This broader set of relevant characteristics suggests that oil price shocks operate through a wider range of channels than monetary policy shocks, spanning production costs, demand, and financing conditions.

At the aggregate level, the covariance between firm sensitivities and weights plays an important but shock-dependent role in the transmission of aggregate shocks. Following a monetary tightening, heterogeneity dampens the aggregate contraction in investment and debt issuance, as firms with larger weights respond less than the average firm. For sales and employment, instead, firms carrying larger weights are more responsive, so heterogeneity amplifies the aggregate decline. The covariance contribution also evolves differently across margins: it declines for investment, increases for debt issuance, and remains broadly stable for sales and employment. Oil price shocks display a different pattern: firms accounting for larger shares of aggregate activity respond less than the average firm across all outcomes, so heterogeneity consistently attenuates the aggregate response, and this contribution remains broadly stable over the horizon. These contrasting patterns show that the aggregate importance of firm heterogeneity depends jointly on the nature of the shock, the economic margin, and the distribution of firms’ weights.

A series of additional results and robustness checks extend our findings. First, we extend the analysis of firm-level heterogeneity and its drivers to financial market outcomes by studying the response of firms’ equity values to aggregate shocks. Unlike the real outcomes, market-value sensitivities are positively correlated with firms’ aggregate importance, so that the covariance term amplifies, rather than dampens, the aggregate response. Second, we find that the aggregate role of

firm heterogeneity is remarkably stable over time for sales and employment, while for investment a declining covariance term—offset by a rising mean—indicates that cyclical sensitivities have become more homogeneous across firms. Third, we show that the aggregate response to monetary policy tightening is state dependent, with larger effects during expansions than during recessions; this asymmetry is driven almost entirely by the covariance term rather than by differences in firms’ average responses, pointing to a cyclical shift in the joint distribution of firm characteristics. Finally, our results are robust to adding firm age to the characteristic space, to alternative aggregation weights, and to the inclusion of firm fixed effects in estimating firms’ sensitivities. Accounting for firm entry and exit leaves the decomposition largely unchanged, with their contributions generally comparable to or smaller than the covariance channel.

Literature. Our work contributes to the literature that studies the heterogeneity in firm-level sensitivity to aggregate shocks and its determinants. Among others, [Ottonello and Winberry \(2020\)](#), [Jeenas \(2023\)](#), [Gertler and Gilchrist \(1994\)](#) and [Jungherr et al. \(2024\)](#) study the role of leverage and distance to default, liquidity, size, and debt maturity for the response of firm investment to monetary policy shocks, respectively; [Gurkaynak et al. \(2022\)](#) investigate how liquidity and leverage influence the response of market value; [Crouzet and Mehrotra \(2020\)](#) examine the role of size for the response of sales and investments to business cycle fluctuations and monetary policy; [Covas and Haan \(2011\)](#) and [Begenau and Salomao \(2019\)](#) study debt issuance by firm size over the business cycle.¹ Prior research typically examined a single firm characteristic at a time and imposes linearity in how firm characteristics influence responses to aggregate shocks, a specification that [David et al. \(2026\)](#) show generally fails to recover causally interpretable state-dependent responses, motivating a nonparametric alternative. We depart from the standard linear panel regression approach and apply machine learning methods to flexibly incorporate a large set of firm characteristics simultaneously. We find evidence that firm sensitivities are highly nonlinear in several interacting characteristics, and that the relevant dimensions are shock- and outcome-specific.

Our paper also contributes to the literature on the macroeconomic implications of micro-level heterogeneity. A large body of work relies on quantitative structural models to quantify how specific sources of firm heterogeneity shape aggregate dynamics ([Cooley and Quadrini, 2001](#); [Buera and Moll, 2015](#); [Crouzet, 2018](#); [Ottonello and Winberry, 2020](#); [Deng and Fang, 2022](#); [Krusell et al., 2023](#)). These models provide a clean structural interpretation and quantitative discipline, but often focus on a limited set of dimensions of heterogeneity. Our empirical, reduced-form estimates instead allow

¹The macroeconomic literature has devoted limited attention to heterogeneity in firm-level sensitivity to oil shocks, despite these being key drivers of macroeconomic fluctuations ([Känzig, 2021](#)). [Narayan and Sharma \(2011\)](#) and [Tsai \(2015\)](#) study how a firm’s market value reacts to oil shocks depending on its size and industry scope. We expand this literature by providing novel evidence on the heterogeneity in firm-level sensitivity to oil shocks, its determinants, and its aggregate impact.

firm sensitivities to vary flexibly along multiple dimensions simultaneously, providing quantitative targets for structural models. A complementary approach uses functional and heterogeneous VARs, in which aggregate variables and cross-sectional distributions are jointly endogenous (Chang et al., 2024a,b; Lenza and Savoia, 2026). These methods capture rich dynamic feedback between aggregate outcomes and the firm distribution, but require structure on their joint dynamics and typically abstract from the underlying sources of cross-sectional heterogeneity. By contrast, our bottom-up approach imposes no structure on the data, while still providing natural calibration or estimation targets and a transparent decomposition of aggregate responses into average and heterogeneity components.

Lastly, this paper contributes to the rapidly growing literature applying machine learning techniques to economic analysis, particularly in the estimation of conditional average treatment effects (Athey and Imbens, 2017; Varian, 2014; Breiman, 2001). For instance, machine learning techniques are used to study the heterogeneous dynamics in earnings after job loss (Athey et al., 2026), the effects of 401(k) eligibility on savings decisions (Belloni et al., 2017), or to explore the heterogeneity of house price elasticity of consumption (Khazra, 2021) and, more broadly, latent worker heterogeneity and its aggregate labor-market implications (Ahn et al., 2023; Gregory et al., 2025).² Only a few studies, however, apply these tools to firms' heterogeneous sensitivity to aggregate shocks. Drechsel et al. (2026) use a clustering regression framework to study the full distribution of the response of investment to monetary policy, while Paranhos (2025a) extends local projections nonparametrically to study how default risk shapes the transmission of monetary policy to investment.³ We contribute to this literature by providing a systematic approach that uses micro-level data and machine-learning techniques to study the aggregate response to shocks while accounting for the heterogeneity in sensitivities across firms.⁴

²Machine learning techniques have been explored in macroeconomics in relation to forecasting. For instance, Mueller and Spinnewijn (2025) use an Ensemble Model, a weighted average of the predictions from the LASSO, Gradient-Boosted Decision Trees, and Random Forest algorithms to assess the predictors of long-term unemployment. Paranhos (2025b) and Nakamura (2005) both use neural networks to predict future inflation, while Medeiros et al. (2021) apply random forests; Cui et al. (2026) use firm-level accounting to forecast GDP growth rates, while Freyberger et al. (2020) and Gu et al. (2020) apply adaptive group Lasso and neural networks, respectively, to improve the prediction of asset returns.

³While built on the same honest-forest architecture, the estimator in Paranhos (2025a) differs from GRF in three respects. First, the splitting variable is drawn randomly, and only the threshold is optimized, with the local-projection coefficient re-estimated in each candidate child; GRF instead selects the best variable-threshold pair from a random subset of candidates using a gradient-based approximation. Second, each tree estimates a local-projection coefficient within its leaves and the forest averages these tree-level estimates (Wager and Athey, 2018); GRF instead uses the trees only to construct neighborhood weights and solves a single weighted moment condition at each covariate value. Lastly, the outcome and the shock are orthogonalized linearly with respect to controls before the forest is grown, whereas GRF residualizes both nonparametrically using auxiliary forest estimates of the conditional means.

⁴The GRF algorithm is particularly well suited to estimating heterogeneous, individual-level responses—more so than alternative estimators such as the R-learner of Nie and Wager (2021), related meta-learners

The remainder of the paper is organized as follows. Section 2 presents an accounting-based aggregation framework and provides information on the methodology we use. Section 3 applies the framework to study sensitivities to business cycle fluctuations and their aggregate implications on U.S. firms. Section 4 extends the analysis to the dynamic, multi-horizon response of firms to identified monetary policy and oil price shocks. Section 5 extends our analysis along several dimensions. Section 6 concludes.

2 Firm Heterogeneity and Aggregate Fluctuations

We first introduce an accounting-based aggregation framework that maps firm-level responses to macroeconomic shocks into the response of aggregate variables. We then discuss how the Generalized Random Forest algorithm of [Athey et al. \(2019\)](#) can be used to estimate firms' sensitivities in the data.

2.1 Decomposing Aggregate Dynamics

Consider the set I_{t+h} of firms operating continuously from $t-1$ through $t+h$.⁵ Let G_{t+h} and $g_{i,t+h}$ denote the aggregate and the firm-level response of variable Y following an aggregate shock W_t , respectively:

$$G_{t+h}(W_t) = \frac{Y_{t+h} - Y_{t-1}}{Y_{t-1}} \quad g_{i,t+h}(W_t) = \frac{Y_{i,t+h} - Y_{i,t-1}}{Y_{i,t-1}} \equiv \beta_{i,t+h} W_t, \quad (1)$$

where $\beta_{i,t+h} = \frac{\partial g_{i,t+h}}{\partial W_t}$ is the sensitivity of firm i 's outcome at $t+h$ to the aggregate shock W_t . Let $\omega_{i,t-1}$ be the share of Y_{t-1} accounted for by firm i :

$$\omega_{i,t-1} = \frac{Y_{i,t-1}}{Y_{t-1}} \quad \text{where } Y_{t-1} = \sum_{i \in I_{t+h}} Y_{i,t-1}. \quad (2)$$

It follows that the h -period-ahead aggregate response of variable Y to an aggregate shock at time t is:

$$G_{t+h} = \sum_{i \in I_{t+h}} \omega_{i,t-1} \cdot g_{i,t+h}. \quad (3)$$

([Künzel et al., 2019](#); [Kennedy, 2023](#)), and Bayesian additive regression trees ([Hahn et al., 2020](#))—because it grows trees to maximize the contrast in estimated sensitivity rather than to predict the outcome accurately.

⁵We extend the accounting framework to allow for firms' entry and exit in Section 5.

The aggregate response G_{t+h} is a weighted average of firm-level responses $g_{i,t+h}$, with weights $\omega_{i,t-1}$ given by each firm's share of the aggregate in the pre-shock period. We decompose Equation (3) to understand how firm heterogeneity contributes to the aggregate response:

$$G_{t+h} = \overline{g_{t+h}} + \text{Cov}(\omega_{i,t-1}, g_{i,t+h}), \quad (4)$$

where the first term is the unweighted average response across firms, $\overline{g_{t+h}} = \frac{1}{|I_{t+h}|} \sum_{i \in I_{t+h}} g_{i,t+h}$, and the second term is the covariance between firm sensitivities and firms' relevance in the aggregate, $\sum_{i \in I_{t+h}} \left(\omega_{i,t-1} - \frac{1}{|I_{t+h}|} \right) (g_{i,t+h} - \overline{g_{t+h}})$. The first term captures how, on average, firms respond to aggregate fluctuations without considering their relative relevance in the economy. The second term captures how heterogeneity in firms' sensitivities interacts with the heterogeneity in their weights.

Equation (4) shows that heterogeneity matters for the aggregate response only when firms' weights are systematically related to their responses. For example, suppose Firm A produces 80 percent of output and responds weakly to a shock ($g_A = 2\%$), while Firm B produces 20 percent and responds strongly ($g_B = 12\%$). The unweighted average response is 7%, but the aggregate response is only 4%. The aggregate response is lower than the unweighted average because the firm contributing most to output is less responsive than the average: weights and responses are negatively correlated. Conversely, if the same firm were the more responsive of the two, the covariance term would be positive, and the aggregate response would exceed the unweighted average. Thus, a covariance that shares the sign of the average response amplifies the aggregate response, moving it further from zero, whereas one of the opposite sign dampens it, pulling it toward zero.

2.2 Estimating Firm-level Sensitivity

While the firm's share in the aggregate, $\omega_{i,t-1}$, is readily measured in the data, the firm-level responses $g_{i,t+h}$ depend on the underlying sensitivities $\beta_{i,t+h}$, which are not directly observed and must be estimated. Assessing the importance of firm heterogeneity for the aggregate response hinges on estimating the full distribution of firm sensitivities, which may depend on many characteristics through complex, nonlinear interactions of unknown form.

Generalized Random Forest. We use the Generalized Random Forest algorithm of [Athey et al. \(2019\)](#) to estimate the dynamic firm-level sensitivities as a flexible, nonparametric function of observable firm characteristics. Consider an empirical setting in which we observe outcomes and characteristics of firms, indexed by i , over multiple periods, indexed by t . The outcome $Y_{i,t+h}$ is a firm-level growth rate in sales, investment, or other key indicators, measured $h \geq 0$ periods ahead. Let W_t denote the aggregate shock common to all firms, and let $\mathbf{X}_{i,t-1} \in \mathbb{R}^M$ collect M lagged firm characteristics that may determine the sensitivity of $Y_{i,t+h}$ to W_t . GRF allows us to estimate the

dynamic heterogeneous response of firm outcomes ($Y_{i,t+h}$) to an aggregate shock (W_t), conditional on a high-dimensional set of firm-level characteristics ($\mathbf{X}_{i,t-1}$), in a nonparametric way:

$$Y_{i,t+h} = \beta_h(\mathbf{X}_{i,t-1}) \cdot W_t + \varepsilon_{i,t+h}, \quad h \geq 0 \quad (5)$$

where $\varepsilon_{i,t+h}$ is an i.i.d. error term. For each horizon h , $\beta_h(\cdot)$ is a scalar-valued function of lagged firm characteristics; thus, the sequence $\{\beta_h(\cdot)\}_{h \geq 0}$ traces the dynamic firm-level response to W_t .⁶

GRF estimates the object of interest $\beta_h(x)$, i.e. the effect of the aggregate shock on the horizon- h outcome for firms with characteristics $\mathbf{X}_{i,t-1} = x$, in two steps. First, it grows a forest of honest trees that recursively partition the space of firm characteristics spanned by $\mathbf{X}_{i,t-1}$. A regression tree repeatedly splits the sample by cutting one characteristic at a time (e.g., size above/below a threshold, then leverage, and so on), carving the characteristic space into rectangular cells called leaves. Crucially, unlike standard regression trees, which choose each split to make the outcome as homogeneous as possible within the resulting cells (Breiman, 2001), GRF chooses splits that best separate high-sensitivity from low-sensitivity firms, making it particularly well suited to capturing how the effects of shocks vary across firms. A forest grows many such trees, each on a different random subsample of the data and considering a random subset of characteristics at each split. Each tree is grown “honestly”: one subsample is used to determine the splits, and a separate subsample to populate the leaves and compute the estimates.⁷

Second, GRF estimates the sensitivity function $\beta_h(x)$ by aggregating information across trees. Rather than averaging each tree’s β estimate, GRF uses the forest only to define similarity weights. For each x , the forest defines weights $\alpha_{i,t}^h(x) \in [0, 1]$, with $\sum_{i,t} \alpha_{i,t}^h(x) = 1$, equal to the frequency with which observation (i, t) falls in the same terminal leaf as x across trees. These weights define a data-adaptive neighborhood around x : they are larger for firms that repeatedly share leaves with x , and averaging over trees makes the estimate less sensitive to any single partition.⁸ Given the weights, $\hat{\beta}_h(x)$ is the solution to a locally weighted version of the moment condition implied by Equation (5). Specifically, at each x , $\hat{\beta}_h(x)$ is the value that sets the α -weighted moment over the neighborhood of x to zero, which takes the form of the following weighted regression slope, applied to variables centered on their conditional means given $\mathbf{X}$ (see Appendix A.1):

⁶As is common in empirical work, we condition the sensitivity on lagged characteristics to avoid endogeneity concerns with respect to the aggregate shock. The number of lags is set to one for simplicity and can be extended as necessary.

⁷Appendix A.1 includes additional technical details on the GRF algorithm.

⁸Classical local methods, e.g. kernel regression, also weight nearby observations, but they measure “nearby” by raw distance in characteristic space with a fixed bandwidth, which collapses once there are more than a few characteristics. The forest instead learns which characteristics matter and adapts the weights accordingly. A separate forest is grown for each outcome-horizon pair, so the weights are specific to each estimation.

$$\hat{\beta}_h(x) = \frac{\sum_{t=1}^T \sum_i \alpha_{i,t}^h(x) (W_t - \bar{W}_\alpha) (Y_{i,t+h} - \bar{Y}_{\alpha,h})}{\sum_{t=1}^T \sum_i \alpha_{i,t}^h(x) (W_t - \bar{W}_\alpha)^2}, \quad (6)$$

where $\bar{W}_\alpha = \sum_{t,i} \alpha_{i,t}^h(x) W_t$ and $\bar{Y}_{\alpha,h} = \sum_{t,i} \alpha_{i,t}^h(x) Y_{i,t+h}$ are the weighted means of the shock and the horizon- h outcome. Evaluating $\hat{\beta}_h(\cdot)$ at each observation's characteristics yields a fitted sensitivity $\hat{\beta}_{i,t,h} \equiv \hat{\beta}_h(\mathbf{X}_{i,t-1})$ for every firm-period and horizon. Their empirical distribution across firms and periods gives the cross-sectional distribution of estimated sensitivities that we leverage in our accounting framework of Section 2.1.

Figure 1 illustrates the algorithm in a simple case with a two-dimensional characteristic space, e.g., firm size and leverage. The blue cross denotes a target observation (i, t) with characteristics x . Each tree in the top row partitions the size-leverage space differently, so the set of observations falling in the same leaf as the target observation varies across trees. The bottom row shows the aggregation step: observations that often share the target's leaf receive larger weights $\alpha_{i,t}^h(x)$, while those that rarely do receive little weight, with each dot's size proportional to its weight. Averaging over trees, therefore, produces a data-driven local comparison group around x . The weighted regression in Equation (6) then effectively estimates the slope of the outcome with respect to W_t within this group, so that $\hat{\beta}_h(x)$ measures how the outcomes of firms with characteristics similar to x respond to the aggregate shock.

2.3 Advantages over Alternative Approaches

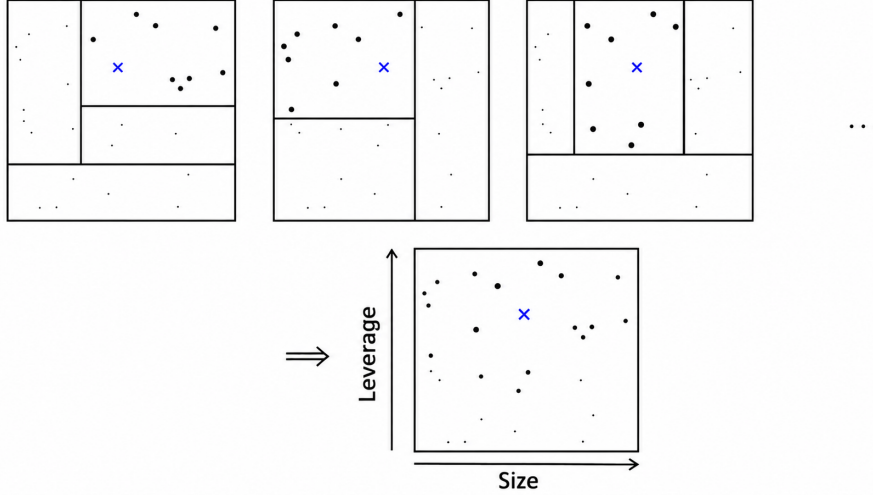
Linear Panel Model. The standard econometric framework for estimating heterogeneity in firms' sensitivity to aggregate shocks is a linear panel model (LPM henceforth) (Jordà, 2005). In an LPM, the heterogeneous response of firms' outcomes to an aggregate shock, conditional on a set of predetermined firm-level characteristics, is estimated as follows:

$$Y_{i,t+h} = \alpha^h + \beta_0^h \cdot W_t + \beta_1^{h'} \cdot \mathbf{X}_{i,t-1} + \beta_2^{h'} (W_t \cdot \mathbf{X}_{i,t-1}) + \varepsilon_{i,t+h}, \quad (7)$$

where $\varepsilon_{i,t+h}$ is an i.i.d. error term, and firm characteristics are predetermined at $t - 1$. The parameter vector of interest, β_2^h , captures how firms' sensitivity to aggregate shocks varies with firm characteristics at horizon h . The marginal effect of the aggregate shock W_t on firm outcomes is given by $\beta_0^h + \beta_2^{h'} \mathbf{X}_{i,t-1}$, which depends linearly on $\mathbf{X}_{i,t-1}$.

The GRF algorithm in Equation (5) offers two advantages over the standard linear panel regression in Equation (7), making it particularly well-suited for estimating the heterogeneous effects of aggregate shocks. First, the linear panel regression model assumes that firms' characteristics linearly influence firm sensitivity to aggregate shocks. However, this linearity assumption may be restrictive and could lead to misspecification if the nonlinear component of heterogeneity is quan-

Figure 1: Illustration of Generalized Random Forest Algorithm



Notes: The figure illustrates how forest weights are constructed in a GRF. The upper panels show example trees that partition the covariate space, represented here by firm size and leverage. The blue cross denotes the target observation (i, t) with characteristics x . Observations that fall in the same leaf as the target observation receive positive tree-specific weight. Averaging these co-leaf assignments across trees yields the forest weights shown in the lower panel, where larger circles indicate larger weights. The Figure is adapted from Figure 1 in [Athey et al. \(2019\)](#).

titatively significant ([David et al., 2026](#)). While the linear panel regression model can incorporate more complex forms of heterogeneity by including polynomials or interaction terms in firm characteristics, it remains a parametric approach that requires the econometrician to take a stance on the unknown forms of nonlinearities, making the LPM vulnerable to model misspecification. In contrast, GRF partitions the covariate space nonparametrically, adaptively detecting intricate relationships without requiring a pre-specified form.

Second, GRF can efficiently handle a high-dimensional characteristics space, automatically putting more weight on the most important covariates. This feature of the GRF mitigates the curse of dimensionality inherent in models with large sets of covariates and interactions. Enumerating all possible pairwise or higher-order interactions in a linear model quickly leads to over-parameterization, while GRF adaptively partitions the data, freeing the researcher from having to ex ante specify functional forms or interactions. Thus, the linearity and parametric features of the linear panel regression model become more restrictive in the presence of a high-dimensional characteristics space.

Alternative ML Methods. A growing literature in econometrics and machine learning proposes methods for estimating heterogeneous treatment effects, including meta-learners — such as

the R-learner (Nie and Wager, 2021) and the broader family of S-, T-, X-, and DR-learners (Künzel et al., 2019; Kennedy, 2023) — and Bayesian additive regression trees (BART) (Chipman et al., 2010; Hill, 2011; Hahn et al., 2020).^{9 10}

The R-learner of Nie and Wager (2021) is one of the closest alternatives to GRF: it estimates heterogeneous effects (i.e. $\beta(x)$ in our setting) by first removing the variation in the outcome and in the treatment (i.e. the aggregate shock W_t in our setting) that observable characteristics can explain, and then recovering the heterogeneous component from the residualized problem. Following Robinson (1988)’s decomposition, it first estimates nonparametrically the conditional expectations of the outcome and of the treatment given characteristics (the so-called nuisance functions), and uses them to form the residualized outcome and treatment. The heterogeneous effect is then obtained by regressing the residualized outcome on the residualized treatment with a flexible nonparametric estimator, which can be implemented with any machine-learning method, such as a random forest or a neural network.¹¹ The GRF of Athey et al. (2019) shares this principle, combining an initial residualization step with a causal-forest framework. After estimating the nuisance functions, it builds adaptive neighborhoods of observationally similar firms from an ensemble of decision trees and estimates local effects using forest weights that solve a local moment condition.

These alternatives provide useful benchmarks, as they accommodate nonlinearities and high-dimensional interactions without strong functional-form assumptions. In practice, no single estimator dominates across settings. The appropriate choice depends on the estimand, the structure and signal-to-noise ratio of the data, the strength and sparsity of the underlying heterogeneity,

⁹Bayesian additive regression trees approximate the conditional expectation of the outcome as the sum of many shallow regression trees, each acting as a weak learner. Prior distributions regularize the contribution of individual trees, producing a flexible yet parsimonious estimate of the outcome function together with posterior credible intervals. Heterogeneous effects are then recovered by differencing predicted outcomes under treatment and control (Chipman et al., 2010; Hill, 2011). Bayesian causal forests refine this approach by modeling the baseline and treatment-effect components of the outcome separately, assigning each its own prior and shrinking the treatment-effect component toward homogeneous effects to mitigate regularization-induced bias (Hahn et al., 2020). Bayesian methods rely on prior distributions to regularize estimation and typically use MCMC algorithms to characterize the posterior distribution, making them computationally more demanding than forest-based methods.

¹⁰Alternative meta-learning approaches estimate heterogeneous effects by combining standard supervised learning algorithms with different identification strategies. S-learners fit a single outcome model that includes the treatment as a covariate, T-learners estimate separate outcome models for the treated and control groups and recover heterogeneous effects by differencing their predictions, X-learners impute individual treatment effects and combine them using propensity-score weights, which can improve performance in imbalanced designs (Künzel et al., 2019), while DR-learners exploit doubly robust scores to reduce sensitivity to nuisance estimation errors and achieve robust estimation under appropriate regularity conditions (Kennedy, 2023).

¹¹Robinson (1988) assumes that the outcome is conditionally linear in the shock with a heterogeneous slope, $Y = b(\mathbf{X}) + \beta(\mathbf{X})W + \varepsilon$ with $\mathbb{E}[\varepsilon | \mathbf{X}, W] = 0$. Taking expectations given $\mathbf{X}$ and subtracting cancels the baseline $b(\mathbf{X})$ and yields $\tilde{Y} = \beta(\mathbf{X})\tilde{W} + \varepsilon$, where $\tilde{Y} = Y - \mathbb{E}[Y | \mathbf{X}]$ and $\tilde{W} = W - \mathbb{E}[W | \mathbf{X}]$ are the residualized outcome and shock. Minimizing the R-loss $\mathbb{E}[(\tilde{Y} - \beta(\mathbf{X})\tilde{W})^2]$ over $\beta(\cdot)$ is exactly the heterogeneous sensitivity. Allowing β to vary with $\mathbf{X}$ is what makes the loss target heterogeneous.

and each method’s tuning requirements and computational cost. We take GRF as our primary estimator for two reasons specific to measuring firm heterogeneity. First, GRF grows its trees by splitting directly on heterogeneity in the sensitivity, so its neighborhoods are designed to resolve how β varies across firm characteristics; a generic R-learner instead recovers β by fitting a flexible regression to the residualized problem, whose splits are chosen to predict that transformed outcome accurately rather than to maximize the contrast in β across firms. Second, through honest sample-splitting and the forest’s variance estimator, GRF delivers asymptotically valid confidence intervals (Athey et al., 2019), which are essential for testing whether — and along which characteristics — sensitivities differ across firms; the R-learner yields accurate point estimates, with quasi-oracle guarantees, but not turnkey inference.

Appendix A.2 compares these methods in a Monte Carlo exercise with a nonlinear data-generating process. A linear model fails to recover the heterogeneity because of misspecification; GRF and the neural-network R-learner perform similarly well, while the random-forest R-learner is considerably less accurate.

2.4 Discussion

We conclude this section by discussing the main limitations of our theoretical and econometric framework.

First, our estimand is the conditional average partial effect, a reduced-form object that the Generalized Random Forest estimates nonparametrically and, under standard exogeneity assumptions, consistently (Athey et al., 2019). By construction, however, this object does not identify the economic mechanisms generating the estimated heterogeneity in firm sensitivities. The same variation in $\beta(x)$ may arise from distinct primitives, including adjustment costs in capital and labor (Cooper and Haltiwanger, 2006; Cooper et al., 2015), heterogeneous access to finance through net worth and collateral constraints (Bernanke et al., 1999; Kiyotaki and Moore, 1997), or differences in markups and market power (Baqee and Farhi, 2020). Because multiple structural environments can generate the same conditional response function, we do not interpret $\beta(x)$ as evidence for any particular mechanism or friction. Instead, we view the estimated sensitivity function as a model-free empirical moment that can discipline structural models and provide natural calibration or estimation targets (Drechsel et al., 2026).

Second, our aggregation framework is deliberately agnostic about the underlying economic structure and therefore does not uniquely determine how firm-level responses should be aggregated. We weight firms by their lagged shares in the aggregate outcome because this follows directly from the accounting identity underlying our decomposition: the aggregate outcome is, by construction, the share-weighted sum of firm-level outcomes, so each firm’s lagged share measures its mechanical

contribution to the aggregate response. The approach is shared by [Crouzet and Mehrotra \(2020\)](#), who use lagged shares based on firm size to assess the role of heterogeneity at the aggregate level. Alternative structural environments, however, may imply different aggregation schemes. For example, Domar weights are appropriate for mapping firm-level productivity shocks into aggregate output ([Domar, 1961](#); [Hulten, 1978](#)), cost shares for aggregating markups ([Edmond et al., 2023](#)), and sales shares for aggregating productivity ([Olley and Pakes, 1996](#)). We assess the robustness of our results to alternative weighting schemes in [Section 5](#).

Finally, our aggregation and econometric framework treats firms’ estimated sensitivities, $\beta(x)$, and their lagged aggregate-outcome shares as separate primitives, abstracting from dynamic feedback between cross-sectional heterogeneity and aggregate outcomes. In particular, we take the cross-sectional distribution of firm characteristics as given when measuring aggregate responses and do not model how aggregate shocks may themselves reshape that distribution over time. Alternative approaches, such as functional vector autoregressions or distributional state-space models, instead treat the cross-sectional distribution as an aggregate state variable whose endogenous evolution influences the dynamics of aggregate variables ([Chang et al., 2024a](#); [Lenza and Savoia, 2026](#)).

Overall, our framework is best viewed as quantifying how observed heterogeneity maps into aggregate responses in a partial-equilibrium, accounting sense, rather than as a model of the joint evolution of aggregate outcomes and the cross-sectional distribution of firms.

We now turn to two applications for the U.S. economy. First, we study the contemporaneous sensitivity of firm sales, employment, investment, and debt issuance to business cycle fluctuations; then we assess the dynamic response of firms’ outcomes to identified exogenous shocks.

3 The Cyclical Sensitivity of U.S. Firms

We estimate the contemporaneous sensitivity of firms’ outcomes to business cycle fluctuations, conditional on a rich set of firm-level characteristics, applying the GRF algorithm to U.S. data. We begin by describing the data, then analyze the estimated firm-level sensitivities and their implications for the aggregate response.

Data. Our primary data source is Compustat, which provides comprehensive financial statement information for publicly listed companies in the U.S. over the period 1990–2019. All variables are deflated using the implied price index of gross value added in the U.S. non-farm business sector. The firm outcomes are sales growth, employment growth, the investment rate (the growth rate of the real capital stock, computed by the perpetual inventory method), and debt issuance (the growth rate of short- and long-term debt)—all computed as one-year growth rates with the [Davis and Haltiwanger](#)

(1992) formula.¹² To capture heterogeneity in firm sensitivities, we consider eight balance-sheet characteristics, grouped into non-financial and financial variables. Non-financial characteristics are firm size (the logarithm of real total assets), industry scope (5-digit NAICS codes), ten-year sales-growth volatility, and profitability (return on assets). Financial characteristics are the liquidity ratio (cash to total assets), leverage (total debt to total assets), distance to default (Merton, 1974), and the share of short-term debt (short-term to total debt). We select these characteristics following the literature on the heterogeneous transmission of aggregate shocks to firm outcomes.¹³ We measure business cycle fluctuations as the one-year growth rate of real GDP, following Crouzet and Mehrotra (2020).

Appendix B provides further detail on the data, variable definitions, and sample construction, with descriptive statistics in Table B.1 and distributions in Figures B.2 and B.3. We also report the pairwise correlations among the firm characteristics (Table B.2), which suggest that, although some correlation exists, they provide distinct information along different dimensions. Figure B.1 in Appendix B plots the time series for business cycle fluctuations.

Estimation Details. We estimate the contemporaneous sensitivity to business cycle by setting $h = 0$ in Equation (5). In the GRF, we set the number of trees to 1,000 with equal weighting. We use honest splitting for sub-sample partitioning, allocating 50 percent of the data to build each tree and requiring a minimum of five observations per leaf. Observations are clustered at the firm level with equal weight, so firms with more observations receive greater weight, reducing the influence of entry and exit. Splitting is allowed across all characteristics, with the parameter controlling the maximum imbalance of a split set to 0.05. The GRF orthogonalizes the outcome and the shock with respect to firm characteristics before estimating heterogeneity (Robinson, 1988). Should macroeconomic confounding be a concern, additional macroeconomic variables can be residualized before estimation.

The linear counterpart to the GRF in Equation (7), which we refer to as the LPM, restricts the sensitivity to be linear in firm characteristics and is estimated by OLS. Specifically, we regress each outcome on the firm characteristics and on their interactions with the aggregate shock W_t , so that the interaction coefficients trace out a linear approximation to $\beta(x)$. The specification thus mirrors the GRF, replacing its flexible sensitivity function with a linear one.

We also estimate two alternative ML methods. For the neural-network R-learner, both the

¹²Notice that, while sales, investment, and debt are reported quarterly, employment is reported only at annual frequency.

¹³For instance, Ottonello and Winberry (2020) and Jeenas (2023) study the role of distance to default, leverage, and liquidity in the transmission of monetary policy shocks to investment; Alfaro et al. (2024) study the effects of uncertainty on financial variables such as liquidity and leverage; and Crouzet and Mehrotra (2020) focus on how size and industry scope shape the response to business-cycle fluctuations.

first-stage nuisance functions and the second-stage sensitivity are fit with feed-forward networks of three hidden layers of 128, 64, and 32 units. The nuisances are estimated by five-fold cross-fitting with folds assigned at the firm level, and the sensitivity is obtained by minimizing the residual-on-residual R-loss, with one fold held out for early stopping, and we train with the Adam optimizer.¹⁴

The random-forest learner instead reuses the same cross-fitted nuisances and estimates the sensitivity with a forest of 2,000 trees, making all characteristics available at each split and requiring a minimum of five observations per leaf. Each observation is weighted by $\tilde{W}_{i,t}^2$, and $\beta(x)$ is read off the out-of-bag predictions of the pseudo-outcome $\tilde{Y}_{i,t}/\tilde{W}_{i,t}$.

3.1 Firm-level Sensitivity

We estimate rich cross-sectional heterogeneity in firms' sensitivity to business cycle fluctuations. Figure 2 shows the distribution of firms' sensitivities to business cycle fluctuations for each outcome variable estimated using the GRF algorithm. A 1 percent increase in real GDP growth is associated, on average, with a 2.1 percent increase in firms' sales, somewhat below the 2.5–3.1 percent range reported by Crouzet and Mehrotra (2020) for QFR firms, consistent with our Compustat sample covering larger, less cyclical firms. Similarly, investment increases by 0.9 percent, while employment and debt issuance increase by approximately 1.2 percent, in line with economic intuition. However, the average sensitivity masks substantial cross-sectional heterogeneity in how firm outcomes co-move with aggregate fluctuations. A large share of firms exhibit sensitivities that are more than twice as large as the median; in the case of investment and debt issuance, some even carry the opposite sign from the average, falling as low as -1 and -2 percent, respectively. We use the Chernozhukov et al. (2023) test to assess the statistical significance of the heterogeneity in β by comparing the prediction from the forest to the predictions that consider only the average effect. We strongly reject the null hypothesis of no heterogeneity for all outcome variables (Appendix Figure C.1).

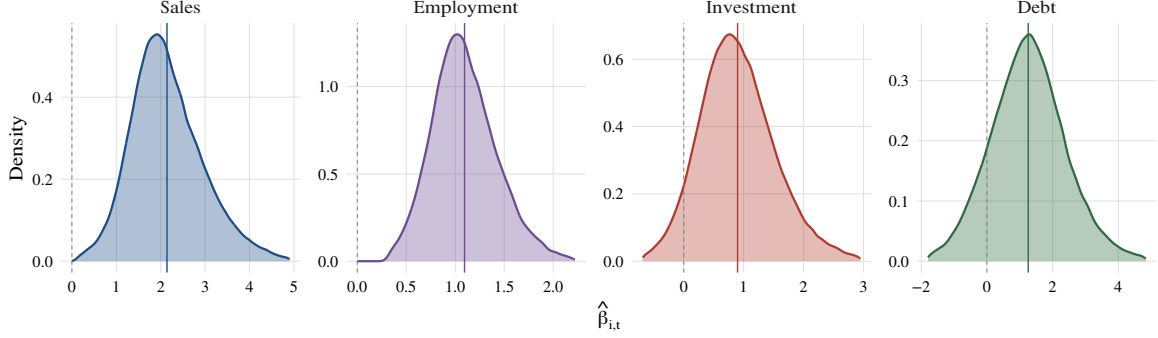
We compute the rank correlation between the estimated firm-level sensitivities across outcomes, to assess whether firms that are more cyclical along one margin are also more cyclical along others.

¹⁴Following standard practice, the networks use ELU activations, batch normalization, and dropout rates between 0.025 and 0.075 depending on the outcome, and all inputs and targets are standardized. The optimizer uses a learning rate of 10^{-3} and L_2 weight decay of 10^{-5} for the nuisances and 10^{-4} for the sensitivity, halving the learning rate on validation-loss plateaus, with early stopping (up to 300 epochs for the nuisances and 500 for the sensitivity) and mini-batches of 1,024 to 4,096 observations. The R-loss is defined as

$$\frac{1}{n} \sum_{i,t} \tilde{W}_{i,t}^2 \left(\tilde{Y}_{i,t} - \beta(X_{i,t-1}) \tilde{W}_{i,t} \right)^2,$$

where $\tilde{Y}_{i,t} = Y_{i,t} - \hat{\mathbb{E}}[Y \mid X_{i,t-1}]$ and $\tilde{W}_{i,t} = W_t - \hat{\mathbb{E}}[W \mid X_{i,t-1}]$ are the residualized outcome and shock and n is the number of firm–period observations.

Figure 2: Distribution of Firm-level Sensitivity



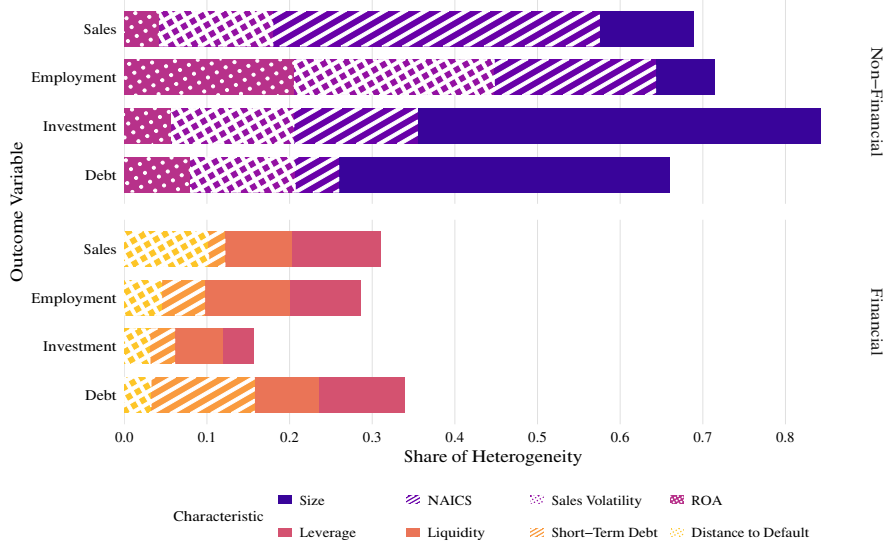
Notes: The figure shows the distribution of firm-level sensitivities to business cycle fluctuations estimated using the GRF algorithm. Each subplot represents a specific outcome variable. The vertical solid line indicates the average sensitivity. Firm-level sensitivities are trimmed at the 0.5% level on both tails.

The correlations are positive across all pairs but modest in magnitude, averaging roughly 20 percent and never exceeding 37 percent (Appendix Table C.2). A firm’s cyclicality along one margin is therefore only weakly informative about its cyclicality along another. This suggests that the mapping between firms’ characteristics and outcomes, $\beta(\mathbf{X})$ in Equation (5), differs across margins, reflecting distinct sources of heterogeneity and nonlinearities.

Drivers of Heterogeneity in β . We assess the role of firm characteristics in explaining the heterogeneity in business cycle sensitivities. Unlike traditional parametric models, the GRF algorithm estimates firm-level sensitivities without imposing a predetermined functional form, assigning greater weight to the most relevant covariates in the splitting stage. We measure the relevance of each characteristic by how often the algorithm relies on it when partitioning the data, captured by the depth-weighted frequency of splits in which the characteristic is used (Breiman, 2001). A characteristic that accounts for a larger share of splits contributes more to the estimated heterogeneity in β . Figure 3 reports the importance shares of each characteristic for each outcome.

Non-financial characteristics—especially industry scope and firm size—are the primary drivers of the heterogeneity in firm sensitivities to business cycle fluctuations. Taken together, non-financial characteristics guide most of the algorithm’s partitioning decisions—roughly 70 percent for sales and employment, 84 percent for investment, and 66 percent for debt issuance. Within non-financial characteristics, industry scope is the dominant driver of the heterogeneity in sales sensitivity, accounting for about 40 percent of the algorithm’s splits, while firm size accounts for nearly half of the splits explaining investment and about 40 percent of those for debt issuance. Non-financial characteristics still account for most of the heterogeneity in the sensitivity of employment, but their relative importance is more evenly distributed across sales volatility, profitability, and indus-

Figure 3: Importance of Firms' Characteristics



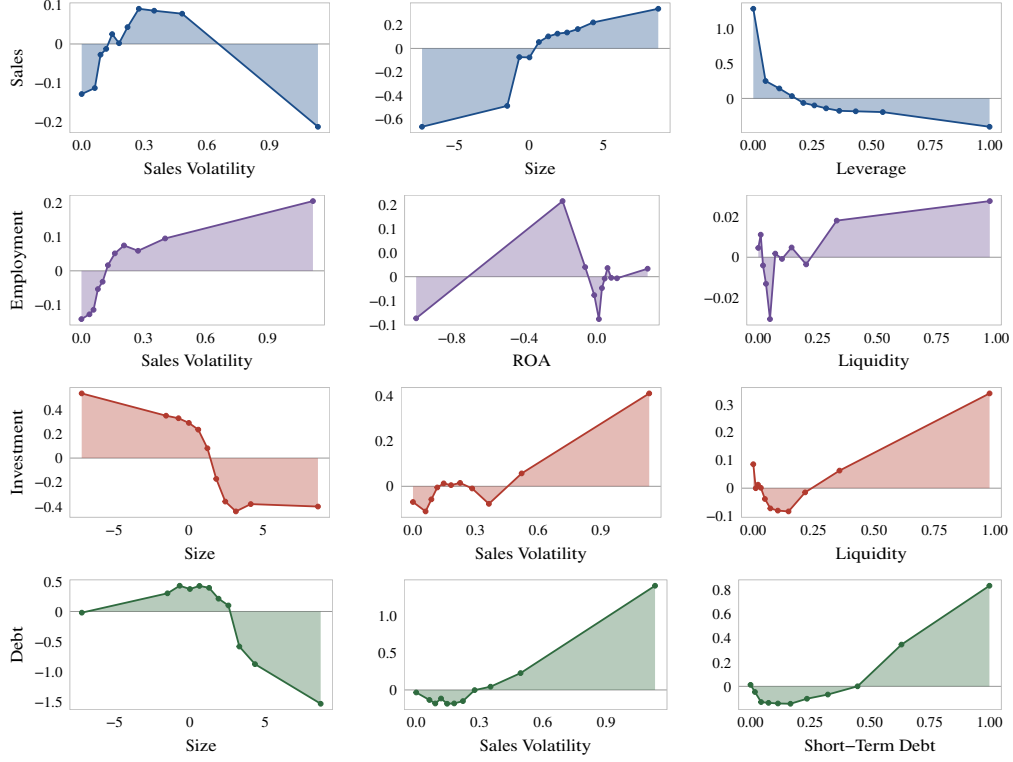
Notes: The figure shows the share of heterogeneity explained by each characteristic across different outcome variables. The share of heterogeneity explained by each characteristic is computed as the depth-weighted frequency of splits in the forest where the characteristic is used. Financial characteristics are shown in warm colors (reds to yellows), non-financial characteristics in cool colors (blues and pink). The x-axis shows the importance share, where a value of 0.01 corresponds to 1% of total heterogeneity.

try scope. By contrast, financial variables play only a marginal role, with individual shares below 15 percent.

Relationship between β and Firm Characteristics. We use Accumulated Local Effects (ALE) plots to illustrate how the marginal effect of each characteristic on predicted firm sensitivities varies across its distribution. A key advantage of the GRF algorithm over traditional linear panel regressions is that it estimates firm-level sensitivities without imposing a predetermined functional form on how firm characteristics influence them, thereby allowing for nonlinearities and interactions among X_{it} . ALE plots provide a way to visualize these relationships while accounting for correlations among characteristics. They display how predicted sensitivities change locally as each characteristic varies, averaging the model's local changes over the conditional distribution of the remaining characteristics within each interval.¹⁵ Steeper ALE slopes indicate stronger marginal

¹⁵ALE plots offer a more reliable alternative to the commonly used Partial Dependence Plots (PDP). PDPs average predictions over the marginal distribution of the other characteristics, which assumes the feature of interest is independent of them—an assumption that does not hold in our empirical setting and can lead the model to be evaluated at unrealistic characteristic combinations. ALE instead computes local effects within small intervals of a characteristic's observed range, conditioning on the local distribution of

Figure 4: Marginal Effects of Firm Characteristics



Notes: The figure presents the Accumulated Local Effects (ALE) plots estimated for each outcome and the three most important characteristics based on their share of importance. Each row corresponds to one outcome variable. The x-axis represents the value of the characteristic, and the y-axis shows the average deviation in the predicted firm sensitivity from the overall mean sensitivity to business-cycle fluctuations. Positive (negative) values indicate firms that are predicted to be more (less) sensitive than the average firm at that point in the distribution of the characteristic.

effects, whereas flatter segments denote regions where the characteristic has little influence on firm sensitivity. Figure 4 presents the ALE plots for the three most important characteristics by importance share for each outcome, excluding industry.

Firm size substantially influences firms' sensitivities of sales, investment, and debt, all else equal. Large firms are more cyclical in terms of sales than the average-sized firm, but they are the ones that increase investment and issue debt less during booms. Firms with more volatile sales tend to expand borrowing, investment, and employment more strongly during booms. Financial characteristics also play an important role: more financially sound firms (i.e., those with low leverage

the other variables, and thus better captures the marginal effect on predicted sensitivity in the presence of correlated covariates.

and greater liquidity) exhibit larger increases in sales, investment, employment, and debt than the average firm during expansions. Finally, firms with shorter debt maturities and lower leverage tend to expand borrowing more strongly in booms. Appendix Figure C.3 shows the ALE plots for all the other characteristics.

Firm characteristics influence business-cycle sensitivity in ways that a linear specification cannot easily capture, along two distinct dimensions. First, the relationship is nonlinear in individual characteristics. The ALE plots in Figure 4 display several kinks and slope changes, indicating that the marginal effect of a characteristic varies substantially over its distribution. For instance, the marginal effect of firm size on the sensitivity of all outcome variables is steep in the middle of the size distribution but becomes much weaker at the tails, indicating that changes in size have the largest effect on cyclical sensitivity for medium-sized firms. Similarly, the marginal effect of leverage on sales sensitivity declines sharply at low levels of leverage before flattening over the remainder of the distribution. In contrast, the marginal effects of sales volatility and the share of short-term debt on debt sensitivity exhibit a non-monotonic pattern, with firms at intermediate values of these characteristics displaying lower cyclical sensitivity than those at either end of the distribution. These nonlinearities are confirmed by several statistical tests—the Generalized Additive Model (GAM) and RESET tests—reported in Appendix C.2.

Second, characteristics interact with one another. Appendix Figure C.8 shows that interactions among firm characteristics account for up to 40 percent of the variance in predicted sensitivity explained by a given characteristic. We quantify the strength of these interactions using Friedman’s H-statistic, which measures the extent to which the joint contribution of two or more characteristics exceeds the sum of their individual effects.¹⁶ Firm size has the highest or second-highest H-statistic across outcomes, suggesting that much of its relevance operates through its influence on the effects of other characteristics. More broadly, non-financial characteristics display stronger interactions than financial ones, consistent with their key role in driving the heterogeneity of β . Taken together, these nonlinearities and interactions illustrate why a high-dimensional characteristic space and flexible methods are necessary to recover and understand the drivers of β .

3.2 Discussion

We discuss the interpretation of our results, relate them to the existing literature, and compare the GRF estimates with those obtained using alternative methods.

Interpreting the Results in the Context of the Literature. Despite their different microfoundations, models of credit market frictions share the prediction that financial constraints,

¹⁶The H-statistic ranges from zero (purely additive) to one (entirely joint), and compares the share of variance in sensitivities explained by interactions with the total variance explained by the characteristics.

often measured empirically using proxies such as firm size, generate heterogeneity in firms' responses to business-cycle fluctuations, implying that smaller firms are more cyclical than larger ones (Gertler and Gilchrist, 1994).

The estimated marginal effects in Figure 4 indicate that sales cyclicalities increase with firm size, measured by total assets, apparently at odds with the standard prediction that smaller firms are more cyclical.¹⁷ Appendix Figure C.7 reconciles this apparent discrepancy by showing that the relationship between size and sales sensitivity is highly nonlinear: cyclicalities rise with size across most of the distribution but decline among the very largest firms. Restricting attention to the largest firms reproduces the findings in Crouzet and Mehrotra (2020): firms in the top 1 percent of the size distribution are less cyclical than the remaining 99 percent. However, this comparison masks substantial heterogeneity within both groups. Even among the largest firms, cyclical sensitivities vary considerably, reflecting the influence of other characteristics. Moreover, among the bottom 99 percent, the relationship is strongly non-monotonic, with firms in the bottom quartile exhibiting lower cyclicalities than firms of intermediate size. These findings illustrate one of the key advantages of GRF and machine-learning methods more generally. Rather than requiring the researcher to define arbitrary size groups *ex ante*, the method flexibly uncovers nonlinearities over the entire size distribution while simultaneously accounting for interactions with other firm characteristics.

We find that employment is less cyclical among firms in the right tail of the size distribution, in contrast to Moscarini and Postel-Vinay (2012), who emphasize the role of large firms in driving employment fluctuations. Instead, it is more consistent with Sahin et al. (2011), who document that employment declined less among small firms during the 2007–09 recession. More broadly, however, firm size explains relatively little of the cross-sectional heterogeneity in employment sensitivity. Figure 3 shows that sales volatility is by far the most important non-financial determinant of employment cyclicalities, whereas firm size contributes the least. Together with the limited role of leverage and liquidity, these findings suggest that heterogeneity in firms' employment responses is driven primarily by differences in demand conditions rather than by financing constraints.¹⁸

Appendix Figure C.7 shows that firms in the right tail of the size distribution also exhibit less cyclical investment and debt issuance, consistent with Crouzet and Mehrotra (2020) and Coval and Haan (2011). As with sales, however, the binary comparison again conceals substantial

¹⁷Several studies also find little systematic difference in the cyclicalities of sales between small and large firms Chari et al. (2013); Kudlyak and Sanchez (2017).

¹⁸Using U.S. data from the 1960s through the 1980s, Sharpe (1994) finds that highly leveraged firms exhibit more cyclical employment. Similarly, Dinlersoz et al. (2024) and Giroud and Mueller (2015) document qualitatively similar patterns during the Great Recession using Orbis data and a combination of Compustat and establishment-level data, respectively. In contrast, we find little systematic heterogeneity in employment cyclicalities across the distributions of leverage and liquidity. The only notable exception is that firms in the bottom 20 percent of the leverage distribution appear to be more sensitive than average.

nonlinearities across the rest of the distribution. For investment, the marginal effect of firm size declines sharply throughout the bottom 99 percent of the distribution before flattening among the very largest firms, whereas for debt issuance it is relatively flat outside the extreme upper tail. These patterns suggest that classifying firms solely according to the top of the size distribution overlooks important variation in cyclical sensitivities among the remaining firms. The literature also emphasizes the role of financing conditions in shaping investment dynamics. For example, [Kalemli-Özcan et al. \(2022\)](#) show that European firms entering the 2008 financial crisis with higher leverage reduced investment more sharply, particularly when they relied heavily on short-term debt and were therefore more exposed to rollover risk. By contrast, we find little evidence that either leverage or the share of short-term debt systematically explains heterogeneity in investment cyclicality, possibly reflecting differences in the sample, period, or identification strategy.¹⁹

Comparison to Alternative Methods. One advantage of GRF relative to an LPM is that it relaxes the assumption that firm characteristics affect sensitivities to aggregate shocks linearly. Table 1 shows that, across all outcomes, the average sensitivities estimated by GRF and the LPM are statistically indistinguishable, suggesting that both approaches recover the same average effect of business-cycle fluctuations. The two methods differ substantially, however, in their firm-level estimates. Figure 5 reports the distribution of the discrepancy between the GRF and LPM estimates, expressed as a percentage of the average sensitivity for each outcome. These differences can be as large as 100 percent in absolute value and, for some firms, even differ in sign, indicating that firm characteristics shape responses to business-cycle fluctuations in complex and nonlinear ways that a linear specification cannot capture.

Appendix Figure C.2 illustrates the complexity of the decision trees underlying the GRF estimates by reporting both the total number of splits (tree depth) and the number of distinct characteristics used to partition the sample. The typical terminal leaf is defined by approximately six to eight different characteristics, indicating that most observations are classified through high-order interactions among firm characteristics. Moreover, the large number of splits relative to the number of characteristics implies that the algorithm repeatedly partitions the same variables, uncovering nonlinear relationships by approximating them with fine step functions. Replicating this structure in a linear panel model would require the researcher to specify ex ante which interaction terms and higher-order polynomials to include, a task that is both difficult and arbitrary.

We compare the GRF micro-sensitivity estimates with those from two alternative R-learners ([Nie and Wager, 2021](#))—one based on a random forest (RF) and one on a neural network (NN)—

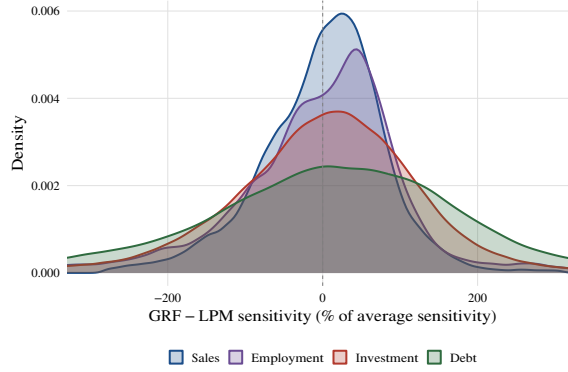
¹⁹[Begenau and Salomao \(2019\)](#) show that, during expansions, large firms rely relatively more on debt financing while distributing equity payouts, whereas small firms issue both debt and equity. By contrast, we find that the largest firms exhibit less cyclical debt issuance than average-sized firms. Instead, stronger debt responses are concentrated among firms with a higher share of short-term debt and more volatile sales.

Table 1: Comparison GRF - LPM: Distribution Moments

Outcome variable	GRF				Linear Panel Model			
	Mean	St. Dev.	Skewness	Kurtosis	Mean	St. Dev.	Skewness	Kurtosis
Business Cycle								
Sales	2.14	0.80	0.49	3.24	2.13	1.82	0.44	3.76
Employment	1.09	0.33	0.41	3.11	1.13	1.30	0.52	6.26
Investment	0.90	0.62	0.36	3.09	0.88	1.18	0.27	5.93
Debt	1.26	1.12	0.14	3.01	1.13	2.41	-0.04	5.35

Notes: The table presents the summary statistics of the estimated firm-level sensitivities obtained from the GRF and the Linear Panel Model (LPM) regression model across different outcome variables. Metrics include the mean, standard deviation, skewness, and kurtosis for each method.

Figure 5: Comparison GRF - LPM: Firm level



Notes: The figure presents the kernel density estimates of the difference between GRF and LPM firm-level sensitivities as a percentage of the average sensitivity across the four dependent variables: Sales, Employment, Investment, and Debt. The x- axis represents the percentage difference between GRF and LPM estimates, calculated as $(\text{GRF} - \text{LPM})/D \times 100$, where D is the average GRF sensitivity. The densities highlight the distribution of deviations for each dependent variable, with colors indicating the specific variable.

to assess the robustness of the results across estimation methods. Appendix Figure C.6 reports the Spearman rank correlation between the GRF estimates and each alternative, by outcome. The alternatives recover similar rankings of firms, but to different extents. The GRF–NN rank correlation is substantial, between 40 and 60 percent across outcomes—roughly twice the GRF–LPM correlation—while the GRF–RF correlation lies in between, above the LPM but below the NN. The stronger agreement between GRF and the NN suggests that the choice of second-stage estimator matters for recovering heterogeneous effects. GRF and NN agree more closely because both methods produce smooth, heavily regularized surfaces and are less sensitive to the high variance of the local targets that a plain forest fits.²⁰ The difference is not only in the loss function. Although both GRF

²⁰All three share the same residualization and differ only in how $\beta(x)$ is fit. The RF R-learner regresses the pseudo-outcome $\hat{Y}/\hat{W}$, which is high-variance wherever the residualized shock $\hat{W}$ is close to zero, and a greedy, non-honest forest partly fits that noise. The NN minimizes the same R-loss without forming the

and RF are forest-based methods, GRF chooses splits to maximize treatment-effect heterogeneity and employs honest sample splitting, whereas the RF R-learner chooses splits to minimize prediction error of the pseudo-outcome and does not use honesty. As a result, the RF estimates tend to be more dispersed and exhibit more extreme values of $\hat{\beta}$.

3.3 Aggregate Response to Business Cycle

With the set of firm-level sensitivities estimated using GRF, we can now apply the accounting framework presented in Section 2 to gauge the relevance of firm heterogeneity for the aggregate economy. Appendix Figure B.4 shows that the distribution of firms' weights, ω_i , is highly unequal, with a small number of firms accounting for a disproportionately large share. Appendix Table B.3 shows that the rank correlation of firm weights across outcome variables is above 75 percent, indicating that the firms most relevant for aggregate fluctuations are largely the same across different outcome variables. Depending on whether larger firms—those accounting for a greater share of the outcome variable—tend to be more or less sensitive than the average firm, the aggregate impact of business cycle fluctuations can be muted or amplified by firm heterogeneity.

To quantify the aggregate implications of heterogeneity in firm-level sensitivities, we first construct the aggregate response $\hat{G}_t$ by weighting the predicted firm-level responses $\hat{g}_{i,t}$ —based on the estimated sensitivities $\hat{\beta}$ —by each firm's lagged share $\omega_{i,t-1}$, as in Equation (3). We then estimate the average aggregate effect of business cycle fluctuations with the time-series regression

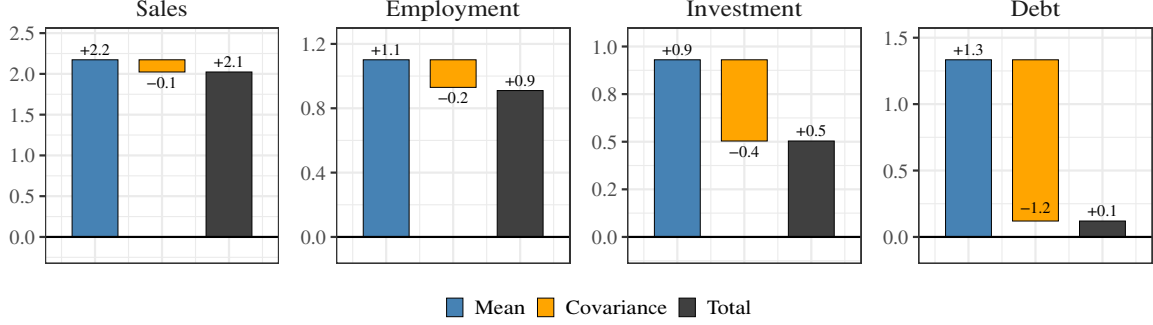
$$\hat{G}_t = \alpha + \gamma W_t + \varepsilon_t, \quad (8)$$

where γ measures the average aggregate response of a given outcome to a one-percentage-point change in real GDP growth, reflecting both the average firm sensitivity and the contribution of the covariance between sensitivities and weights. To separate the two, we apply the decomposition in Equation (4) and estimate Equation (8) separately for the mean and covariance terms. Figure 6 shows the aggregate response and its decomposition for each outcome.

We find that firm heterogeneity dampens the aggregate response to business cycle fluctuations, but its magnitude depends on the outcome variable. The unweighted average response is positive for all outcomes, in line with economic intuition: sales, investment, debt issuance, and employment all rise when the economy expands. However, for all variables, the covariance between firms' sensitivities to the business cycle and their economic weights is negative, so the aggregate response falls short of the average firm response as more relevant firms are less sensitive to business cycle fluctuations. The negative covariance term reduces the aggregate response of sales, employment,

ratio explicitly and under heavy regularization, so high-variance individual targets are smoothed out; GRF solves a locally-weighted moment condition that is likewise far more stable than averaging $\tilde{Y}/\tilde{W}$ leaf by leaf.

Figure 6: Aggregate Response to Business Cycle



Notes: The figure illustrates the decomposition of aggregate responses into mean and covariance terms for each outcome variable. The black bar denotes the total average aggregate response. We estimate Equation (8) using the mean and covariance terms in Equation (4) as the dependent variable. The mean and covariance terms are constructed using the set of firm-level sensitivities estimated with the GRF algorithm.

and investment by about 5, 18, and 44 percent, respectively, and almost completely offsets it for debt issuance. The modest dampening in sales and employment indicates that firms that account for most sales and employment are only slightly less cyclical than the average firm. The more pronounced dampening in investment plausibly reflects that firms with larger weights are typically closer to their optimal scale, which reduces the marginal benefit of additional investment and limits its cyclical fluctuations (Cooper and Haltiwanger, 2006; Winberry, 2021). For debt, highly leveraged firms are less likely to issue new liabilities, since high debt levels reduce the marginal value of borrowing even under favorable conditions (Hennessy and Whited, 2005).

Within and Across Sector Heterogeneity. We show that both within-sector and across-sector heterogeneity equally contribute to the dampening of the aggregate response due to firms' heterogeneity. To assess this, we consider a counterfactual exercise where we assign each firm the median sensitivity of firms in its 5-digit NAICS sector and quarter, construct the implied aggregate response, and re-estimate Equation (8). This counterfactual preserves across-sector variation in sensitivities while removing within-sector variation. Appendix Figure C.10 shows that the mean component is virtually unchanged, while the covariance term falls by roughly one-half across outcomes. Thus, the dampening in Figure 6 reflects both across-sector heterogeneity—large sectors are less sensitive on average—and within-sector heterogeneity—larger firms within sectors are less sensitive than their sector peers. The fact that both margins of heterogeneity are equally significant for the aggregate response underscores the importance of accounting for both dimensions of heterogeneity.

Aggregate Role of Nonlinearities. We find that nonlinearities in firm-level sensitivities are not only prevalent at the micro level but may also influence the aggregate response of outcomes to macroeconomic fluctuations. To evaluate their aggregate impact, we construct the economy-wide response using firm-level sensitivities estimated from both GRF and the LPM, and compare the average aggregate effect γ from Equation (8) across the two methods. Panel A of Appendix Table C.4 shows that the point estimates differ by approximately 25 basis points for sales, and approximately 10 for employment and investment; the difference is instead small and not statistically significant for debt. Appendix Figure C.11 shows that the differences in the aggregate response of sales and investment are primarily driven by the covariance term, as both methods deliver similar average sensitivities, but the LPM understates how sensitivities covary with firm weights. These findings indicate that nonlinearities in firm sensitivities can influence the aggregate role of firm heterogeneity, underscoring the importance of estimating firm-level sensitivities without imposing linearity in characteristics.

Additional Robustness. We assess the robustness of the aggregate results using alternative weights to the outcome-based shares implied by the accounting framework in Section 2.1. Appendix Figure C.9 reports the same decomposition using firm size, measured by total assets, as the aggregation weight. The results are quantitatively unchanged, suggesting that our findings are not driven by the particular weighting scheme implied by the accounting framework but instead reflect systematic differences in firms' cyclical sensitivities.

We also assess the robustness of the results to the inclusion of firm fixed effects in the estimating equation, following Ottonello and Winberry (2020) among others. Appendix Figure C.13 and Table C.3 show that the distributions of the estimated firm-level sensitivities obtained from both the GRF and the LPM remain very similar across specifications, as reflected in the high rank correlations between the baseline and firm fixed-effects estimates. Appendix Figure C.14 further shows that firm heterogeneity continues to dampen the aggregate response to business-cycle fluctuations, although the magnitude of the dampening is somewhat smaller when firm fixed effects are included. This reduction is consistent with the dampening arising from between-firm differences: by absorbing the between-firm variation in sensitivities, firm fixed effects attenuate the covariance between firms' shares and their sensitivities.

4 The Dynamic Response to Exogenous Shocks

As a second application, we study the dynamic responses of firm-level and aggregate outcomes to exogenous shocks, specifically monetary policy shocks and oil price shocks.

Estimation Details. We measure monetary policy shocks using interest-rate surprises around Federal Reserve announcements, identified from high-frequency variation in 3-month federal funds rate futures and orthogonalized with respect to past macroeconomic and financial information following [Bauer and Swanson \(2023\)](#). The oil price shocks are structural oil supply news shocks identified using high-frequency OPEC announcement surprises following [Känzig \(2021\)](#) in an augmented VAR that incorporates financial variables, as proposed by [Mori and Peersman \(2024\)](#). Before estimation, we normalize each shock using the unit-effect procedure of [Stock and Watson \(2018\)](#). Specifically, we instrument an endogenous variable with the corresponding shock and use the fitted values from the first-stage regression as the normalized shock. This expresses each shock as a one-unit change in the corresponding endogenous variable. The normalized shock is then used as the treatment variable in both the GRF and the LPM. For monetary policy shocks, the endogenous variable is the one-year change in the one-year US government bond yield, while for oil price shocks it is the one-year change in the oil price index (WTI).

We estimate the horizon-specific sensitivity functions $\beta_h(x)$ of Equations (5) and (7) for $h = 0, 1, \dots, 4$, where the outcome at horizon h is the cumulative growth of the firm-level variable between $t - 1$ and $t + h$, computed with the [Davis and Haltiwanger \(1992\)](#) formula. Following standard convention, $h = 0$ denotes the response on impact. In the aggregation step, for each horizon h , we weight firm-level responses by the pre-shock shares $\omega_{i,t-1}$ and decompose the aggregate impulse response into its mean and covariance components using the framework of Section 2.

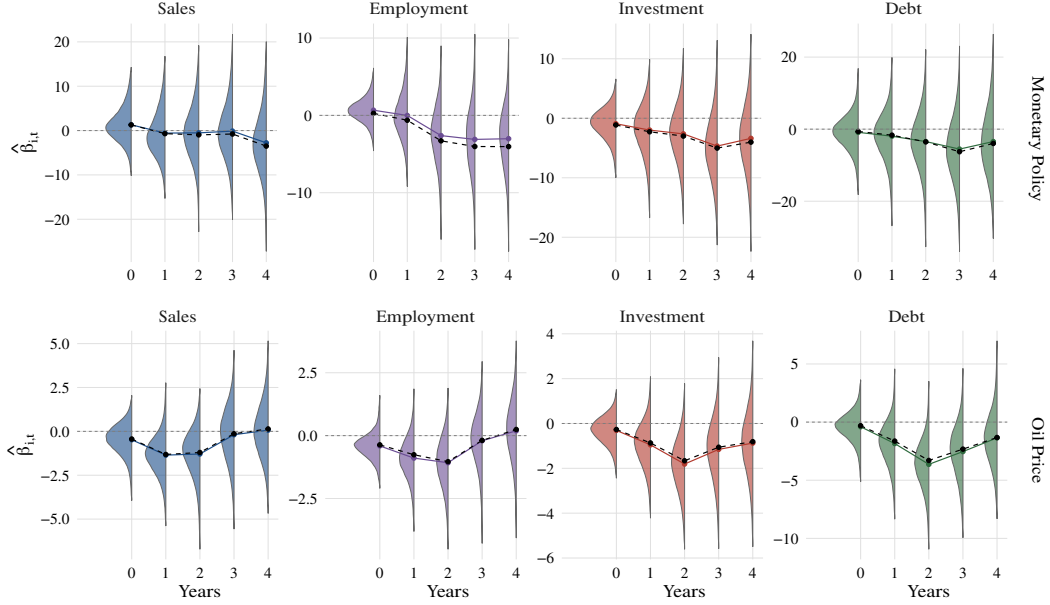
4.1 Firm-level Sensitivities

Figure 7 reports the full cross-sectional distribution of firm-level sensitivities estimated by the GRF at each horizon h .

Monetary Policy. A one-percentage-point increase in interest rates progressively reduces the average firm’s activity across all outcomes, with the magnitude of the response reaching between 2.5 and 3.5 percent at longer horizons. Despite the broadly similar long-run magnitude, the speed of adjustment differs across margins: investment and debt issuance respond more quickly, while sales and employment adjust more gradually, with their responses building over a longer horizon. The slower adjustment of output and employment is consistent with the gradual transmission of monetary policy to real activity and prices ([Gertler and Karadi, 2015](#); [Christiano et al., 2005](#)).

Around these average responses, firm-level heterogeneity is substantial and increases with the horizon, with 20 to 35 percent of firms displaying sensitivities of the opposite sign to that of the average firm. Understanding the sources of this heterogeneity has been central to a large literature on the transmission of monetary policy. Appendix Figure C.15 shows that financial characteristics are substantially more important than non-financial characteristics in explaining

Figure 7: Distribution of Firm-level Sensitivity



Notes: The figure reports the distribution of firm-level GRF sensitivities $\hat{\beta}_h(x_{it})$ by forecast horizon, for each shock and outcome. The top panel refers to monetary policy shocks, the bottom panel to oil price shocks. Points and connecting lines mark the average estimated response at each horizon. The dashed line reports the average response from a standard local projection estimated by OLS with firm characteristics as controls.

firms' heterogeneous responses to monetary policy. Among financial characteristics, default risk is the most important across outcomes, followed by liquidity for investment and leverage for debt issuance. These findings provide data-driven support for the emphasis of the monetary policy literature on heterogeneity in firms' financial conditions (Ottonello and Winberry, 2020; Jeenas, 2023; Jungherr et al., 2024; Durante et al., 2022). By contrast, industry scope and firm size—key drivers of heterogeneity in business-cycle sensitivities (Crouzet and Mehrotra, 2020)—account for little of the heterogeneity in responses to monetary policy shocks. The limited role of size stands in contrast to Gertler and Gilchrist (1994), though our Compustat sample excludes the small private manufacturers that drive their result.

Appendix Figure C.19 plots the average sensitivity of investment to a monetary tightening across percentiles of the main firm characteristics studied in the literature, revealing patterns that are broadly in line with, though not identical to, previous findings.²¹ Firms with lower leverage and greater distance to default experience larger declines in investment following an interest rate increase, consistent with Ottonello and Winberry (2020) and Paranhos (2025a). However, the

²¹ALE plots for all outcomes and characteristics are reported in Appendix Figure C.4.

GRF points to an inverted U-shaped relationship over the distance-to-default distribution, with the marginal effect flattening in the upper tail, similar to [David et al. \(2026\)](#).²² Consistent with [Gertler and Gilchrist \(1994\)](#), investment also responds more strongly among smaller firms, although the quantitative differences are modest, in line with [Crouzet and Mehrotra \(2020\)](#). In contrast to [Jeenas \(2023\)](#), we find larger investment declines among more liquid firms, with the difference particularly pronounced at the upper end of the liquidity distribution. Finally, consistent with [Jungherr et al. \(2024\)](#), investment responses become more negative with the share of short-term debt, although the effect is concentrated in the upper tail of the distribution. Appendix Figure C.20 performs a similar exercise for employment; while [Bahaj et al. \(2022\)](#) find that employment responses to monetary policy are strongest among younger, more leveraged firms in the UK, our US sample shows little systematic heterogeneity in employment sensitivities along the leverage distribution. Overall, the GRF recovers several established cross-sectional patterns while revealing that financial conditions shape firms' responses to monetary policy in a nonlinear and margin-specific way.

Oil Price. An unexpected increase in oil prices temporarily reduces the average firm's real activity and debt issuance. The average firm-level sensitivity is negative on impact, peaks about three years after the shock, and gradually reverts toward zero in the long run, consistent with the interpretation of oil price increases as adverse supply shocks ([Känzig, 2021](#)). At the peak, a 10 percent increase in the oil price lowers the average firm's sales by approximately 1.2 percent, employment by 1.1 percent, investment by 1.7 percent, and debt issuance by 3.5 percent.

A long-standing literature attributes the incidence of oil shocks to sectoral characteristics: employment losses are concentrated in energy- and capital-intensive industries ([Davis and Haltiwanger, 2001](#)), with oil-intensive sectors affected primarily through supply channels and other sectors through demand channels ([Lee and Ni, 2002](#); [Jo et al., 2019](#)). In our sample, however, grouping firms into oil-intensive and other sectors reveals little systematic difference in their employment responses (Appendix Figure C.21). Instead, we find substantial time variation in the average effect, which turns positive after 2011, coinciding with the expansion of U.S. fracking and shale production that made higher oil prices expansionary for parts of the U.S. economy ([Bjørnland and Skretting, 2024](#)) (Appendix Figure C.16).²³

The drivers of heterogeneity in the response to oil shocks differ markedly from those of monetary policy shocks, highlighting the distinct features of each aggregate shock. While

²²In the implementation, [David et al. \(2026\)](#) use a sieve-based nonparametric LP estimator to estimate the unknown state-dependent coefficient function.

²³Notably, the GRF detects this structural change in the employment response without requiring the timing of the break to be specified *ex ante*. We do not find a similar break in sales or investment, suggesting that the change in the employment response reflects a distinct adjustment in firms' labor demand rather than a broad-based shift in the response of real activity to oil price shocks.

heterogeneity in the response to monetary policy is primarily driven by firms’ financial conditions, oil price sensitivities reflect a more balanced mix of financial and non-financial characteristics, with firm size emerging as the single most important characteristic on average; only sales volatility and distance to default for some outcomes account for more than 30 percent of the heterogeneity (Appendix Figure C.15). This broader set of relevant characteristics suggests that oil price shocks operate through a wider range of firm-level channels than monetary policy shocks, spanning production costs, demand, and financing conditions.²⁴

Importantly, for both exogenous shocks, the average of the estimated $\beta_h(x)$ closely tracks the average effect from a standard local projection estimated by OLS with characteristics as controls (Figure 7, dashed line). This mirrors our findings for business-cycle fluctuations in Section 3.2, where the GRF and the linear benchmark deliver statistically identical average sensitivities but diverge markedly in the cross-section. Around this common mean, the GRF uncovers substantial heterogeneity in firm-level responses without imposing parametric restrictions on the underlying data-generating process. This heterogeneity increases with the horizon, with the dispersion in sensitivities roughly doubling relative to the contemporaneous response. Despite this increasing dispersion, we find no evidence of rank reversals across firms. Figure 8 assesses the persistence of firm-level sensitivities by grouping firms into quintiles based on their contemporaneous sensitivity. Across all cases, firms that are most responsive on impact remain the most responsive at longer horizons, although the magnitude of their responses varies substantially across horizons. This persistence appears consistent with the stability of the underlying drivers of heterogeneity: the relative importance of firm characteristics changes little across horizons (Appendix Figure C.15). The horizon therefore changes the magnitude of firms’ responses more than the characteristics determining their relative exposure to aggregate shocks.

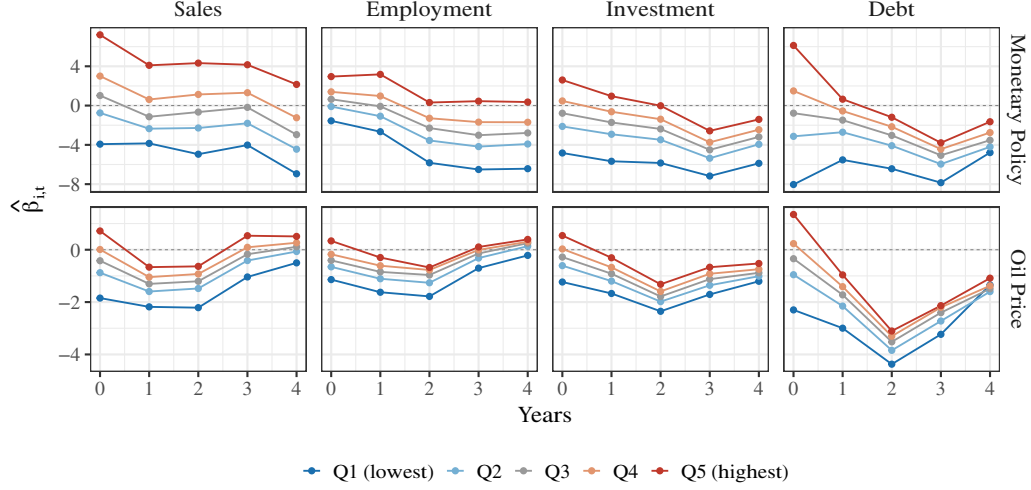
4.2 Aggregate Response

Leveraging Equations (3) and (4), we estimate Equation (8) separately for the two components at each horizon, allowing us to trace the aggregate response alongside the contribution of firm heterogeneity. Figure 9 reports the resulting decomposition for each shock–outcome pair.

Following a monetary tightening, aggregate responses build gradually: investment and debt decline on impact, while sales and employment adjust only with a lag. The magnitude of the

²⁴The role of firm size varies across margins. Larger firms exhibit greater sensitivity in sales and employment, plausibly reflecting their greater scale and broader exposure to changes in aggregate demand and input costs. Their greater size may also entail broader geographic or product-market exposure, amplifying their sensitivity to the effects of oil price shocks. In contrast, larger firms exhibit lower sensitivity in investment and debt issuance, consistent with greater access to internal funds and external financing that allows them to smooth these margins in response to fluctuations in oil prices (Appendix Figure C.5).

Figure 8: Persistence in Sensitivity over Quintiles

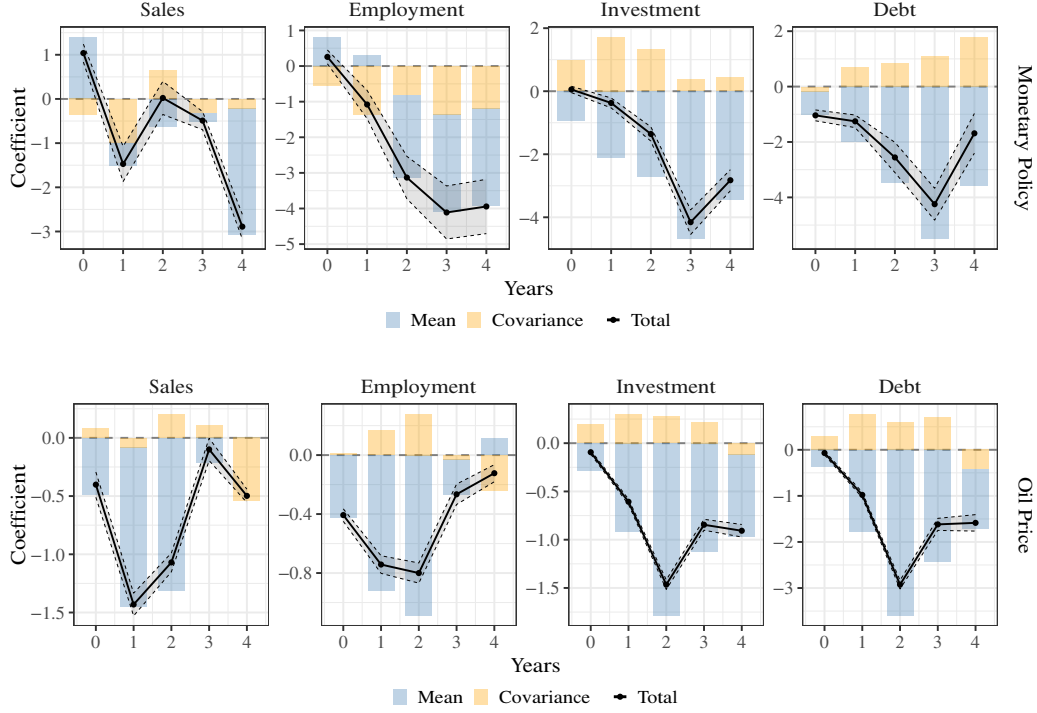


Notes: The figure reports the persistence of firm-level GRF estimates across the distribution of initial exposure. Firms are sorted into quintiles based on their estimated response at the initial horizon, and the lines trace the average estimated response for each quintile over subsequent years.

response increases with the horizon across all outcomes, consistent with the gradual transmission of monetary policy to economic activity and prices documented in the literature (Ramey, 2016). While the average firm-level response accounts for most of the aggregate response, the covariance between firms' shares and their responses operates in opposite directions across margins. On average, it dampens the contraction in investment and debt, as firms with larger shares respond less, but amplifies the decline in sales and employment, as firms with larger shares respond more. These effects also evolve differently over the horizon: the magnitude of the covariance term declines for investment, increases for debt, and remains broadly stable for sales and employment. Thus, heterogeneity systematically dampens monetary transmission through investment and debt but amplifies it through sales and employment, with its importance evolving over the horizon. The opposite signs across margins point to systematic differences in how monetary policy propagates through firms' investment, financing, sales, and employment decisions.

Following an exogenous increase in oil prices, the aggregate responses are negative across all outcomes on impact or shortly thereafter, build in magnitude over time, and peak around two to three years after the shock before gradually reverting toward zero. In contrast to a monetary policy shock, the covariance term is positive across all outcomes and remains broadly stable in magnitude over the horizon. Thus, cross-sectional heterogeneity consistently dampens the aggregate response to oil price shocks, with its contribution stable across margins and horizons. This suggests that the firms accounting for a larger share of aggregate activity are better able to absorb or pass through

Figure 9: Aggregate Response to Exogenous Shocks



Notes: The figure reports the horizon-specific decomposition of the aggregate response to monetary policy and oil price shocks based on the GRF-estimated sensitivities. For each outcome and horizon, the aggregate response is decomposed into the mean and covariance terms as in Equation (4). The black line denotes the total aggregate response, equal to the sum of the two components. Shaded gray areas and dashed lines delimit 95 percent confidence intervals.

oil-price shocks, so the composition of economic activity attenuates their aggregate transmission.²⁵

Comparison with time series methods. Prior works that studied the dynamic responses to monetary policy and oil price shocks focused on aggregate time series, mainly in VAR settings. Despite using a distinct methodology that aggregates responses entirely from micro-level estimates,

²⁵Appendix Figure C.17 shows that, over the past three decades, the aggregate response to oil price shocks has declined in magnitude, driven primarily by a decline in average firm-level sensitivity. This pattern is consistent with structural changes that have made firms better able to absorb supply shocks. In contrast, the responses of investment and debt to monetary policy have increased in magnitude since the Great Financial Crisis. In this case, the change reflects a combination of lower average sensitivity and a higher covariance term, with the two components contributing in opposite directions to the evolution of the aggregate response. Finally, Appendix Figure C.18 shows that, as for business cycle fluctuations in Section 3.3, most of the differences in aggregate responses between the GRF and LPM-OLS estimates are accounted for by the covariance term.

we obtain aggregate responses that are remarkably similar in both shape and magnitude. For monetary policy shocks, the negative and persistent response of sales matches similar dynamics of industrial production (both proxies for output in the US) as in [Gertler and Karadi \(2015\)](#), while the employment response is consistent with [Caldara and Herbst \(2019\)](#).²⁶ On the investment side, [Jeenas \(2019\)](#) extends the VAR framework of [Gertler and Karadi \(2015\)](#) and finds that an increase in the one-year rate causes nonresidential investment to begin declining approximately four quarters after the shock, and reach about -2 percent after twelve quarters, again in line with our findings.

Our aggregate results are in line with the empirical findings from the literature on oil supply shocks. Aggregate sales drop by 1 percent after two years, in line with [Känzig \(2021\)](#), who uses industrial production, as well as the delayed dynamics for aggregate investment (reduction around 1 percent within the first two years after the shock). We confirm the hump-shaped pattern for the response of aggregate employment, although [Känzig \(2021\)](#) uses the unemployment rate. More broadly, our results are in line with the magnitudes and dynamic responses documented in aggregate time-series evidence based on identified oil supply shocks ([Kilian, 2008, 2009](#); [Baumeister and Peersman, 2013](#); [Baumeister and Hamilton, 2019](#)).²⁷

While delivering aggregate responses broadly consistent with those identified in the time-series literature, our bottom-up approach offers the additional advantage of tracing their underlying drivers and quantifying how firm-level heterogeneity shapes aggregate transmission.

5 Additional Results

This section presents further discussion and additional results. Appendix D provides details.

Firm Age. Several works have highlighted the importance of firm age for the sensitivity of investment and employment to monetary policy shocks ([Cloyne et al., 2023](#); [Singh et al., 2022](#)) and business cycle fluctuations ([Fort et al., 2013](#)). For a subset of observations, we augment the baseline specification with firm age, measured as the log of years since founding, with founding dates from Ritter’s IPO data supplemented by web scraping. Adding firm age to the characteristic space produces some variation in the estimated firm-level sensitivities, particularly for monetary policy shocks (Appendix Figure D.4). Nevertheless, most observations continue to cluster around

²⁶[Caldara and Herbst \(2019\)](#) use the unemployment rate instead of employment growth; however, the delayed response, peaking around two to three years after the shock, is consistent with our results. Magnitudes are consistent as well: they find a peak of 5 percent in the unemployment rate, while our results indicate a 3-4 percent drop in employment growth.

²⁷Direct evidence on how firm debt responds to oil shocks is scarce in the existing VAR literature; the debt response documented here therefore extends that work by linking oil supply shocks to firms’ outcomes and balance-sheet characteristics.

the 45-degree line across outcomes and shocks, indicating that the overall pattern of heterogeneity is largely preserved.

We find that firm age accounts for a modest share of the heterogeneity in firm-level sensitivities—between 5 and 10 percent on average across most outcome–shock pairs. The main exception is the response of investment to business-cycle fluctuations, for which age explains more than 20 percent of the heterogeneity across firms (Appendix Figure D.5). Appendix Figure D.6 reports the accumulated local effects of age and shows that investment is more cyclical for younger firms. However, this relationship is nonlinear, with the marginal effect of age flattening in the right tail of the age distribution.

We do not find a systematic pattern in how the importance of the other firm characteristics changes when age is added to the characteristic space, suggesting that the role of age is outcome- and shock-specific and interacts with different firm characteristics across specifications (Haltiwanger et al., 2013; Dinlersoz et al., 2024). While the estimated marginal effects are broadly consistent with previous findings (Fort et al., 2013; Cloyne et al., 2023; Singh et al., 2022), age is generally not among the most important determinants of heterogeneity. At the aggregate level, Appendix Figure D.7 shows that including firm age has only a minor effect on the mean–covariance decomposition. The small differences that do emerge are driven almost entirely by changes in the covariance component.

Overall, while firm age contains useful information for explaining some dimensions of firm-level heterogeneity, it does not alter our main conclusions on the sources and aggregate relevance of heterogeneity. This likely reflects the composition of our sample, which is skewed toward relatively large and mature firms.

Market Value. We extend the analysis of firm-level heterogeneity and its drivers to financial market outcomes by studying the response of firms’ equity values to aggregate shocks, a topic that has received considerable attention in the literature, particularly in the context of monetary policy (Rigobon and Sack, 2003; Bernanke and Kuttner, 2005; Kaplan et al., 2018). Specifically, we estimate firm-level sensitivities of one-year changes in market value to business-cycle fluctuations, monetary policy shocks, and oil price shocks.

Consistent with the existing literature, firms’ market values respond strongly to aggregate fluctuations. On average, a one-percentage-point increase in real GDP growth raises market value by 4.3 percent. A contractionary monetary policy shock reduces market value by approximately 9 percent on impact, although this effect gradually dissipates over time. Similarly, oil price shocks lower market value by about 4 percent contemporaneously, but the response becomes positive after four years (Appendix Figure D.1).

Figure D.2 shows that the drivers of heterogeneity differ markedly across aggregate shocks. For business-cycle fluctuations and oil price shocks, non-financial characteristics account for most

of the cross-sectional variation in firm-level sensitivities. In particular, profitability and firm size explain roughly 30 and 20 percent of the heterogeneity in business-cycle responses, respectively, while firm size alone explains between 40 and 60 percent of the heterogeneity in responses to oil price shocks. By contrast, the contemporaneous response to monetary policy depends primarily on financial characteristics, with distance to default emerging as the single most important determinant, although financial and non-financial characteristics contribute roughly equally overall.²⁸ Since market values reflect both expected future cash flows and discount rates, these results suggest that differences in firms' operating exposure dominate the responses to business-cycle and oil price shocks, whereas financing conditions and risk premia play a comparatively larger role in the transmission of monetary policy.

Unlike the real outcomes considered previously, heterogeneity in market-value sensitivities is positively correlated with firms' aggregate importance, so that the covariance term amplifies, rather than dampens, the aggregate response. Appendix Figure D.3 reports the rolling mean–covariance decomposition of the contemporaneous aggregate effect. For both business-cycle fluctuations and monetary policy shocks, the covariance component has the same sign as the mean effect, amplifying aggregate market-value responses. This finding suggests that aggregate equity valuations are disproportionately driven by the largest firms: while these firms may be relatively insulated in terms of employment and investment responses, they account for a large share of aggregate market capitalization and exhibit stronger market-value responses to aggregate fluctuations.²⁹

Comparison across Shocks and Outcomes. Appendix Figure D.8 reports the Spearman rank correlation matrix of firm-level sensitivities across all outcome variables (sales, employment, investment, debt issuance, and market value) and aggregate fluctuations (business-cycle, monetary policy, and oil price shocks). This exercise assesses whether firms that are most sensitive for one outcome–shock pair also tend to be the most sensitive for another.³⁰

Overall, the rank correlations are modest, never exceeding 40 percent in absolute value, indicating that the cross-sectional ranking of firms depends substantially on the specific outcome

²⁸Appendix Figures C.3, C.4, and C.5 show that the marginal effects of firm characteristics on market-value sensitivities are also highly nonlinear.

²⁹Over the last three decades, market values have become increasingly sensitive to business-cycle fluctuations, largely because the average firm-level sensitivity has risen. By contrast, the aggregate response to monetary policy has remained broadly stable: the increase in average sensitivity has been largely offset by a decline in the covariance component, reflecting the fact that larger firms have become less responsive to interest-rate shocks. Finally, the aggregate response to oil price shocks has remained stable over time, with the covariance term remaining close to zero.

³⁰To facilitate the comparison across shocks, we redefine the exogenous shocks so that all correspond to expansionary events by reversing the sign of monetary policy and oil price shocks. Consequently, a positive shock represents either an expansionary monetary policy, a decline in oil prices, or an improvement in business-cycle conditions.

and aggregate shock under consideration. For a given aggregate fluctuation, sensitivities of sales and employment are relatively more strongly correlated with one another, as are sensitivities of investment and debt issuance, suggesting that these margins reflect similar underlying firm characteristics. By contrast, the sensitivity of market value is positively correlated with the sensitivities of sales and employment, but negatively correlated with those of investment and debt issuance.

Across aggregate shocks, sensitivities to business-cycle fluctuations are positively, although only weakly, correlated with sensitivities to negative oil price shocks. In contrast, sensitivities to monetary policy shocks exhibit little correlation with sensitivities to either business-cycle or oil price shocks. Appendix Figure D.9 further reports the rank correlations of the importance shares of firm characteristics across outcome–shock pairs. Consistent with the results in Section 4, the characteristics that are most important for explaining heterogeneity in responses to monetary policy shocks differ markedly from those driving heterogeneity in responses to business-cycle and oil price shocks. By contrast, business-cycle and oil price shocks display relatively similar rankings of characteristic importance, suggesting that they operate through more closely related channels.

Evolution Over Time. We examine whether the aggregate responses and their mean–covariance decomposition remain stable over time, given the rise of large firms (De Loecker et al., 2020) and the changes in the distribution of firm characteristics. We re-estimate Equation (8) using five-year rolling windows and decompose the estimated responses into mean and covariance components. Appendix Figure C.12 shows that both are remarkably stable for sales and employment, whereas investment and debt exhibit more pronounced long-run dynamics.

For investment, the covariance component has gradually declined, indicating that the alignment between firms’ cyclical sensitivities and their economic importance has weakened over the last three decades. An offsetting increase in the mean component of similar magnitude leaves the aggregate response largely unchanged. Investment sensitivities have thus become more homogeneous across firms, a pattern consistent with evidence that firm-level investment volatility has declined even as aggregate volatility has remained broadly stable (McConnell and Perez-Quiros, 2000; Stock and Watson, 2002; Davis and Kahn, 2008; Bachmann and Bayer, 2014; De Veirman and Levin, 2018). Debt issuance displays a different pattern: the covariance component remains broadly stable while the mean sensitivity has declined noticeably over the last fifteen years, making the aggregate response of debt issuance less cyclical over time.

State Dependency. A growing literature documents that the effects of monetary policy are asymmetric and state dependent, both empirically (Tenreyro and Thwaites, 2016; Ascari and Haber, 2022) and theoretically (Santoro et al., 2014; Bernstein, 2021; Eichenbaum et al., 2025). We estimate the difference in the aggregate response to monetary policy shocks between recessions and

expansions, decomposing it into mean and covariance components. Appendix Figure D.10 shows that, consistent with this literature, a monetary tightening has larger effects during expansions than during recessions across all outcomes—and that this state dependence is driven almost entirely by the covariance component. The average firm response is statistically indistinguishable across states for all outcomes except employment (Kurt, 2025), and the recession–boom gap is concentrated in the covariance term at every horizon.

Across horizons, the recession–boom gap in the aggregate response is concentrated almost entirely in the covariance term. Because a firm’s sensitivity is a fixed function of its characteristics, a stable mean term implies that the average firm’s responsiveness does not vary over the cycle; what moves is the covariance between firms’ economic weight and their sensitivity. State dependence in monetary transmission thus reflects a cyclical shift in the joint distribution of firm characteristics — in how responsiveness is aligned with the firms that carry the most aggregate weight — rather than a change in the response of a representative firm.

Firm Entry and Exit. Our baseline decomposition holds the set of firms fixed. Extending the framework to entry and exit does not alter the role of the mean and covariance terms, for a simple reason: the firm-level response $g_{i,t} = \beta(\mathbf{X}_{i,t-1}) \cdot W_t$ is defined only for firms present in both periods. It requires the lagged characteristics $\mathbf{X}_{i,t-1}$ to form the sensitivity and a realized outcome at t against which that sensitivity operates; entrants lack the former and exiters the latter. The mean and covariance are moments of $g_{i,t}$ across firms, so they are mechanically restricted to the set of continuing firms and are undefined for entrants and exiters by construction.

We extend the accounting framework of Section 2.1 to assess the cyclicity of entry and exit. Appendix Equation (D.16) shows that the extensive-margin effect is additively separable, with the mean and covariance defined exactly as in the fixed-firm case—over surviving firms, with weights summing to one—so the covariance channel retains the same interpretation once turnover is accounted for. The exit contribution always carries the opposite sign of the aggregate response and thus mechanically attenuates it; the sign of the entry contribution is governed by the covariance of entrants’ activity with the shock, and is positive when entry is procyclical.

Appendix Figure D.11 quantifies these contributions for business-cycle fluctuations. Exit acts as a drag on the aggregate response, but a quantitatively small one, amounting to less than 0.3 percentage points across all outcomes. Entry is procyclical for sales and investment, amplifying their aggregate responses, but dampens those of employment and debt, as entrants issue more debt during recessions. Overall, the contributions of entry and exit are comparable to or smaller than the role of firm heterogeneity captured by the covariance term, and the aggregate responses are broadly unchanged relative to the fixed-firm decomposition—with the exception of debt, where entrants’ countercyclical issuance is large enough to reverse the sign of the aggregate response.

6 Conclusions

This paper studies how firm heterogeneity shapes the response of the macroeconomy to aggregate shocks. We combine an accounting framework—which decomposes the aggregate response into the average firm response and the covariance between firms’ sensitivities and their economic weights—with the GRF algorithm of [Athey et al. \(2019\)](#), which estimates firm-level sensitivities as flexible, nonparametric functions of observable characteristics.

We apply this framework to firms’ sensitivities to business cycle fluctuations and to identified monetary policy and oil price shocks. Sensitivities vary substantially across firms, and the characteristics driving this heterogeneity are shock-specific: industry scope and firm size dominate for business cycle fluctuations, with financial characteristics playing only a limited role; default risk, liquidity, and debt maturity dominate for monetary policy shocks; for oil price shocks, heterogeneity reflects a more balanced mix of financial and non-financial characteristics. In the aggregate, a common pattern emerges. For most outcomes, shocks, and horizons, firms with larger aggregate weights tend to be less sensitive than the average firm, generating a covariance term that offsets the average sensitivity. Firm heterogeneity therefore generally dampens aggregate responses.

Our framework provides a transparent characterization of how heterogeneity in firm-level responses shapes aggregate outcomes, while abstracting from the underlying structural mechanisms and from feedback between the firm distribution and aggregate dynamics. We view the two as complements: the rich patterns of heterogeneity we document provide quantitative empirical targets for structural models of firm heterogeneity, and the framework offers a flexible way to uncover such targets as micro data become richer and machine-learning methods more widely used in macroeconomics.

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Appendix

A Methodological Details

A.1 Generalized Random Forest - Algorithm

The Generalized Random Forest (GRF) estimator of [Athey et al. \(2019\)](#) builds on the random forest methodology of [Breiman \(2001\)](#). Standard random forests use recursive partitioning and subsampling to estimate conditional expectations. In particular, a random forest prediction $\hat{m}(x)$ for an outcome $Y_{i,t}$ can be written as a forest-weighted average,

$$\hat{m}(x) = \sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) Y_{i,t}. \quad (\text{A.1})$$

The weights $\alpha_{i,t}(x)$ are large for observations that frequently fall in the same terminal leaves as the target point x across the trees in the forest.

GRF generalizes this logic from prediction to parameter estimation. Rather than using forest weights only to estimate a conditional mean, GRF uses these weights to solve local moment equations. Let $O_{i,t} = (Y_{i,t}, W_t, \mathbf{X}_{i,t-1})$ denote the observables, where $Y_{i,t}$ is the firm-level outcome, W_t is the aggregate shock, and $\mathbf{X}_{i,t-1}$ is the vector of predetermined firm characteristics. In the general GRF framework, the target parameter is denoted by $\theta(x)$ and is defined by the conditional moment restriction

$$\mathbb{E} [\psi_{\theta(x), \nu(x)}(O_{i,t}) \mid \mathbf{X}_{i,t-1} = x] = 0, \quad (\text{A.2})$$

where $\psi_{\theta, \nu}(\cdot)$ is a moment function and $\nu(x)$ collects nuisance functions. In our application, the generic parameter $\theta(x)$ is the conditional sensitivity of firm outcomes to the aggregate shock. We therefore set $\theta(x) \equiv \beta(x)$, where $\beta(x)$ measures the local sensitivity of $Y_{i,t}$ to W_t for firms with characteristics $\mathbf{X}_{i,t-1} = x$.

The empirical analogue of (A.2) is obtained by solving a forest-weighted sample moment condition,

$$(\hat{\theta}(x), \hat{\nu}(x)) \in \arg \min_{\theta, \nu} \left\{ \left\| \sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) \psi_{\theta, \nu}(O_{i,t}) \right\|_2 \right\}. \quad (\text{A.3})$$

Thus, the role of the forest is to construct the weights $\alpha_{i,t}(x)$ that define a local, data-adaptive neighborhood around the target point x . To see how these weights are constructed, consider a forest with B trees. Let $L_b(x)$ denote the terminal leaf of tree b that contains the target point x . Suppressing the distinction between splitting and estimation subsamples for notational simplicity,

the weight assigned by tree b to observation (i, t) is

$$\alpha_{b,i,t}(x) = \frac{\mathbf{1}\{\mathbf{X}_{i,t-1} \in L_b(x)\}}{|L_b(x)|}. \quad (\text{A.4})$$

The forest weight averages these tree-specific weights across trees:

$$\alpha_{i,t}(x) = \frac{1}{B} \sum_{b=1}^B \alpha_{b,i,t}(x). \quad (\text{A.5})$$

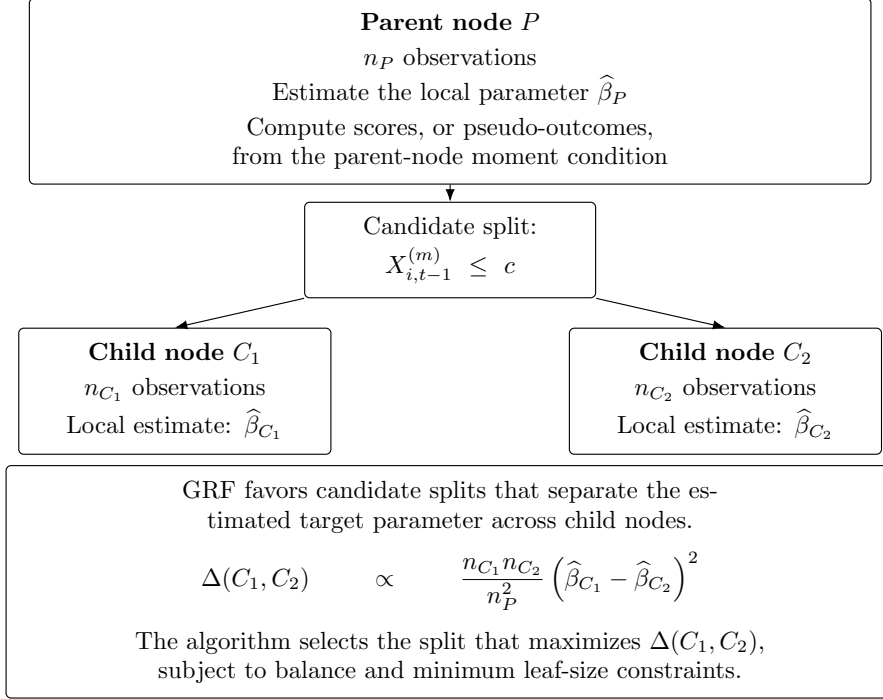
Observations that repeatedly fall in the same leaves as x receive larger weights. Importantly, GRF does not directly observe which firms have similar sensitivities. Instead, it constructs partitions of the covariate space that are informative about heterogeneity in the target parameter, and then estimates the local sensitivity using the observations selected by those partitions. Figure 1 - inspired by Figure 1 of [Athey et al. \(2019\)](#) - illustrates this logic. Each tree partitions the covariate space, here represented by firm size and leverage, in a different way. For a target firm x , observations that fall in the same terminal leaf across several trees receive larger forest weights, shown by larger dots in the bottom panel, and therefore contribute more to the local estimate of $\beta(x)$.

The splitting rule is targeted to heterogeneity in the parameter of interest rather than to the prediction of the outcome. At each parent node, GRF first estimates the local parameter and nuisance components using the observations in that node. It then forms observation-level scores, or pseudo-outcomes, from the gradient of the moment condition evaluated at the parent-node estimate. Candidate splits are evaluated according to how much they separate these scores across the two child nodes. Equivalently, the algorithm favors splits for which the implied estimates of $\beta(x)$ differ more across children, subject to the usual constraints on leaf size and balance. In this way, the tree grows along covariates that are informative about variation in the local shock sensitivity, rather than along covariates that only improve fit for the level of $Y_{i,t}$. A complete treatment of the algorithm is provided in [Athey et al. \(2019\)](#), while Figure A.1 provides a schematic representation of the splitting procedure.

In our setting, the treatment variable is a continuous aggregate shock. Since W_t is common to all firms in period t , identification of the local sensitivity comes from time-series variation in W_t , while heterogeneity in the estimated response comes from the way the forest weights firm-time observations according to their predetermined characteristics. Abstracting from nuisance adjustment, the local sensitivity can be represented as the local regression coefficient

$$\beta(x) = \frac{\text{Cov}(Y_{i,t}, W_t \mid \mathbf{X}_{i,t-1} = x)}{\text{Var}(W_t \mid \mathbf{X}_{i,t-1} = x)}. \quad (\text{A.6})$$

Figure A.1: Split Selection in a Generalized Random Forest



Notes: The figure illustrates the splitting logic in generalized random forests. At each parent node, the algorithm evaluates candidate splits according to the extent to which they separate the estimated local parameter across the two child nodes. In the application, the target parameter is the local sensitivity of firm outcomes to the aggregate shock.

The corresponding forest estimate is a weighted regression slope:

$$\hat{\beta}(x) = \frac{\sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) (W_t - \bar{W}_\alpha(x)) (Y_{i,t} - \bar{Y}_\alpha(x))}{\sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) (W_t - \bar{W}_\alpha(x))^2}, \quad (\text{A.7})$$

where

$$\bar{W}_\alpha(x) = \sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) W_t, \quad \bar{Y}_\alpha(x) = \sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) Y_{i,t}. \quad (\text{A.8})$$

In the empirical implementation, we follow the orthogonalization procedure recommended by [Athey et al. \(2019\)](#). We first remove the component of both the outcome and the aggregate shock that is predictable from firm characteristics:

$$\tilde{Y}_{i,t} = Y_{i,t} - \hat{m}^{(-i,t)}(\mathbf{X}_{i,t-1}), \quad \tilde{W}_{i,t} = W_t - \hat{e}^{(-i,t)}(\mathbf{X}_{i,t-1}), \quad (\text{A.9})$$

where $\hat{m}^{(-i,t)}(\cdot)$ and $\hat{e}^{(-i,t)}(\cdot)$ are out-of-bag predictions of

$$m(x) = \mathbb{E}[Y_{i,t} \mid \mathbf{X}_{i,t-1} = x], \quad e(x) = \mathbb{E}[W_t \mid \mathbf{X}_{i,t-1} = x]. \quad (\text{A.10})$$

Although the underlying shock W_t varies only over time, the residualized shock $\widetilde{W}_{i,t}$ can vary across firms because the fitted component depends on firm characteristics. This step makes the estimator less sensitive to confounding from the direct association between firm characteristics and firm outcomes.

Using the centered variables, the local moment for the continuous-treatment case can be written as

$$\mathbb{E} \left[\left(\widetilde{Y}_{i,t} - \beta(x) \widetilde{W}_{i,t} \right) \widetilde{W}_{i,t} \mid \mathbf{X}_{i,t-1} = x \right] = 0, \quad (\text{A.11})$$

which yields the residualized forest estimator

$$\widehat{\beta}(x) = \frac{\sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) \widetilde{W}_{i,t} \widetilde{Y}_{i,t}}{\sum_{t=1}^T \sum_{i \in \mathcal{I}_t} \alpha_{i,t}(x) \widetilde{W}_{i,t}^2}. \quad (\text{A.12})$$

The estimated object $\widehat{\beta}(x)$ is therefore a local average sensitivity for firms with characteristics close to x . Averaging $\widehat{\beta}_{i,t}$ over the empirical distribution of firm characteristics gives the average sensitivity in the sample, while the full distribution of $\widehat{\beta}_{i,t}$ is used in our aggregation exercise.

A.2 Montecarlo Simulation

We conduct a Monte Carlo experiment to compare the GRF estimator against alternative algorithms in recovering heterogeneous responses to aggregate fluctuations. We specify data-generating processes (DGPs) with non-linear conditional effects driven by multiple covariates. The econometrician does not observe the true DGP and instead estimates the conditional effects with a chosen model; the experiment asks how well each estimator recovers the true firm-level sensitivities.

Data-generating process. The simulated economy consists of N firms, indexed by i , observed over T periods. We denote by $X_{i,t}^j$ the j -th characteristic of firm i at time t , with $j = 1, \dots, J$ and $J = 6$. Covariates are generated through a Gaussian copula: for each j we draw a latent stationary AR(1) process $z_{i,t}^j$ with persistence ρ_x and unit-variance innovations, map it to a uniform via the probability integral transform, and pass it through a Compustat-like marginal quantile function (log-normal firm size, Beta-distributed leverage, cash, and tangibility, and approximately normal profitability and investment). This yields realistic, bounded marginal distributions, alongside within-firm persistence. We set $\rho_x = 0.8$, reflecting the high persistence of balance-sheet

characteristics in the data. The aggregate shock W_t is common to all firms and drawn i.i.d. from a standard normal distribution. The outcome of firm i at time t is

$$Y_{i,t} = f(\{X_{i,t}^j\}_{j=1}^J) + \beta(\{X_{i,t}^j\}_{j=1}^J) W_t + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, \sigma_\varepsilon^2),$$

where $f(\cdot)$ is an additive main effect, $\varepsilon_{i,t}$ is i.i.d. noise with σ_ε , and the firm-level sensitivity to the aggregate shock is

$$\beta(X_{i,t}) = \bar{\beta} + F(\{X_{i,t}^j\}_{j=1}^{J'}),$$

the object the estimators aim to recover. Heterogeneity in the response is governed by the function F , which depends only on a subset $J' \leq J$ of the characteristics. Notably, one characteristic enters the main effect f but not the sensitivity β : a successful estimator must learn that it predicts the level of Y without inducing spurious heterogeneity in the response. Because W_t is drawn independently of the covariates and each X^j evolves only according to its own history, the contemporaneous characteristics are exogenous to the shock, and we model heterogeneity as a function of $X_{i,t}$.

We specify a non-linear form for F . Heterogeneity is expressed in terms of the population-standardized characteristics $s_{i,t}^j \equiv (X_{i,t}^j - \mu_j)/\sigma_j$, where μ_j and σ_j are the analytically derived population mean and standard deviation of characteristic j . Standardizing by population moments makes the magnitude of each component comparable and independent of the particular sample draw. We winsorize the standardized characteristics at ± 3 standard deviations before they enter F ; without this step, rare extreme draws entering the quadratic component generate sensitivities far outside the empirically relevant range. The sensitivity combines four types of departure from linearity, each centred and normalized to contribute one standard deviation of heterogeneity, and weighted by $\lambda_{\text{lin}}, \lambda_{\text{quad}}, \lambda_{\text{inter}}, \lambda_{\text{thr}}$:

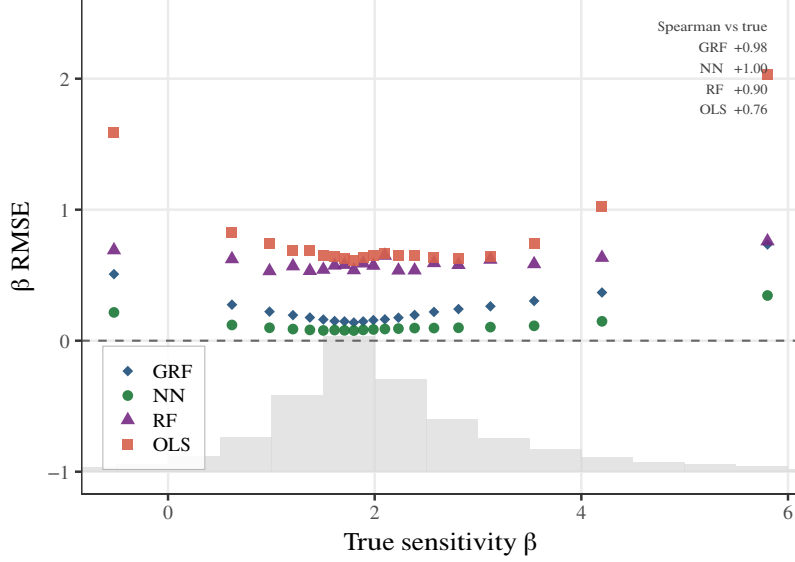
$$\begin{aligned} F(X_{i,t}) = & \lambda_{\text{lin}} \frac{s^1 + s^2 + s^3}{\kappa_{\text{lin}}} + \lambda_{\text{quad}} \frac{((s^4)^2 - 1) + ((s^5)^2 - 1)}{\kappa_{\text{quad}}} \\ & + \lambda_{\text{inter}} \frac{s^1 s^2 + s^3 s^4}{\kappa_{\text{inter}}} + \lambda_{\text{thr}} \frac{g_{\text{thr}}(s^1, s^2) - \bar{g}_{\text{thr}}}{\kappa_{\text{thr}}}, \end{aligned}$$

where the threshold component introduces regime switching in two characteristics,

$$g_{\text{thr}}(s^1, s^2) = (1.2 s^2 \mathbf{1}\{s^2 > 0\} + 0.4 s^2 \mathbf{1}\{s^2 \leq 0\}) + (1.2 s^1 \mathbf{1}\{s^1 > 0\} + 0.4 s^1 \mathbf{1}\{s^1 \leq 0\}),$$

$\bar{g}_{\text{thr}}$ is its unconditional mean, and the constants κ normalize each component to unit standard deviation, so that the λ weights are directly comparable. Table A.2 reports a summary of the parameters used to simulate the data in the Monte Carlo design.

Figure A.2: Root Mean Square Error of the Estimated β



Notes: The figure plots the average RMSE in the estimated β of each method against the true sensitivity.

Results. Table A.1 reports the performance of each estimator in recovering the true firm-level sensitivities $\beta(X)$ in the non-linear design. Three patterns emerge. First, the parametric benchmark fails: OLS attains roughly twice the RMSE of GRF and a much lower correlation with the true sensitivities, since it can represent no more than the best linear approximation to a non-linear $\beta(\cdot)$. Second, GRF and the neural-network R-learner both recover the sensitivity function well and are close to one another, with biases an order of magnitude smaller than their RMSE. Third, and in sharp contrast, the random-forest R-learner breaks down, manufacturing several times the true dispersion in sensitivities. The failure is specific to how this estimator recovers β . It regresses the pseudo-outcome $\tilde{Y}_{i,t}/\tilde{W}_t$, a ratio of two noisy quantities that becomes unstable whenever the residualized shock is close to zero. Because the shock is aggregate, this pathology is not scattered across isolated observations: in periods with $\tilde{W}_t \approx 0$ the pseudo-outcome is unstable for the entire cross-section at once, and non-honest splits fit part of that noise as if it were heterogeneity. GRF and the neural network avoid the problem by construction, since neither forms the ratio explicitly.³¹

Figure A.2 makes this visible, plotting the average estimation error of each method against the true sensitivity: GRF's error is small and flat through the bulk of the distribution but turns systematic in the tails, whereas the linear benchmark errs across the entire support. This regu-

³¹GRF solves a locally weighted moment condition in which the residualized shock enters both numerator and denominator, so observations carrying little shock variation contribute little to either; the neural network minimizes the R-loss directly under heavy regularization. This difference widens with the amount of noise.

Table A.1: Monte Carlo Summary Results

Estimator	Bias	RMSE	RMSE (bulk)	RMSE (tail)	Var. ratio	Corr.
GRF	0.001	0.285	0.175	0.396	0.817	0.981
OLS	0.001	0.877	0.645	1.139	0.595	0.762
R-learner (RF)	0.007	0.595	0.570	0.630	1.140	0.912
R-learner (NN)	0.017	0.129	0.087	0.174	0.952	0.996

Notes: The table reports the performance of each estimator in recovering the true firm-level sensitivities $\beta(X)$ in the non-linear Monte Carlo design. Bias is the absolute mean signed error of the estimated β s; RMSE the root mean squared error over all β s; RMSE (bulk) and RMSE (tail) restrict to the middle 80% and the extreme 20% of the true- β distribution, respectively; Var. ratio is $\text{var}(\hat{\beta})/\text{var}(\beta)$, where values below one indicate under-dispersion; and Corr. is the correlation with the true sensitivity.

larization is deliberate—honest sample splitting and local moment estimation trade tail accuracy for stability—and it has a specific implication for our application. Because the covariance between firm sensitivities and firm weights depends on the dispersion of β , an estimator that compresses that dispersion will, if anything, understate the role of heterogeneity for the aggregate; the effects we report should therefore be read as conservative. Among the flexible estimators, GRF combines near-zero bias, stability in the presence of noise, and—through its honest construction—valid inference, at the cost of some compression in the tails.

Table A.2: Monte Carlo Design Parameters

Parameter	Symbol	Value
<i>Panel dimensions</i>		
Number of firms (grid)	N	{6000}
Number of periods	T	40
Replications per cell	M	1
<i>Data-generating process</i>		
Number of characteristics	J	6
Characteristics driving β	J'	5
Average sensitivity	$\bar{\tau}$	2.10
Covariate persistence (AR(1))	ρ_x	0.8
Aggregate-shock persistence	ρ_w	0 (i.i.d.)
Idiosyncratic noise SD	σ_ε	2
Standardized covariates clipped at	—	± 3 SD
Main-effect scale	—	0.3
<i>Non-linear scenario weights</i>		
Linear component	λ_{lin}	0.1
Quadratic component	λ_{quad}	0.8
Interaction component	λ_{inter}	0.8
Threshold component	λ_{thr}	0.4

Notes: Parameters of the Monte Carlo design, described in Section [A.2](#).

B Data

We merged macroeconomic and firm-level data from different sources at different frequencies.

B.1 Macroeconomic Variables

Macroeconomic variables for data construction and the business cycle sensitivity analysis are from FRED St. Louis. Business cycle fluctuations are measured as the one-year growth rate of real GDP, computed with the [Davis and Haltiwanger \(1992\)](#) formula. Monetary policy shocks are the quarterly high-frequency interest-rate surprises from [Bauer and Swanson \(2023\)](#), and the oil price shocks are structural oil supply news shocks identified using high-frequency OPEC announcement surprises following [Känzig \(2021\)](#) in an augmented VAR that incorporates financial variables, as proposed by [Mori and Peersman \(2024\)](#). To normalize the size of the latter two, we project the one-year change in the one-year government bond yield and the one-year change in the oil price index from FRED (WTI) onto the respective surprise series, and use the fitted values as rescaled shocks. The GRF is therefore run on the rescaled treatment, imposing a unit-effect normalization ([Stock and Watson, 2018](#)). Figure [B.1](#) plots the aggregate time series of real GDP growth and of the instrumented shocks.

B.2 Firm-level Data and Sample Selection

Firm data are from the quarterly and annual Compustat datasets, covering 1990 to 2019. We process the two datasets separately but apply identical sample restrictions to each, ensuring consistency across panels.

- i) We retain only U.S.-incorporated firms ($fic_{i,t} = \text{"USA"}$).
- ii) We exclude regulated utilities, financial firms, and public administration, whose financial structures are not comparable to those of non-financial corporations. Specifically, we drop observations with a 4-digit SIC code in $[4900, 5000)$ or $[6000, 6800)$, and observations in NAICS 2-digit sectors 22 (utilities), 52–53 (finance, insurance, and real estate), or 99 (public administration).
- iii) To limit the influence of large mergers, we drop observations for which acquisitions ($aqc_{i,t}$) exceed 5% of total assets ($atq_{i,t}$).
- iv) We drop observations with missing or non-positive values of total assets ($atq_{i,t}$), sales ($saleq_{i,t}$), net property, plant, and equipment ($ppentq_{i,t}$), cash and short-term investments ($cheq_{i,t}$), debt in current liabilities ($dlecq_{i,t}$), or long-term debt ($dlttq_{i,t}$).

- v) We drop observations preceding a firm's first reported value of gross property, plant, and equipment ($ppeg_{i,t}$).
- vi) We require each firm to be observed for at least five years (twenty quarters in the quarterly panel).

The two panels differ only in frequency: four-quarter lags, leads, and trailing averages in the quarterly panel are replaced by their one-year counterparts in the annual panel. We use the annual panel only for the employment analysis, since employment (emp) is reported exclusively at annual frequency.

B.3 Firm-Variable Definitions and Transformations

We construct firm-level outcomes and characteristics following standard practice, using the same definitions in the quarterly and annual panels. Sales, total assets, and capital are expressed in real terms, deflated by the implied price index (ipd) of gross value added in the U.S. non-farm business sector.

Outcomes. All outcome variables are growth rates, computed with the [Davis and Haltiwanger \(1992\)](#) formula, which is standard in the firm-dynamics literature and bounds growth in $[-2, 2]$. Sales growth is computed on real sales ($saleq/ipd$). Debt issuance is the growth rate of total debt, defined as the sum of debt in current liabilities ($dlcq$) and long-term debt ($dlttq$). The investment rate is the growth rate of the real capital stock, where capital is the book value computed by the perpetual inventory method: the initial value is the firm's earliest reported gross property, plant, and equipment ($ppeg_{i,t}$), and subsequent values are obtained by iterating on net property, plant, and equipment ($ppentq$). Market value is the stock price ($prccq$) times shares outstanding ($cshoq$).

Characteristics. Firm characteristics are expressed in levels. Size is the log of real total assets (atq). Leverage is total debt ($dlcq + dlttq$) over total assets (atq). The liquidity ratio is cash and short-term investments ($cheq$) over total assets (atq). The short-term debt ratio is current debt ($dlcq$) over total debt. Return on assets is net income (niq) over total assets (atq). Sales-growth volatility is the standard deviation of annual real sales growth over a 10-year rolling window. Distance to default is computed following the [Merton \(1974\)](#) model. Industry scope is the firm's 5-digit NAICS classification. Leverage, the liquidity ratio, return on assets, and the short-term debt ratio are averaged over a trailing window (four quarters in the quarterly panel, two years in the annual panel) before being lagged in the empirical application. Firm age is the number of years since the firm's founding, taken from the founding-date data of Ritter's website and supplemented

by web scraping, and entered as $\log(1 + \text{age})$. Founding dates are unavailable for a large number of firms, so we include age only in robustness checks.

Before estimation, we trim outliers. Outcome variables are trimmed at the 1.5th and 98.5th percentiles. Distance to default and sales-growth volatility, which are non-negative, are trimmed only at the 98.5th percentile. Return on assets is trimmed at the 1.5th and 98.5th percentiles and further restricted to $[-1, 1]$. Leverage and the liquidity ratio are bounded to $[0, 1]$, and we drop observations with $|\text{size}| > 10$. To reduce missing values, we linearly interpolate each continuous characteristic within firm over time after completing the cleaning steps. Interpolation reduces the loss of firm-quarter observations from sporadically missing characteristics. Table B.1 reports summary statistics for all variables; Tables B.2 and B.3 report correlations among the characteristics and outcomes. The distributions of the variables and weights are in Figures B.2, B.3 and B.4.

Table B.1: Summary Statistics

Variable	Statistics							Obs.
	Mean	Median	St. Dev.	Min	Max	IQR	Skewness	
Panel A. Characteristics								
Size	0.67	0.68	2.40	-9.29	8.67	3.40	0.01	430605
Leverage	0.29	0.25	0.21	0.00	1.00	0.28	0.91	329664
Liquidity	0.14	0.07	0.17	0.00	0.99	0.16	2.05	351868
Distance to Default	5.78	4.75	4.47	0.00	21.02	5.86	1.05	326283
Short-Term Debt	0.29	0.21	0.26	0.00	1.00	0.38	0.91	336427
ROA	-0.04	0.02	0.19	-1.00	0.25	0.13	-2.25	343728
Sales Volatility	0.27	0.19	0.23	0.00	1.13	0.25	1.55	360961
Log Age	3.25	3.18	0.83	0.00	5.39	1.10	0.03	277573
Panel B. Outcome								
Sales Growth	0.04	0.03	0.27	-1.04	1.08	0.22	-0.04	240000
Employment Growth	0.01	0.00	0.18	-0.74	0.66	0.15	-0.27	58336
Investment Rate	0.06	0.02	0.23	-0.66	1.10	0.19	1.10	398296
Debt Rate	0.01	-0.02	0.45	-1.53	1.64	0.35	0.32	227974
Market Value	0.01	0.03	0.51	-1.43	1.32	0.61	-0.19	214622

Notes: The first panel reports summary statistics for the balance-sheet characteristics used as independent variables; the second reports statistics for the outcome variables. The data are from quarterly and annual (employment only) Compustat, covering 1990–2019. Independent variables are linearly interpolated after the cleaning steps. Outcome variables are measured in percentage points, where 0.01 represents 1%. Additional information on variable construction and trimming is provided in Appendix B.

Table B.2: Correlation Matrix

Variable	Size	Leverage	Liquidity	Distance to Default	Short-Term Debt	ROA	Sales Volatility	Log Age
Size	1.00	0.07	-0.19	0.33	-0.43	0.39	-0.40	0.45
Leverage	0.07	1.00	-0.31	-0.38	-0.24	-0.11	0.03	0.07
Liquidity	-0.19	-0.31	1.00	0.14	0.13	-0.28	0.30	-0.33
Distance to Default	0.33	-0.38	0.14	1.00	-0.07	0.31	-0.23	0.18
Short-Term Debt	-0.43	-0.24	0.13	-0.07	1.00	-0.21	0.21	-0.22
ROA	0.39	-0.11	-0.28	0.31	-0.21	1.00	-0.46	0.28
Sales Volatility	-0.40	0.03	0.30	-0.23	0.21	-0.46	1.00	-0.31
Log Age	0.45	0.07	-0.33	0.18	-0.22	0.28	-0.31	1.00

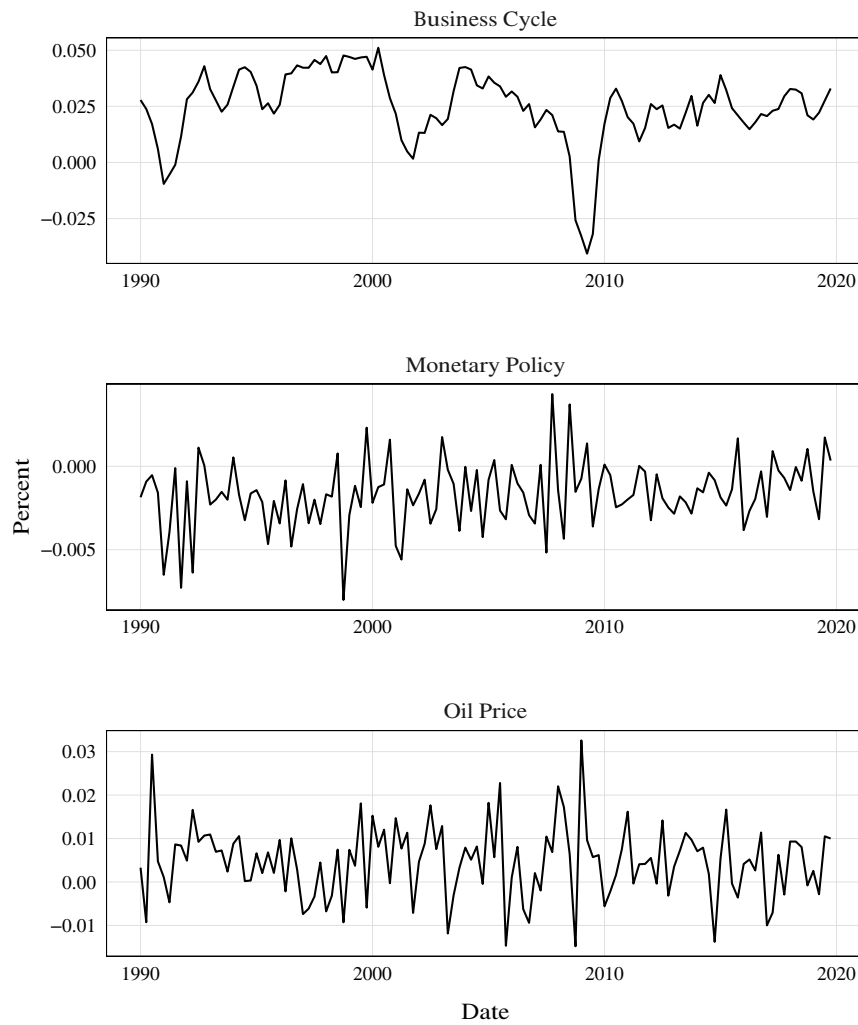
Notes: The table reports pairwise correlations among the quarterly firm balance-sheet characteristics used as independent variables. The data are from quarterly Compustat only. Continuous characteristics are linearly interpolated after the cleaning steps. Additional information on variable construction and trimming is provided in Appendix B.

Table B.3: Weight Rank Correlation

	Sales	Employment	Investment	Debt	Market Value
Sales	1.000	0.936	0.875	0.854	0.830
Employment		1.000	0.851	0.806	0.778
Investment			1.000	0.857	0.796
Debt				1.000	0.728
Market Value					1.000

Notes: The table reports pairwise correlations across the firm weights $\omega_{i,t}$ used in the paper. Sales, assets (investment), debt stock (debt), and market value are from quarterly Compustat, while employment is from annual Compustat.

Figure B.1: Aggregate Time-Series



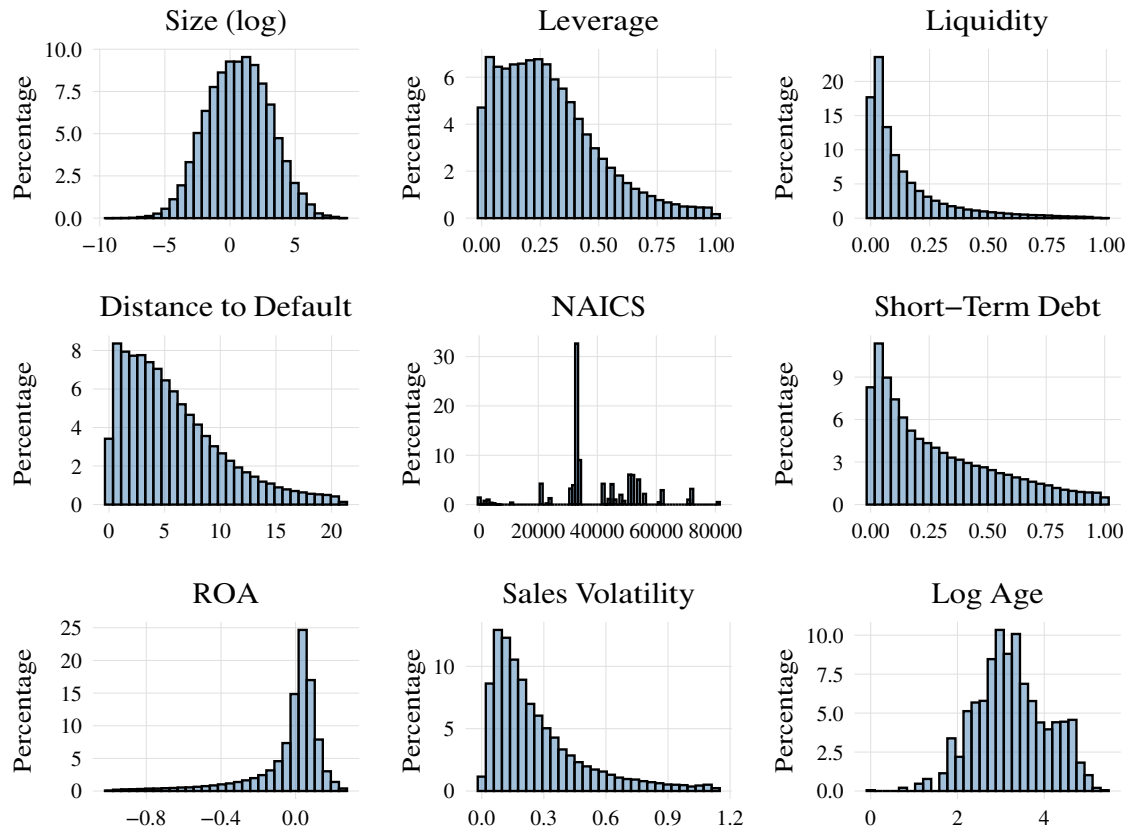
Notes: The figure shows the time series of business cycle and aggregate shocks used in the empirical application. Units of measurement are in percentage points, where 0.01 represents 1%. Additional information on the variable construction can be found in [Appendix B](#).

Figure B.2: Firm Outcomes



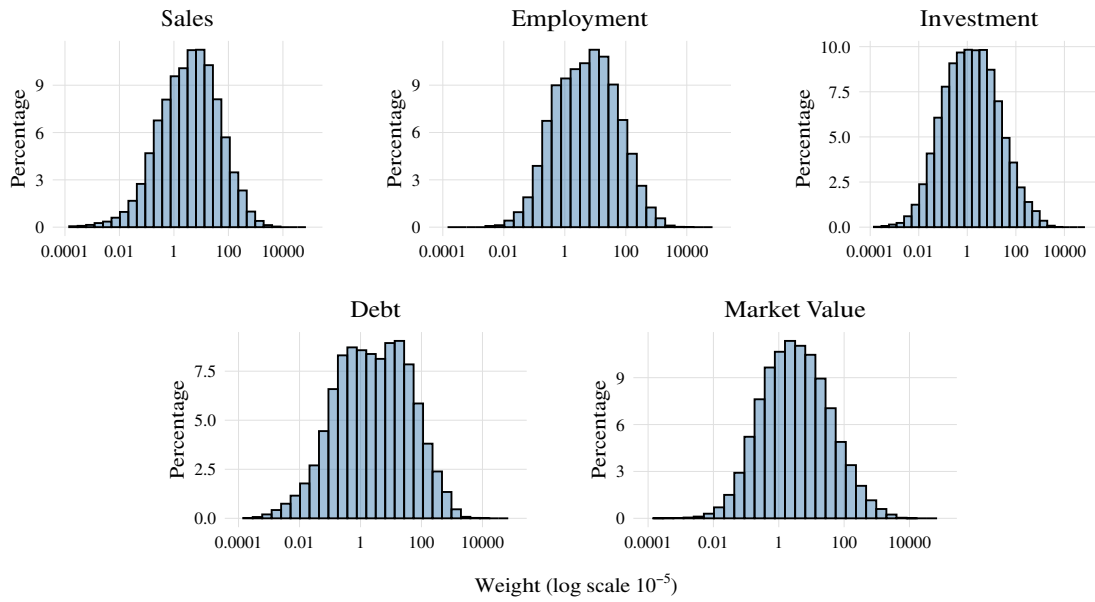
Notes: The figure shows the distribution of the firm outcomes used as dependent variables in the empirical application. The data are from quarterly and annual Compustat, spanning from 1990 to 2019. Growth rates are annual, and they are calculated using the Haltiwanger formula. Units of measurement are in percentage points, where 0.01 represents 1%. Additional details on variable construction and data cleaning are provided in Appendix B.

Figure B.3: Firm Characteristics



Notes: The figure shows the distribution of firm-quarter balance-sheet characteristics used as independent variables in the empirical application. Variables are linearly interpolated before the empirical application. The plot for NAICS shows the percentages at the 4-digit level (i.e. each bin collects 1000 industry codes). Additional details on variable construction and data cleaning are provided in Appendix B.

Figure B.4: Weight Distribution



Notes: The figure presents the distribution of firm-level shares across different outcome variables. The x-axis represents the firm-level share on a 10^{-5} log scale, while the y-axis denotes the percentage.

C Additional Figures and Tables

C.1 Chernozhukov et al. (2023) Test for Heterogeneity

The test creates two synthetic variables, C_i and D_i :

$$C_i = \bar{\beta}(W_i - \hat{W}_i),$$

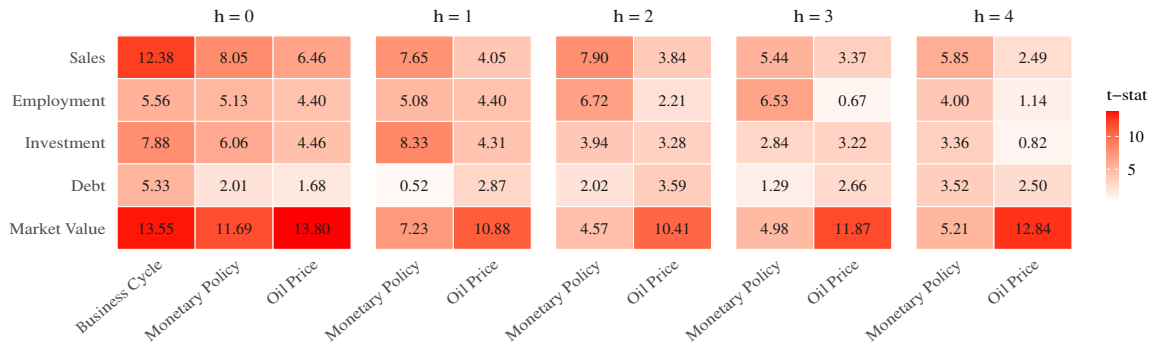
$$D_i = (\hat{\beta}^{GRF} - \bar{\beta})(W_i - \hat{W}_i),$$

where the former uses only the average treatment effect, while the latter is the prediction that takes into account the heterogeneity as predicted by the GRF. The test consists of running the following regression of residuals in treatment on C_i and D_i :

$$Y_i - \hat{Y}_i = \gamma C_i + \delta D_i \quad (\text{C.13})$$

The null hypothesis of the test is $\delta = 0$, which indicates that the GRF does not capture any heterogeneity. We find that we can reject the null hypothesis of no heterogeneity in treatment effects for almost all aggregate shock-outcome variable pairs.

Figure C.1: Test for Heterogeneity in Sensitivity



Notes: The table reports the t-statistic of the Chernozhukov et al. (2023) test for each aggregate variable–outcome variable pair. An absolute t-statistic value below 1.648 indicates no particular degree of heterogeneity, while a value above the threshold of 1.648 suggests a statistically significant level of heterogeneity in firm sensitivity at a 90% confidence interval.

C.2 Non-linearity Tests and Accumulated Local Effect Plots

We complement the machine learning tools by formally testing whether the relationship between the conditional effect estimated using GRF and firms' characteristics is linear. In the LPM, the implied conditional effect of an aggregate shock on firms' outcomes is linear in firms' characteristics, i.e. $\beta(\mathbf{X}_{i,t-1}) = \beta_0 + \sum_{j \in J} \beta_j \cdot \mathbf{X}_{i,t-1}^j$, where J is the set of characteristics. We test whether the estimated GRF sensitivities, $\widehat{\beta(\mathbf{X}_{i,t-1})}$, are linear in the characteristics, leveraging three different statistical measures commonly used in testing for linearity: the estimated degrees of a Generalized Additive Model (GAM henceforth), and the Regression Specification Error Test (RESET henceforth).

We estimate a GAM of the firms' sensitivities on firms' characteristics. In a GAM, a univariate dependent variable depends linearly on unknown smooth functions of some predictor variables. Formally, this translates into estimating the following GAM: $\widehat{\beta(\mathbf{X}_{i,t-1})} = \sum_{j \in J} f_j(\mathbf{X}_{i,t-1}^j)$, where J is the set of characteristics and f_j is a smooth function of characteristic j .³² The effective degrees of freedom estimated by the GAM for each smooth function f_j can be interpreted as a proxy for the degree of non-linearity in the relationship between dependent and predictor variables: an EDF around one indicates a linear relationship, while an EDF larger than one indicates a non-linear relationship. The third row in each panel of Table C.1 reports the minimum estimated degree of freedom across characteristics for each outcome variable - aggregate shock pair. In all cases, the minimum EDF is around six, well above the threshold value of one, indicating the presence of strong non-linearities in firms' characteristics, in line with partial dependence analysis.

As an alternative, we run the RESET test to check for misspecification in a linear OLS regression of $\widehat{\beta(\mathbf{X}_{i,t-1})}$ on the complete set of firms' characteristics as explanatory variables. The test adds higher-order terms or interaction terms of the independent variables to the regression. If these added terms are statistically significant, it suggests that the model may be misspecified. The first two rows of each panel in Table C.1 report the test statistics and the corresponding p-values for each outcome variable - aggregate shock pair, respectively. Also, in this case, linearity is rejected, as no outcome variable - aggregate shock pair accepts the null hypothesis of correct model specification.

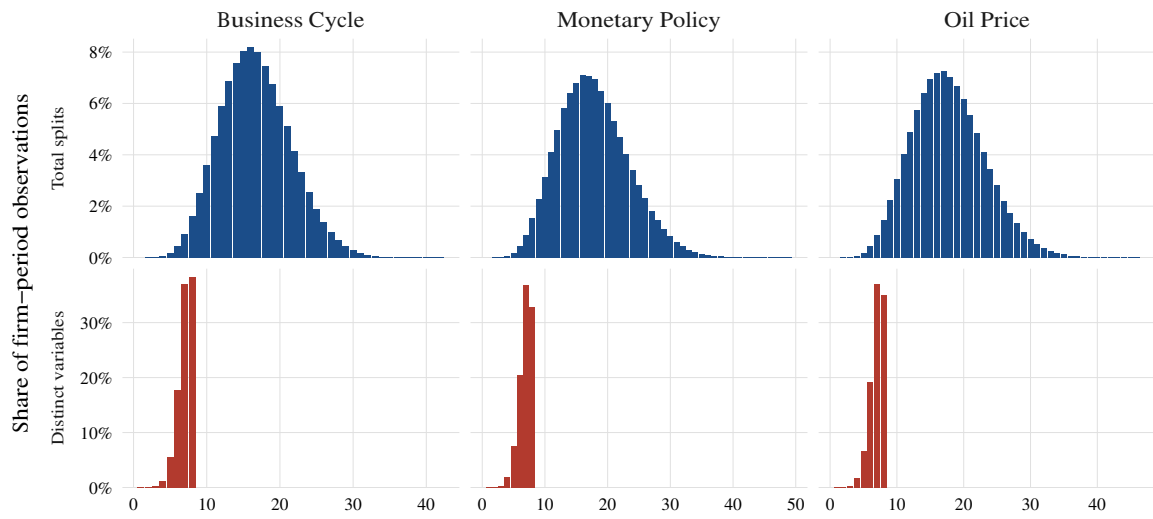
³²We exclude industry scope from the set of characteristics because it is unreasonable to assess whether the conditional effect is non-linear in 5-digit NAICS. We include 5-digit NAICS fixed effects to control for heterogeneity in industry scope.

Table C.1: Statistical Test for Non-linearity

	Business Cycle		Monetary Policy				Oil Price					
	$h=0$		$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
Sales												
RESET Stat.	930.52	9265.02	7721.77	4572.99	3249.91	2783.32	3282.62	426.16	1497.66	881.07	105.14	
p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
GAM Min EDF	6.53	7.70	7.71	7.40	6.27	7.39	7.31	7.75	7.32	7.66	7.36	
Employment												
RESET Stat.	128.58	510.82	1049.08	437.38	122.67	160.21	181.73	191.80	57.73	51.37	102.79	
p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
GAM Min EDF	4.40	7.13	6.53	6.88	2.81	5.58	7.08	7.37	3.92	5.50	5.79	
Investment												
RESET Stat.	1787.76	870.51	831.42	119.42	1657.17	35.94	257.81	465.04	1092.89	2488.67	1569.35	
p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
GAM Min EDF	7.07	7.25	7.64	7.05	7.48	7.44	7.77	7.71	7.75	7.28	7.17	
Debt												
RESET Stat.	2103.26	340.26	66.42	767.20	1790.52	1378.48	80.04	36.57	45.41	129.88	129.33	
p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
GAM Min EDF	6.32	7.73	6.51	6.73	7.72	7.32	6.27	7.18	7.18	7.09	6.97	
Market Value												
RESET Stat.	1941.63	6064.30	2445.16	355.86	750.50	350.13	2637.95	5652.13	467.64	204.95	2330.81	
p -value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
GAM Min EDF	7.69	7.15	6.85	7.11	7.70	5.82	6.32	6.85	7.54	7.53	7.43	

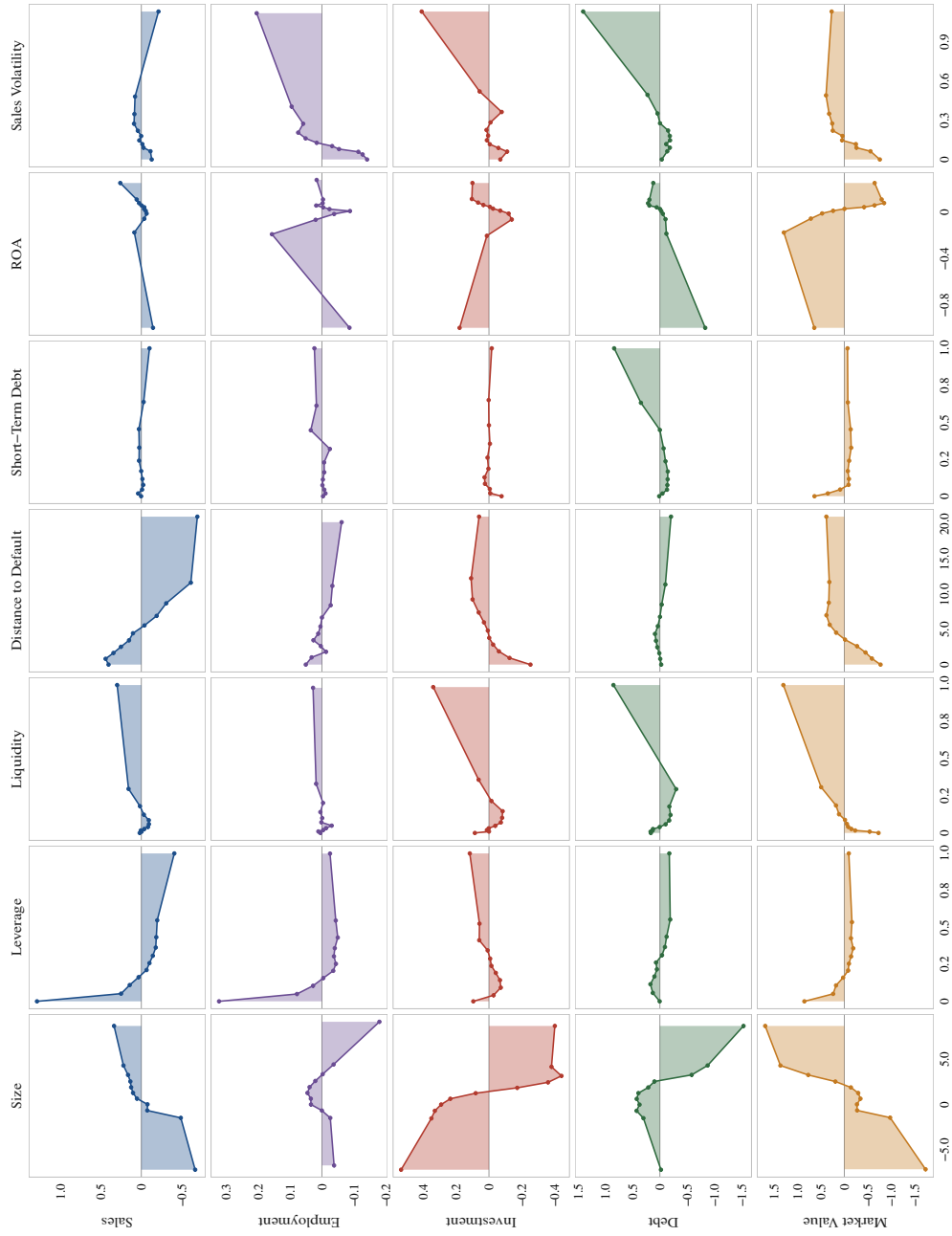
Notes: The table reports the results of two different linear specification tests between covariates and the sensitivities produced by GRF for each outcome variable across shocks. We assess the linearity of the conditional effect of an aggregate shock on firms' outcomes in firms' characteristics, i.e. $\beta(\mathbf{X}_{i,t-1}) = \beta_0 + \sum_{j \in J} \beta_j \cdot \mathbf{X}_{i,t-1}^j$, where J is the set of characteristics. The null hypothesis of the RESET test is that the model is linear, for which we report the test statistics and the p -value. We estimate a GAM model that includes all characteristics. For each characteristic, we estimate the effective degrees of freedom (EDF). We report the minimum effective degrees of freedom among characteristics in each outcome variable - aggregate shock. Results are presented for all outcome variables under each aggregate shock (business cycle, monetary policy, and oil price).

Figure C.2: Tree Complexity



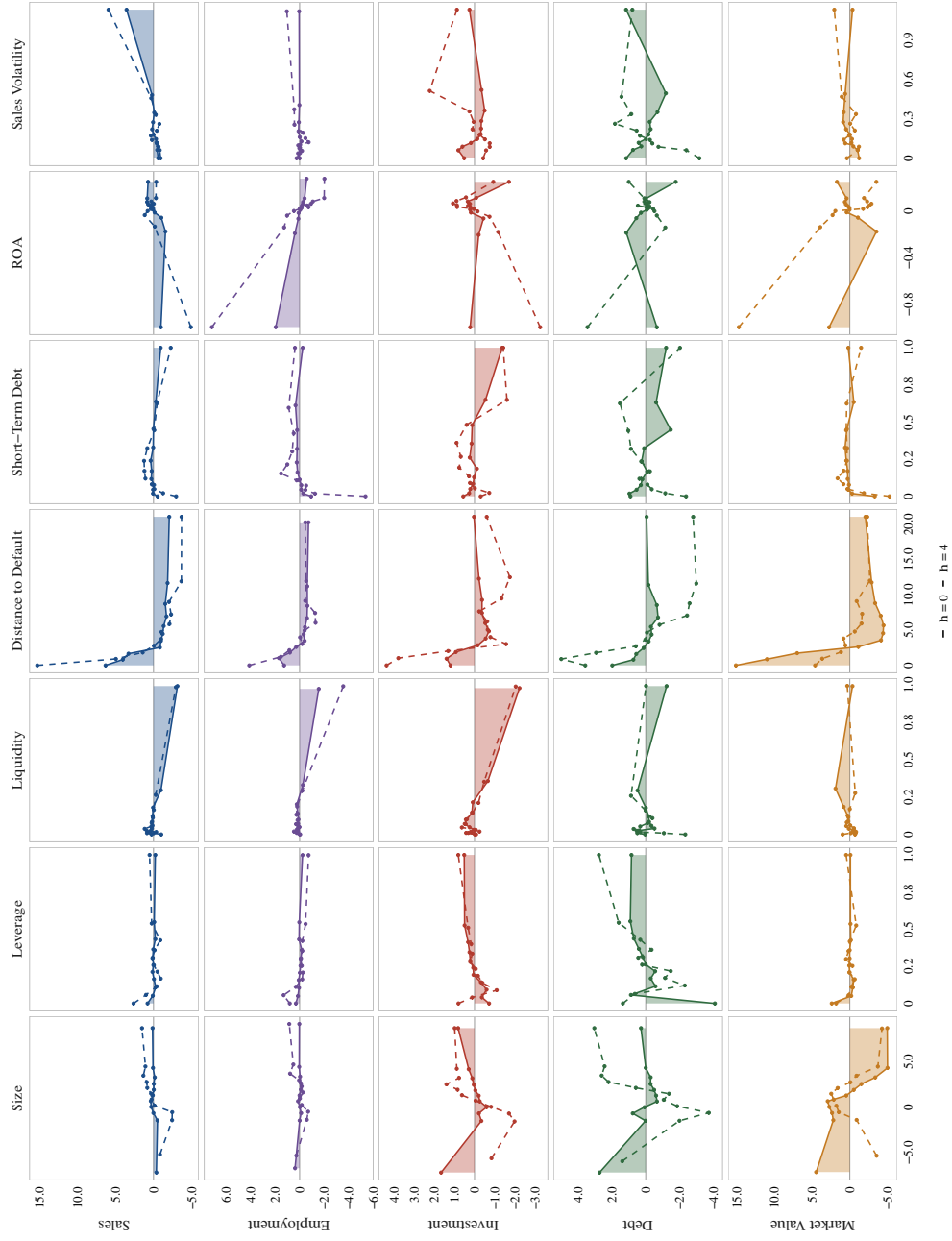
Notes: The figure summarizes the complexity of the GRF partition underlying the estimated heterogeneous effects. The upper panel reports the total number of splits used to form terminal leaves, while the lower panel reports the number of distinct covariate dimensions used along each path to a leaf. The gap between the two measures captures repeated splits along the same dimension, which approximate univariate nonlinearities, such as higher-order terms, beyond the interaction structure that would be required in a linear probability model.

Figure C.3: Accumulated Local Effects - Business Cycle



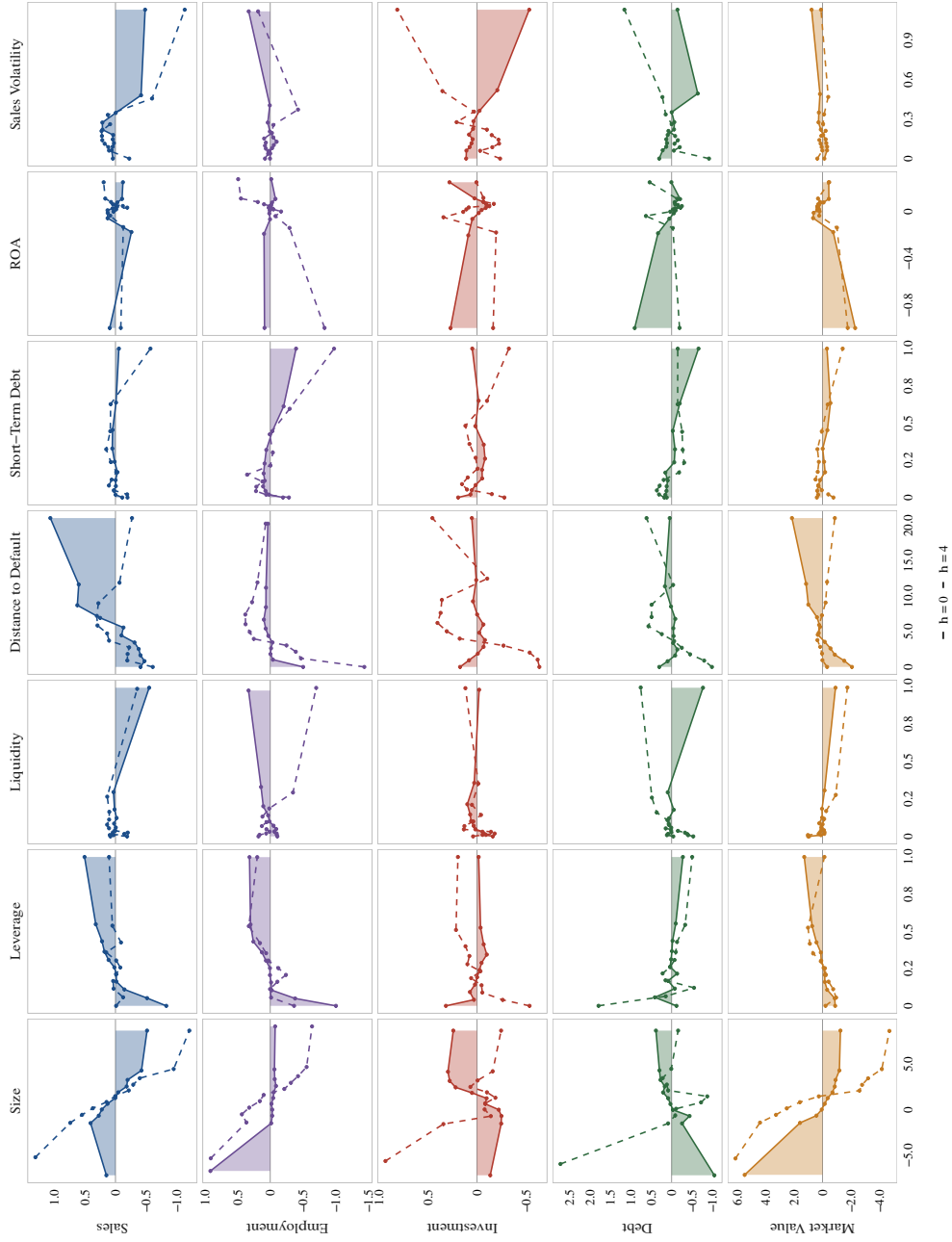
Notes: The figure presents ALE plots estimated for each firm characteristic across different outcome variables over business cycle fluctuations. The y-axis shows the estimated difference in firm sensitivity relative to the average firm sensitivity.

Figure C.4: Accumulated Local Effects - Monetary Policy



Notes: The figure presents ALE plots estimated for each firm characteristic across different outcome variables for the monetary policy shock. The y-axis shows the estimated difference in firm sensitivity relative to the average firm sensitivity. The solid (dashed) line represents the contemporaneous ($t + 4$) specification.

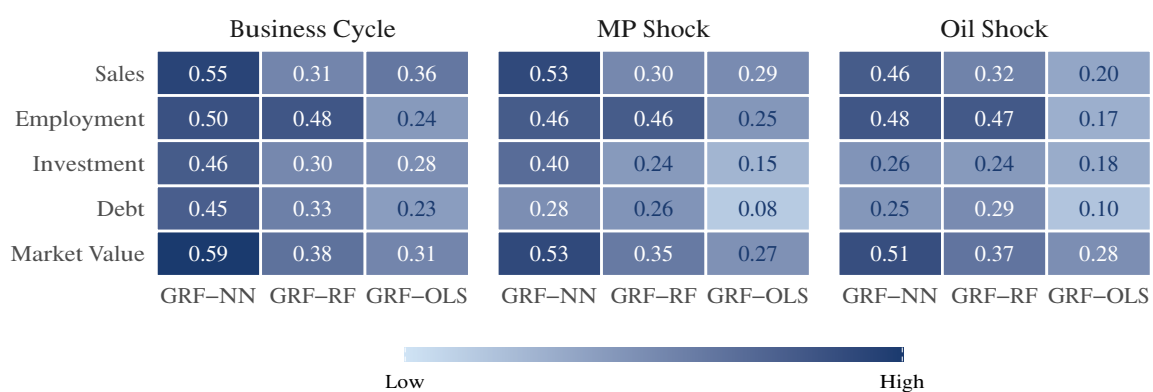
Figure C.5: Accumulated Local Effects - Oil Price



Notes: The figure presents ALE plots estimated for each firm characteristic across different outcome variables for the oil price shock. The y-axis shows the estimated difference in firm sensitivity relative to the average firm sensitivity. The solid (dashed) line represents the contemporaneous ($t + 4$) specification.

C.3 Comparison across ML Algorithms

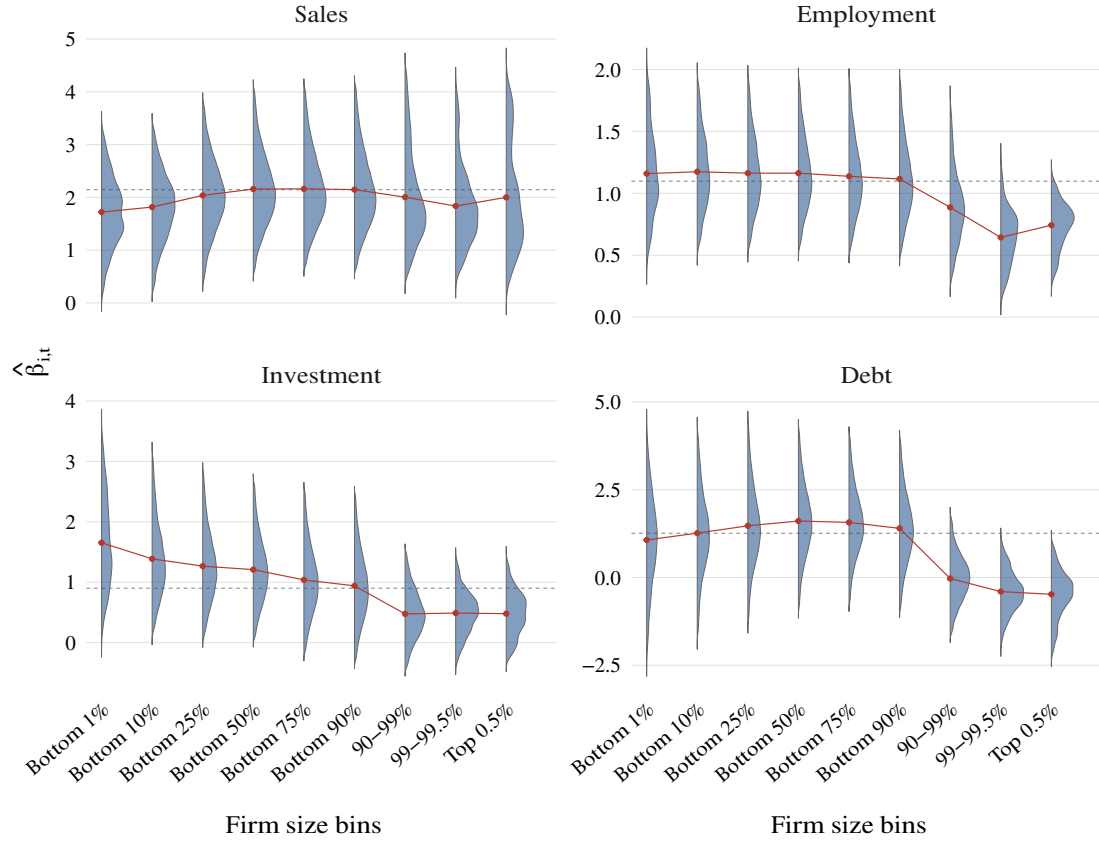
Figure C.6: Comparison across ML Methodologies



Notes: The heatmap reports Spearman rank correlations across firm-level sensitivities obtained from alternative machine-learning methods. Darker cells indicate stronger rank agreement between methods, while lighter cells indicate weaker agreement. Correlations are computed separately by outcome-shock pair.

C.4 Miscellanea Business Cycle

Figure C.7: Firm Sensitivity and Firm Size as in [Crouzet and Mehrotra \(2020\)](#)



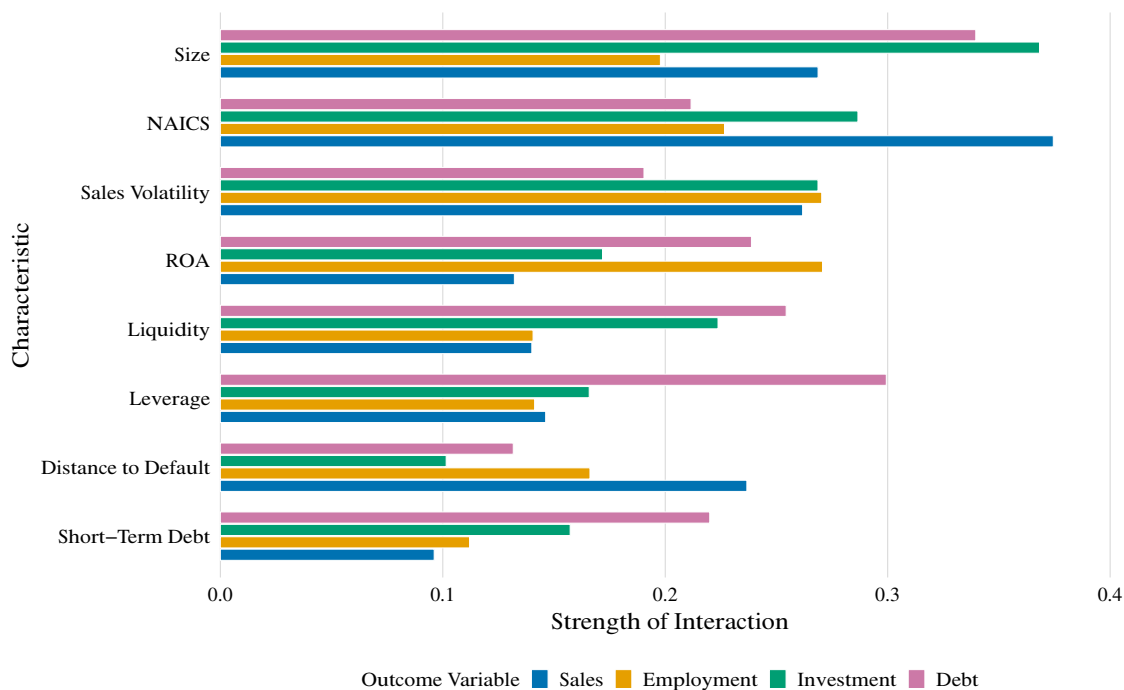
Notes: The figure reports the distribution of firm-level GRF estimates, $\hat{\beta}_{i,t}$, for all outcome variables to business-cycle shocks across firm-size bins. Size bins are constructed using lagged total assets, with lower bins defined cumulatively and the upper tail split separately. The dashed horizontal line reports the overall mean effect, while the red line and markers show the mean estimate within each size bin.

Table C.2: Correlation across $\hat{\beta}$

	Sales	Employment	Investment
<i>Panel A: Whole sample</i>			
Employment	0.238 (0.005)	–	
Investment	0.092 (0.006)	0.268 (0.005)	–
Debt	0.117 (0.006)	0.219 (0.005)	0.374 (0.005)

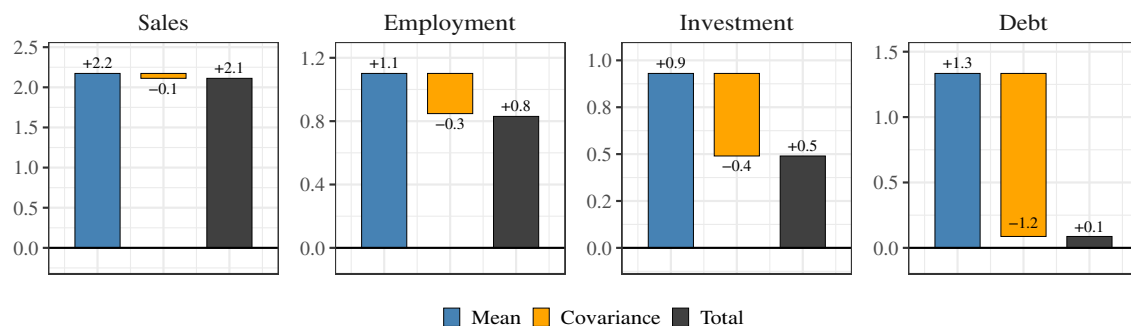
Notes: The table reports pairwise correlations between the firm-level GRF sensitivity estimates $\hat{\beta}$ across outcome variables. Each cell is the Pearson correlation, computed over the pooled sample of firm-year observations for which both sensitivities are estimated. The outcome variables at quarterly frequency are averaged at annual frequency. Standard errors are reported in parentheses.

Figure C.8: Strength of Interactions



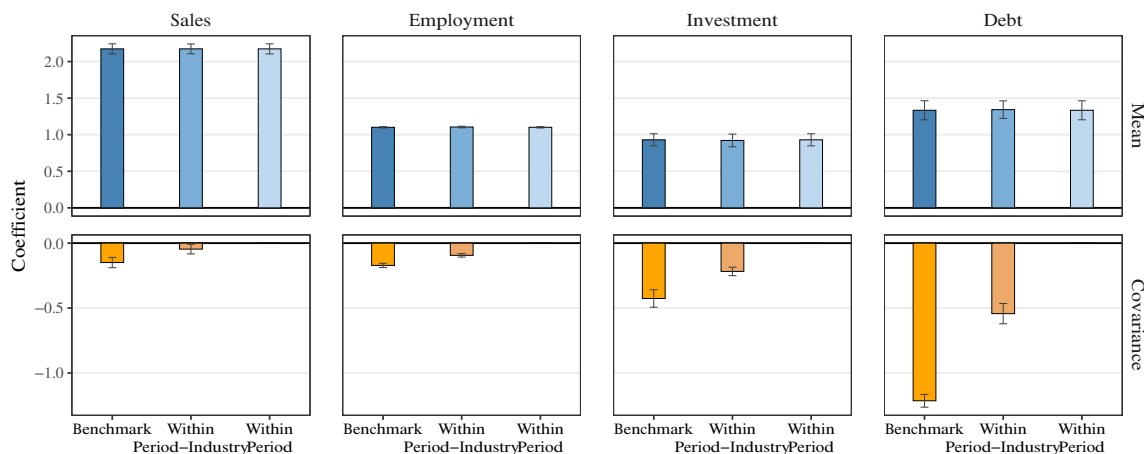
Notes: The figure presents the strength of interaction between firm characteristics for business cycle fluctuations measured using Friedman's H statistic. The characteristics on the y-axis are ordered by their average strength of interaction within each outcome variable. The x-axis reports the interaction strength, where a value of 0.01 corresponds to 1%.

Figure C.9: Aggregate Response with Alternative Weights



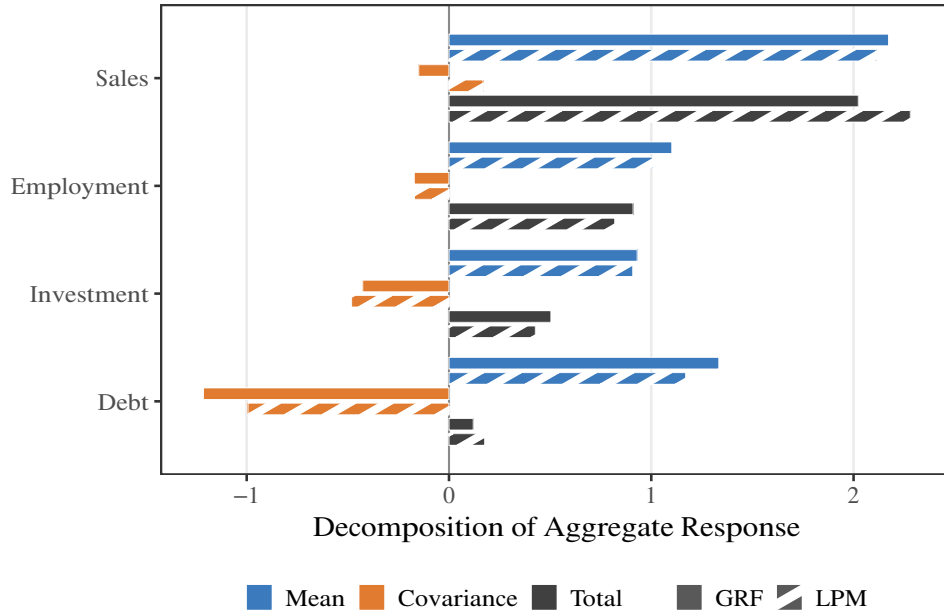
Notes: The figure illustrates the decomposition of aggregate responses into mean and covariance terms for each outcome variable over business cycle fluctuations. The black bar denotes the total average aggregate response. We estimate Equation (8) using the mean and covariance terms in Equation (4) as the dependent variable, using size (total assets) instead of the relative outcome shares. The mean and covariance terms are constructed using the set of firm-level sensitivities estimated with the GRF algorithm.

Figure C.10: Within and Across Sector Heterogeneity



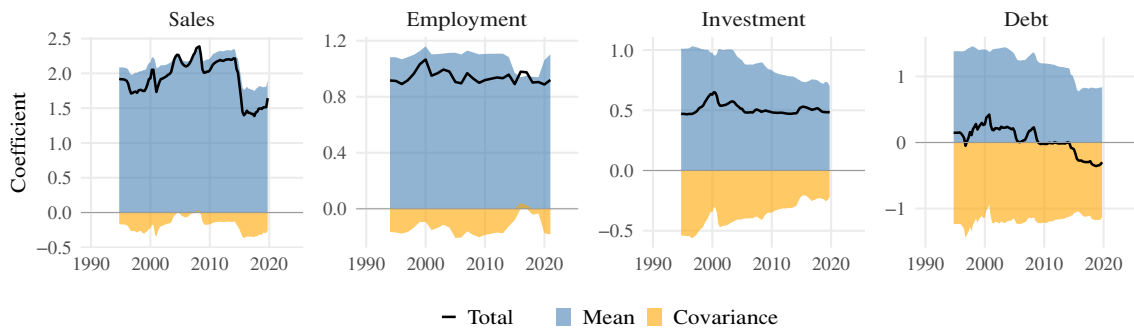
Notes: The figure reports counterfactual decompositions of the business-cycle response by outcome. For each outcome, the benchmark bars use the original firm-level GRF estimates, while the counterfactual bars replace firm-level heterogeneity with period or period-industry medians. Error bars report 95 percent confidence intervals.

Figure C.11: Aggregate Role of Non-linearities



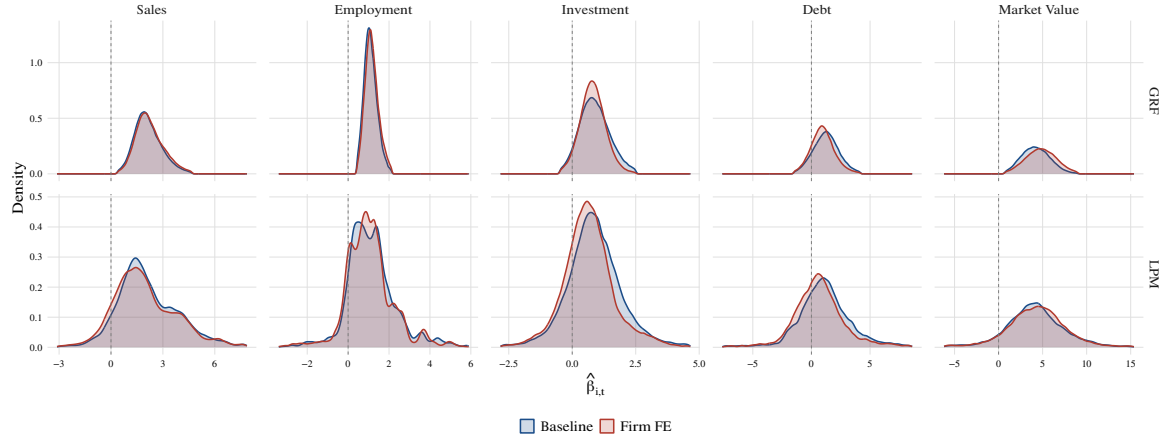
Notes: The figure decomposes the average aggregate response into mean and covariance components for the benchmark estimates and the estimates from the linear panel model. Each bar represents the estimated coefficient from Equation (8), with the mean term capturing the average firm-level sensitivity (blue) and the covariance term reflecting the interaction between firm shares and sensitivities (orange). Filled bars denote estimates from the benchmark (GRF), while striped bars correspond to estimates from the LPM.

Figure C.12: Evolution over Time



Notes: The figure reports rolling estimates of the GRF aggregate response to the business-cycle shock, decomposed into mean and covariance components across outcomes. Quarterly outcomes use a 20-quarter rolling window, while employment uses a 5-year rolling window. The black line denotes the total aggregate response, equal to the sum of the mean and covariance components.

Figure C.13: Role of Firm FE - Distribution of Sensitivities



Notes: The figure reports the distribution of estimated sensitivities using GRF and LPM when firm FEs are introduced in the specification. Firm FEs are introduced in the GRF by partialing out the outcome variable and the aggregate shock.

Table C.3: Role of Firm FE - Distribution Moments

Method	Outcome	Mean of β		SD of β		Rank corr. (base, FE)
		Baseline	Firm FE	Baseline	Firm FE	
GRF	Sales	2.145	2.252	0.847	0.901	0.792
	Employment	1.098	1.171	0.345	0.361	0.730
	Investment	0.901	0.805	0.652	0.537	0.650
	Debt	1.261	0.964	1.185	1.087	0.697
	Market Value	4.221	4.848	1.640	1.849	0.801
LPM	Sales	2.113	1.938	2.771	2.786	0.941
	Employment	1.068	0.977	6.122	6.239	0.865
	Investment	0.896	0.651	1.803	1.738	0.896
	Debt	1.113	0.633	3.508	3.428	0.899
	Market Value	4.157	4.297	5.781	5.752	0.911

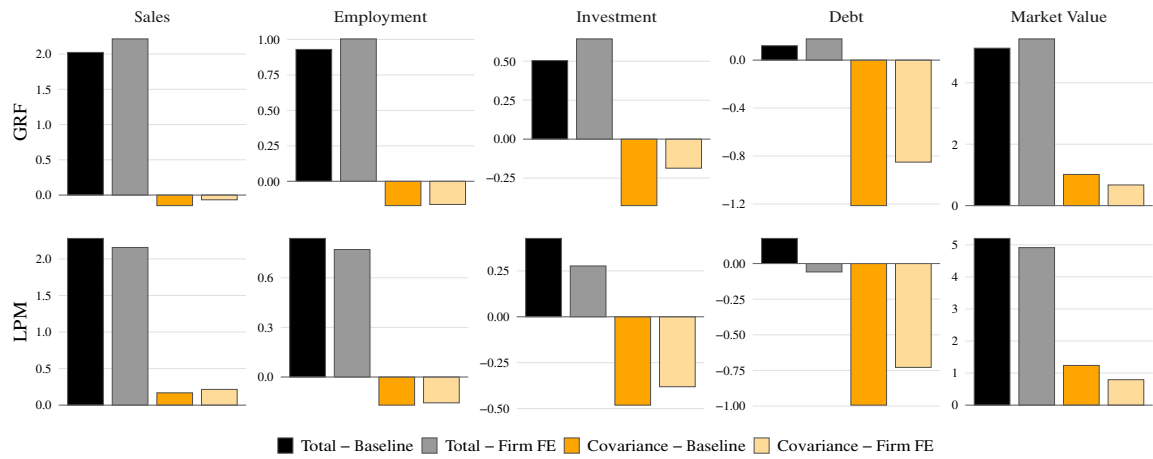
Notes: The table reports moments from the distribution of estimated sensitivities using GRF and LPM when firm FEs are introduced in the specification. Firm FEs are introduced in the GRF by partialing out the outcome variable and the aggregate shock. We report the standard deviation of the estimated β s together with the rank correlation between the baseline set of estimates and the one including firm FEs.

Table C.4: Aggregate Role of Non-linearities

Outcome	Horizon	GRF		Linear Panel Model		Difference
		Coefficient	StD. Error	Coefficient	StD. Error	
Panel A: Business Cycle						
Sales	0	2.023	0.034	2.280	0.044	-0.257***
Employment	0	0.910	0.029	0.819	0.034	0.091**
Investment	0	0.504	0.008	0.427	0.043	0.077*
Debt	0	0.120	0.034	0.176	0.069	-0.057
Market Value	0	5.127	0.072	5.199	0.101	-0.072
Panel B: Monetary Policy						
Sales	0	1.038	0.103	0.700	0.240	0.339**
	1	-1.472	0.203	-2.192	0.467	0.720**
	2	0.022	0.191	0.357	0.617	-0.334
	3	-0.490	0.105	-1.571	0.603	1.081**
Employment	4	-2.890	0.143	-5.661	0.530	2.771***
	0	0.257	0.099	-0.711	0.265	0.968***
	1	-1.080	0.203	-1.953	0.575	0.873**
	2	-3.132	0.304	-3.470	0.936	0.338
Investment	3	-4.112	0.380	-5.202	1.257	1.090
	4	-3.943	0.389	-5.229	1.369	1.285
	0	0.056	0.051	-0.726	0.207	0.782***
	1	-0.373	0.080	-2.230	0.213	1.858***
Debt	2	-1.357	0.115	-2.121	0.312	0.764**
	3	-4.155	0.200	-4.422	0.268	0.267
	4	-2.824	0.171	-4.095	0.277	1.271***
	0	-1.039	0.100	1.004	0.356	-2.042***
Market Value	1	-1.254	0.116	-0.786	0.147	-0.468**
	2	-2.558	0.278	0.142	0.610	-2.700***
	3	-4.246	0.293	-1.892	0.903	-2.353***
	4	-1.687	0.366	-1.867	0.467	0.179
Market Value	0	-16.591	0.231	-17.897	0.394	1.306***
	1	-15.841	0.315	-13.938	0.331	-1.903***
	2	-5.299	0.366	-5.464	0.318	0.165
	3	-12.769	0.374	-14.431	0.438	1.662***
4	-5.591	0.375	-5.948	0.381	0.357	
Panel C: Oil Price						
Sales	0	-0.402	0.055	-0.417	0.059	0.016
	1	-1.431	0.050	-1.792	0.122	0.361***
	2	-1.070	0.041	-1.355	0.061	0.286***
	3	-0.099	0.049	-0.533	0.037	0.434***
Employment	4	-0.498	0.029	-0.900	0.062	0.402***
	0	-0.407	0.022	-0.404	0.040	-0.003
	1	-0.742	0.030	-0.533	0.048	-0.210***
	2	-0.800	0.035	-0.302	0.067	-0.498***
Investment	3	-0.266	0.035	-0.235	0.103	-0.031
	4	-0.123	0.030	-0.607	0.141	0.485***
	0	-0.094	0.010	0.273	0.023	-0.367***
	1	-0.607	0.014	-0.033	0.058	-0.574***
Debt	2	-1.465	0.025	-0.788	0.057	-0.678***
	3	-0.845	0.029	-0.428	0.087	-0.417***
	4	-0.907	0.033	-0.850	0.055	-0.057
	0	-0.070	0.021	0.000	0.000	-0.070***
Market Value	1	-0.979	0.030	0.000	0.000	-0.979***
	2	-2.927	0.048	0.000	0.000	-2.927***
	3	-1.619	0.067	0.000	0.000	-1.619***
	4	-1.585	0.091	0.000	0.000	-1.585***
Market Value	0	-3.736	0.093	-5.234	0.131	1.499***
	1	-4.522	0.074	-6.007	0.153	1.485***
	2	-1.480	0.070	-3.045	0.160	1.565***
	3	0.277	0.093	-2.495	0.251	2.771***
4	-1.193	0.091	-4.068	0.259	2.875***	

Notes: The table reports the aggregate response to business cycle fluctuations (Panel A) and exogenous shocks (Panels B and C) implied by the GRF and LPM sensitivities estimated using Equation (8). Heteroskedasticity-robust standard errors are reported in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.

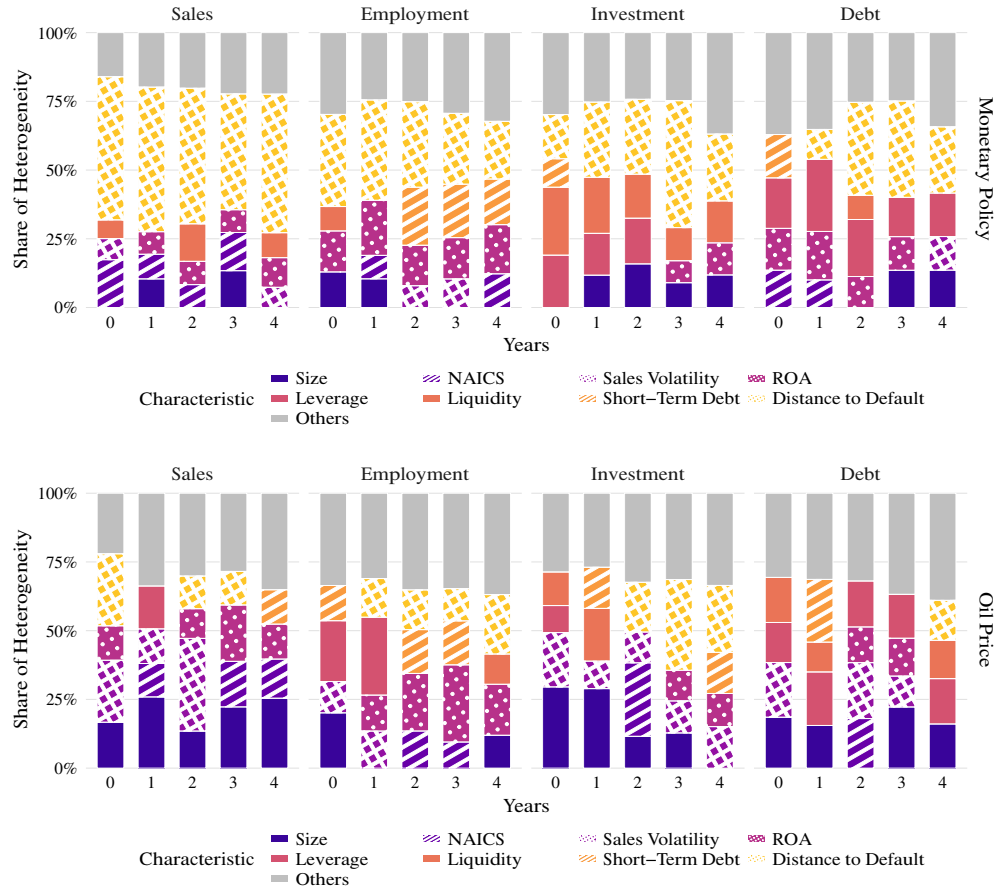
Figure C.14: Role of Firm FE - Aggregate Results



Notes: The figure reports the aggregate response and the covariance terms for each outcome variable to business cycle fluctuations. The dark bar denotes the total average aggregate response, while the orange bar denotes the covariance term. We estimate Equation (8) using the mean and covariance terms in Equation (4) as the dependent variable, using the relative outcome shares as weights. The bars are constructed using the baseline set of firm-level sensitivities estimated with the GRF algorithm and LPM, together with the ones estimated including firm FEs.

C.5 Miscellanea Exogenous Shocks

Figure C.15: Importance of Firms' Characteristics



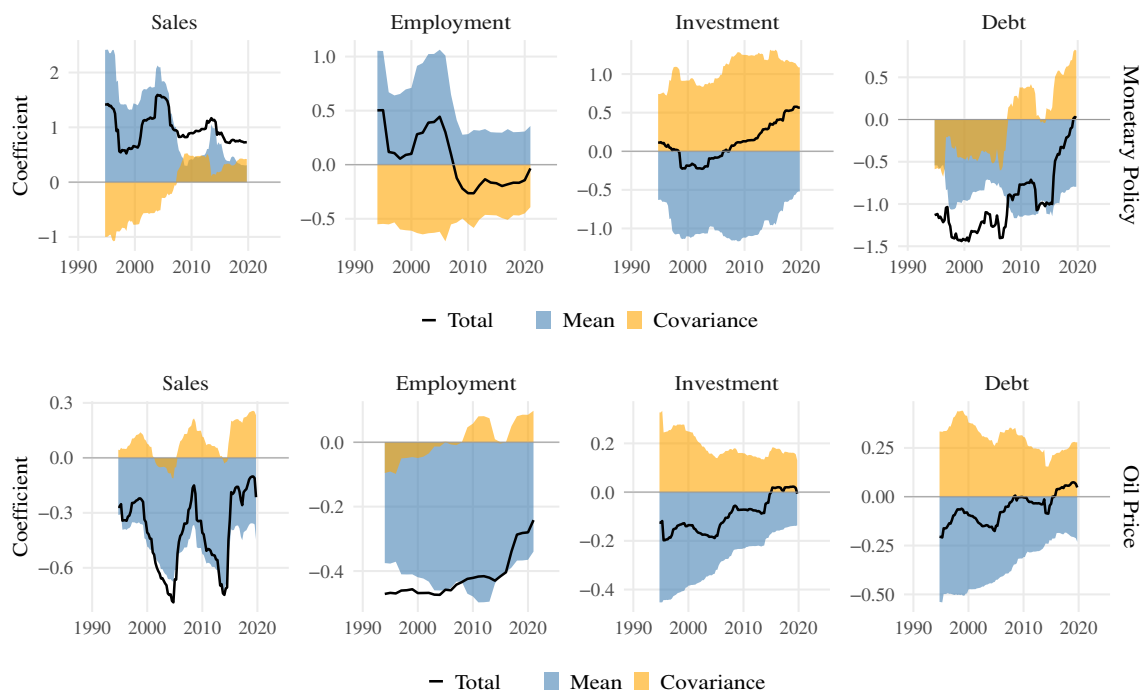
Notes: The figure reports the decomposition of GRF-based heterogeneity shares across firm characteristics for monetary policy and oil price shocks over horizons of one to five years. For each outcome and horizon, the stacked bars show the contribution of the four most important characteristics to the cross-sectional variation in estimated firm-level responses. The residual category, labeled “Others”, aggregates all remaining characteristics.

Figure C.16: Oil Shock on Employment



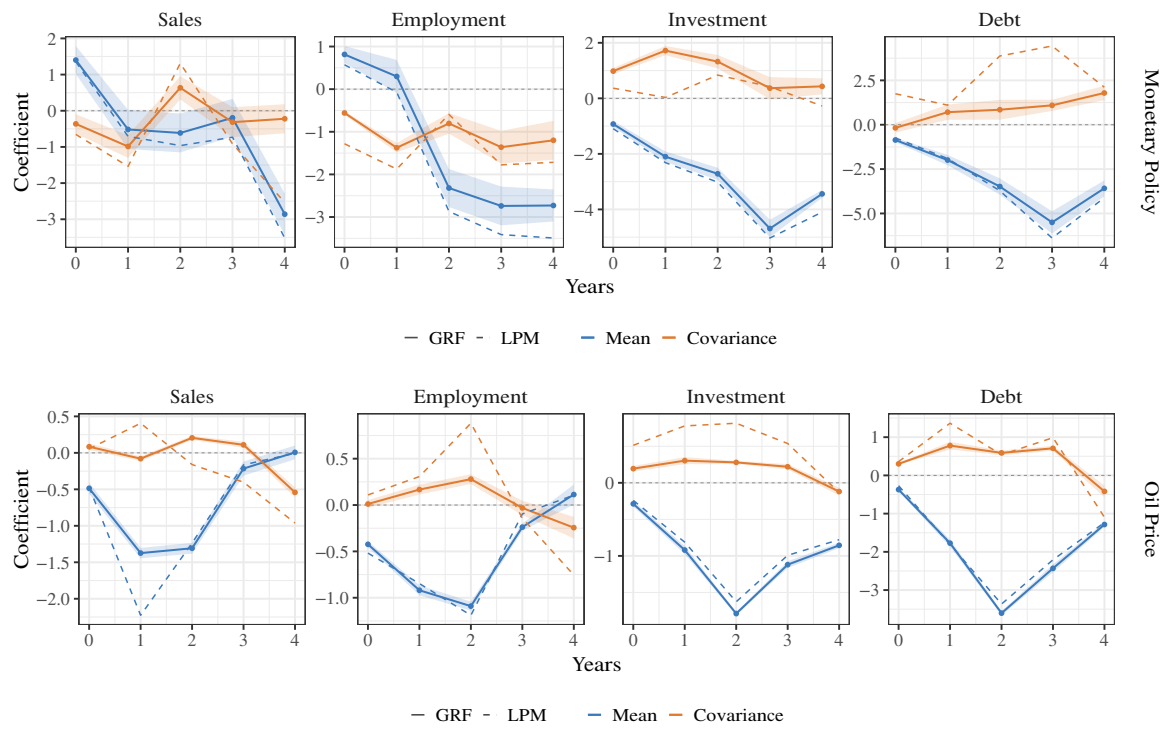
Notes: The figure reports GRF-based average employment responses to oil price shocks separately for firms observed before and after 2011. Estimates are computed using the full-sample forest and then evaluated on the two subsamples, so differences across lines reflect heterogeneity in the estimated response over the two periods. Since the oil shock is a continuous treatment, the GRF estimand corresponds to the doubly robust average partial effect, $\mathbb{E}[\text{Cov}(W, Y \mid \mathbf{X}) / \text{Var}(W \mid \mathbf{X})]$. Shaded areas denote 95 percent confidence intervals.

Figure C.17: Aggregate Effect over Time



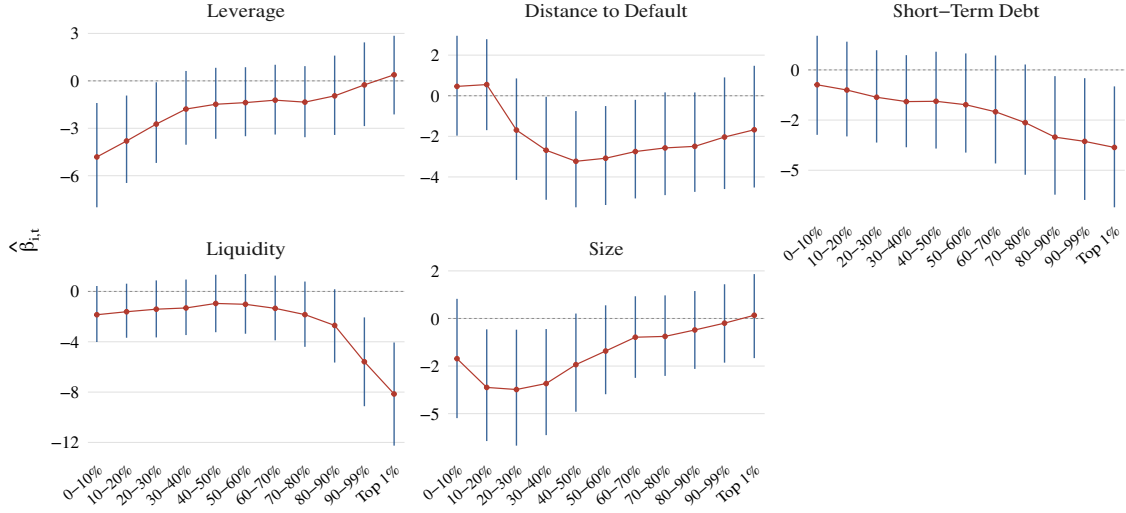
Notes: The figure reports rolling-window estimates of the aggregate contemporaneous response decomposition for each outcome at horizon one. Quarterly outcomes use a 20-quarter rolling window, while employment uses a 5-year rolling window. The black line reports the total aggregate response, equal to the sum of the mean and covariance components.

Figure C.18: Aggregate Effect of Non-Linearities



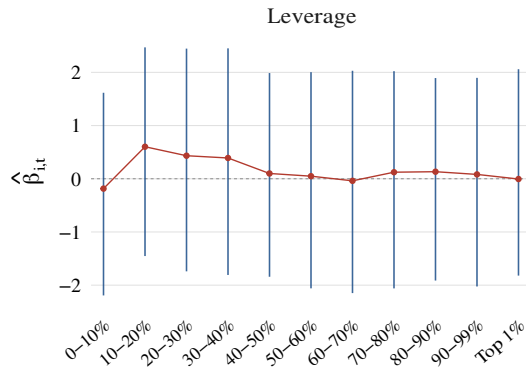
Notes: The figure reports the decomposition of aggregate impulse responses into mean and covariance components over horizons. Solid lines show GRF estimates, while dashed lines show the corresponding LPM estimates. Shaded areas denote 95 percent confidence intervals around the GRF estimates.

Figure C.19: Investment Sensitivities to Monetary Policy Shock by Firm Characteristics



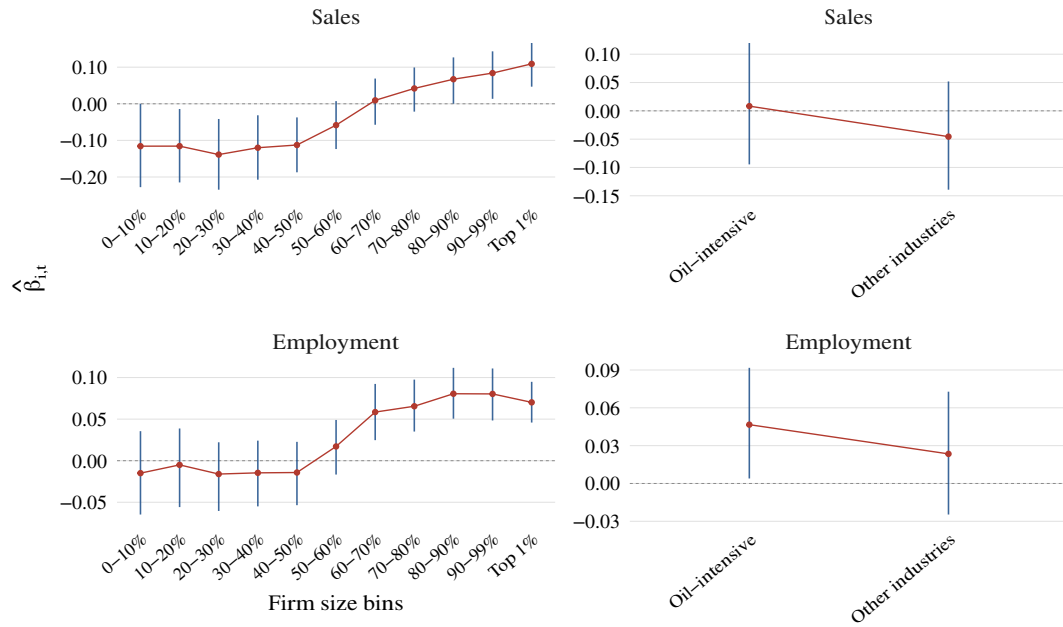
Notes: The figure reports firm-level GRF estimates of investment sensitivities to the monetary policy shock at $h = 2$ (peak of the covariance term). Firms are grouped into deciles by lagged leverage, distance to default, liquidity, and size, with the top decile split into the 90th-99th percentile and the top 1 percent. Points show the mean sensitivity within each bin; vertical lines show the interquartile range.

Figure C.20: Employment Sensitivities to Monetary Policy Shock by Firm Characteristics



Notes: The figure reports firm-level GRF estimates of employment sensitivities to the monetary policy shock at $h = 2$ (peak of the covariance term). Firms are grouped into deciles by lagged leverage with the top decile split into the 90th-99th percentile and top 1 percent. Points show the mean sensitivity within each bin; vertical lines show the interquartile range.

Figure C.21: Sales and Employment Sensitivities to Oil Price Shock by Firm Characteristics

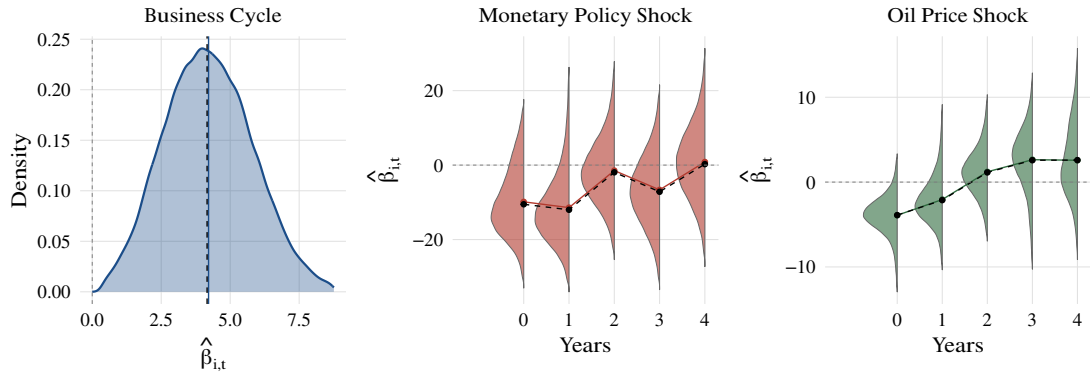


Notes: The figure reports firm-level GRF estimates of sales and employment sensitivities to the oil price shock. Sales sensitivities are shown at $h = 3$, while employment sensitivities are at $h = 2$ (peak of covariance term). Firms are grouped by size deciles, with the top decile split into the 90th-99th percentile and top 1 percent, and by oil-intensive versus other industries based on industry classifications. Points show the mean sensitivity within each bin; vertical lines show the interquartile range.

D Additional Results

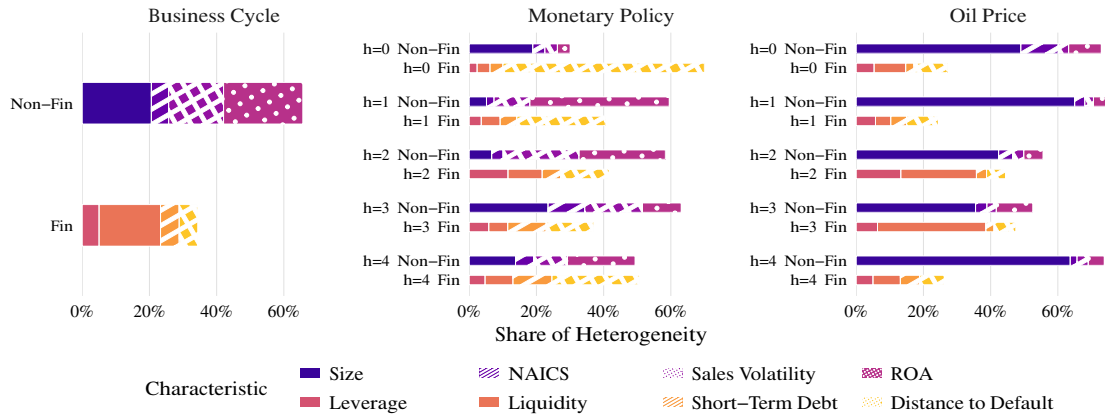
D.1 Financial Variables: Market Value

Figure D.1: Firm-level Sensitivity



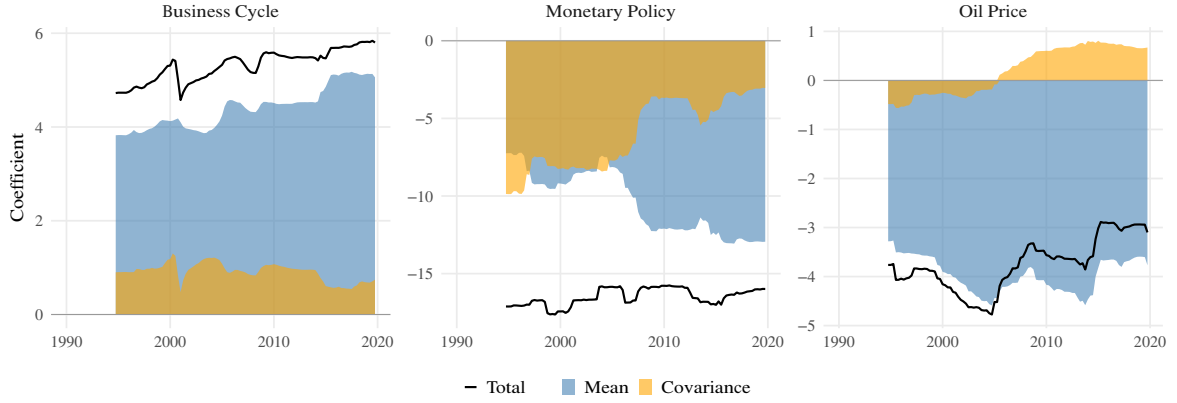
Notes: The figure reports the distribution of estimated firm-level market-value sensitivities, $\hat{\beta}_{it}$, across the three shock specifications. Vertical and connected solid (dashed) lines indicate the average estimated GRF (LPM) sensitivities within each panel or horizon.

Figure D.2: Share of Importance



Notes: The figure reports the contribution of the most important firm characteristics to heterogeneity in market-value responses across shocks. Bars are grouped by shock and split between financial and non-financial characteristics; for monetary policy and oil price shocks, results are shown separately by response horizon.

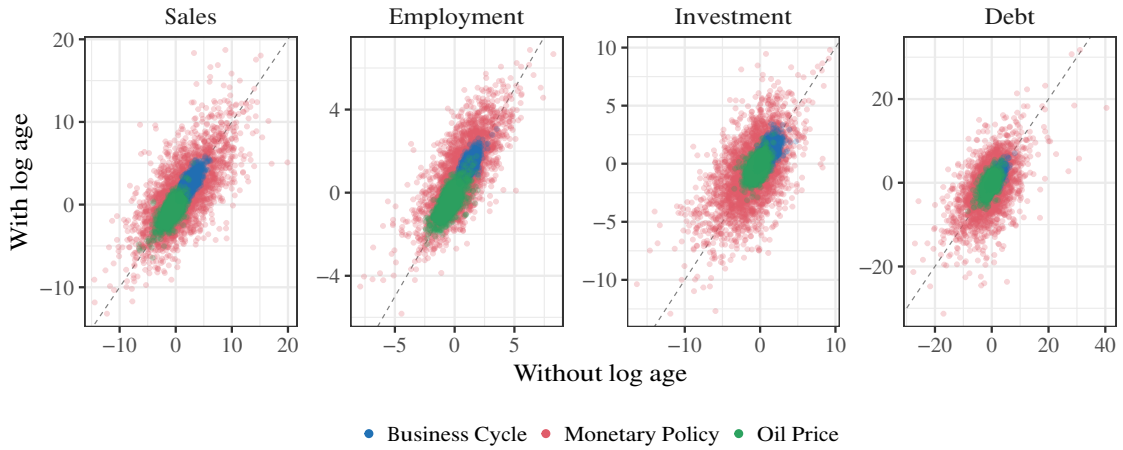
Figure D.3: Aggregate Response of Market Value



Notes: The figure reports rolling-window decompositions of the aggregate contemporaneous market value response across business cycle, monetary policy, and oil price shocks. The black line reports the total aggregate response, equal to the sum of the mean and covariance components. Estimates are computed separately for each shock using a 20-quarter rolling window over the sample period.

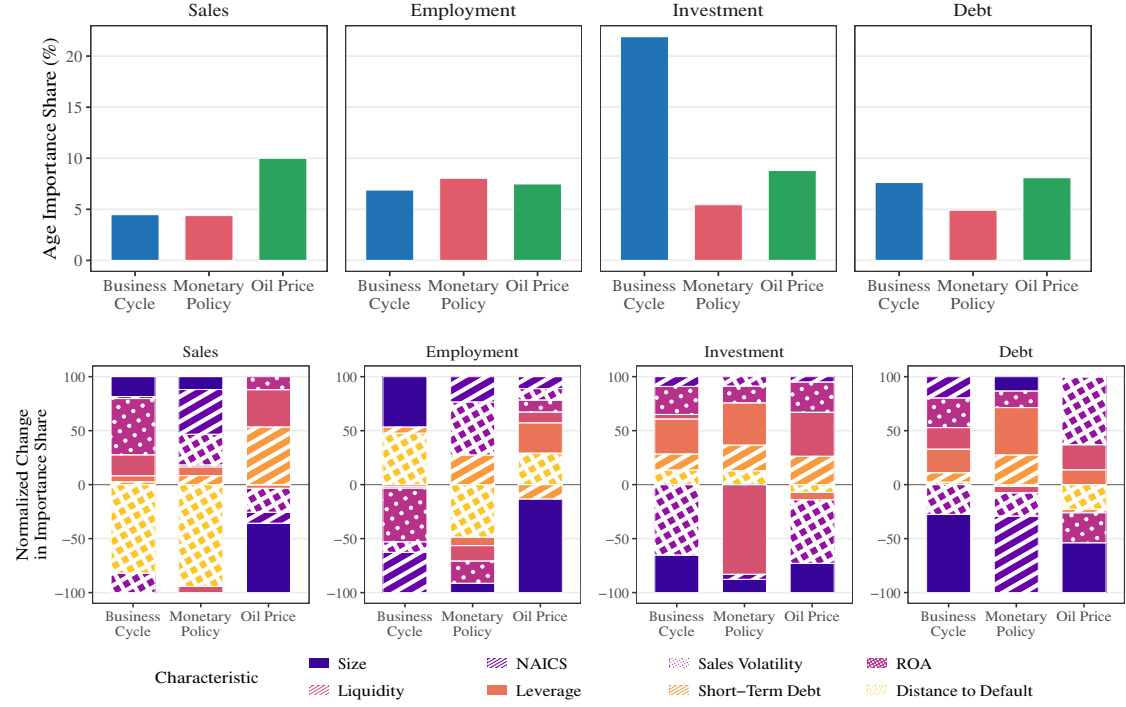
D.2 The Role of Firm Age

Figure D.4: Effect of Age on Distribution of Sensitivities



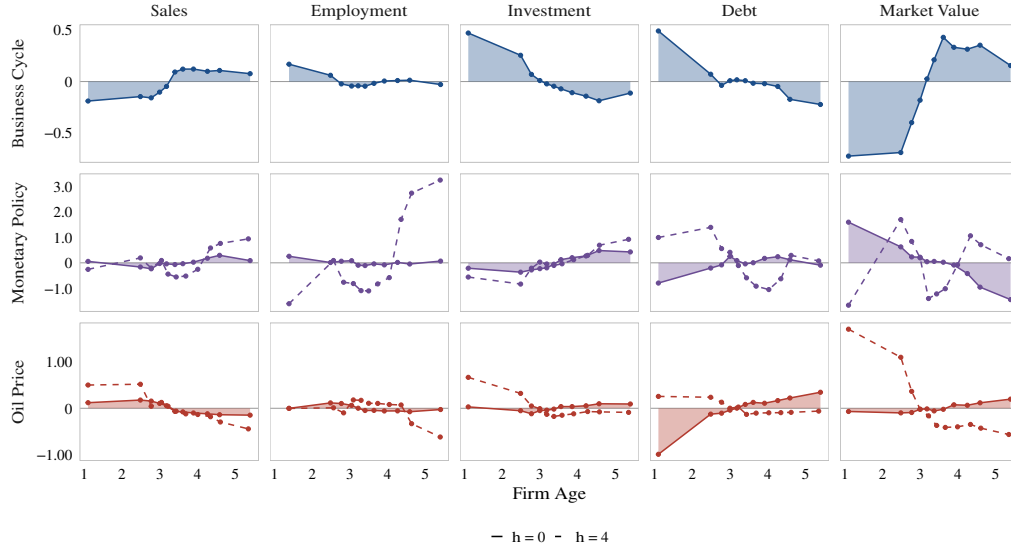
Notes: The figure compares firm-level GRF estimates obtained with and without log firm age included among the conditioning variables. Each point is a sampled firm-period observation at horizon one, plotted separately by outcome, with colors denoting the shock. The 45-degree dashed line indicates equality between the two specifications; deviations from this line show how adding log age changes the estimated firm-level response.

Figure D.5: Share of Importance: Age vs Other Characteristics



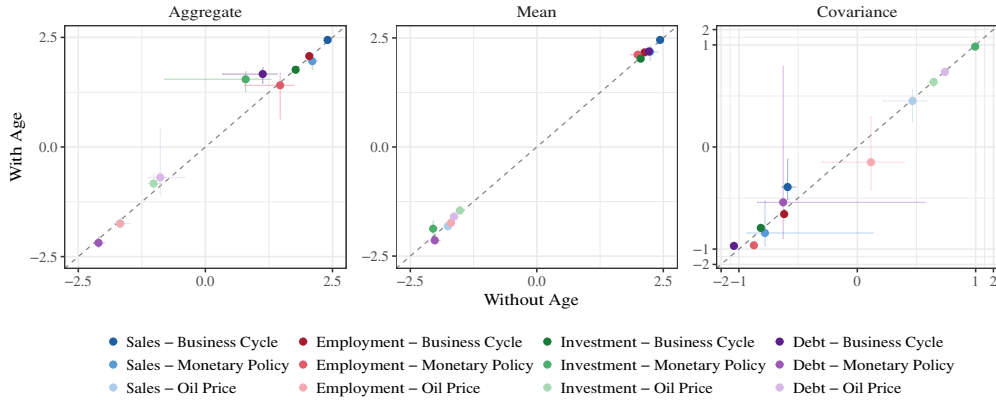
Notes: The top panel figure reports the importance share of firm (log) age in explaining heterogeneity in estimated firm-level responses across outcomes and shocks. Values for monetary policy and oil shock are averaged across horizons. The bottom panel reports how the relative importance of firm characteristics changes when firm age is added to the GRF specification at horizon one. For each outcome and shock, the bars show the normalized change in importance shares among the original firm characteristics, excluding the direct contribution of age itself. Positive values indicate characteristics whose relative importance increases after conditioning on firm age, while negative values indicate characteristics whose relative importance declines.

Figure D.6: Marginal Effect of Age



Notes: The figure reports accumulated local effects of firm age on estimated firm-level responses across outcomes and shocks. Each panel plots the partial effect of log firm age while averaging over the empirical distribution of the remaining firm characteristics. Solid lines report the contemporaneous impact; dashed lines report the four-year horizon for the monetary policy and oil price shocks.

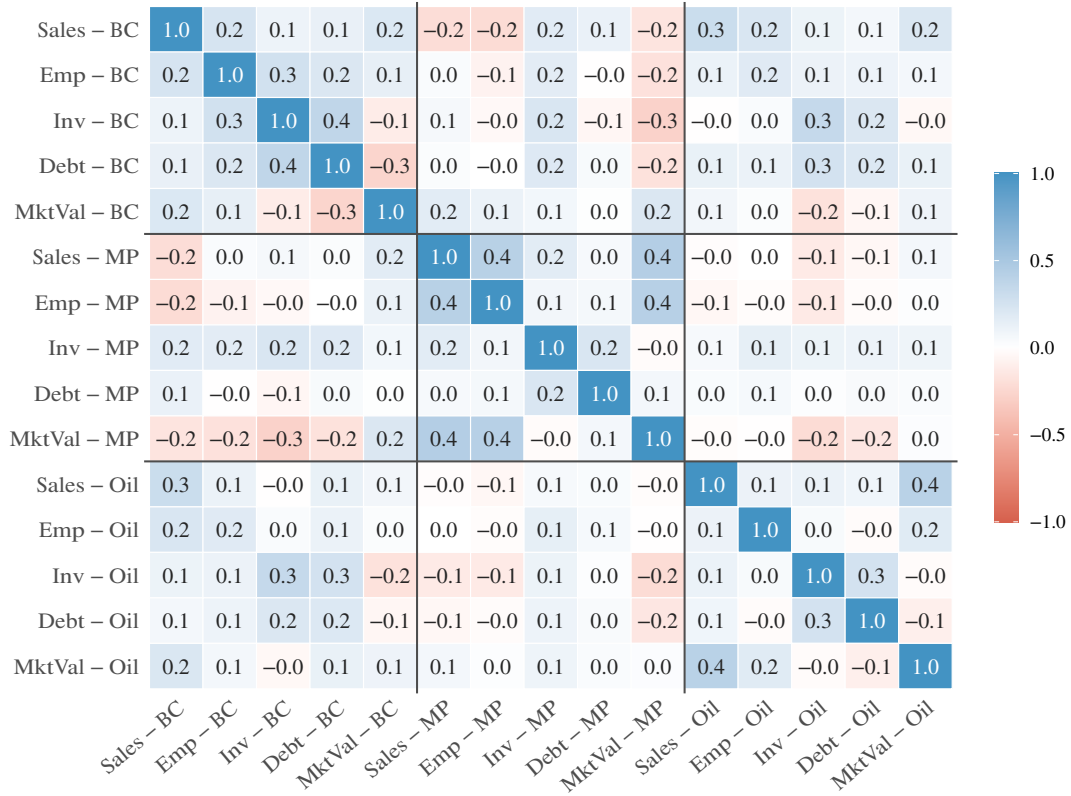
Figure D.7: Effect of Age on Aggregate Decomposition



Notes: The figure compares aggregate response components estimated from the baseline GRF specification without firm age to those from the specification that includes firm age, at horizon one. Each point corresponds to an outcome-shock pair, with colors identifying the pair and panels separating the total aggregate response, the mean, and the covariance component. The dashed 45-degree line marks equality between the two specifications; points close to the line indicate that adding firm age leaves the estimated component largely unchanged. Error bars report 95% confidence intervals around the corresponding estimates.

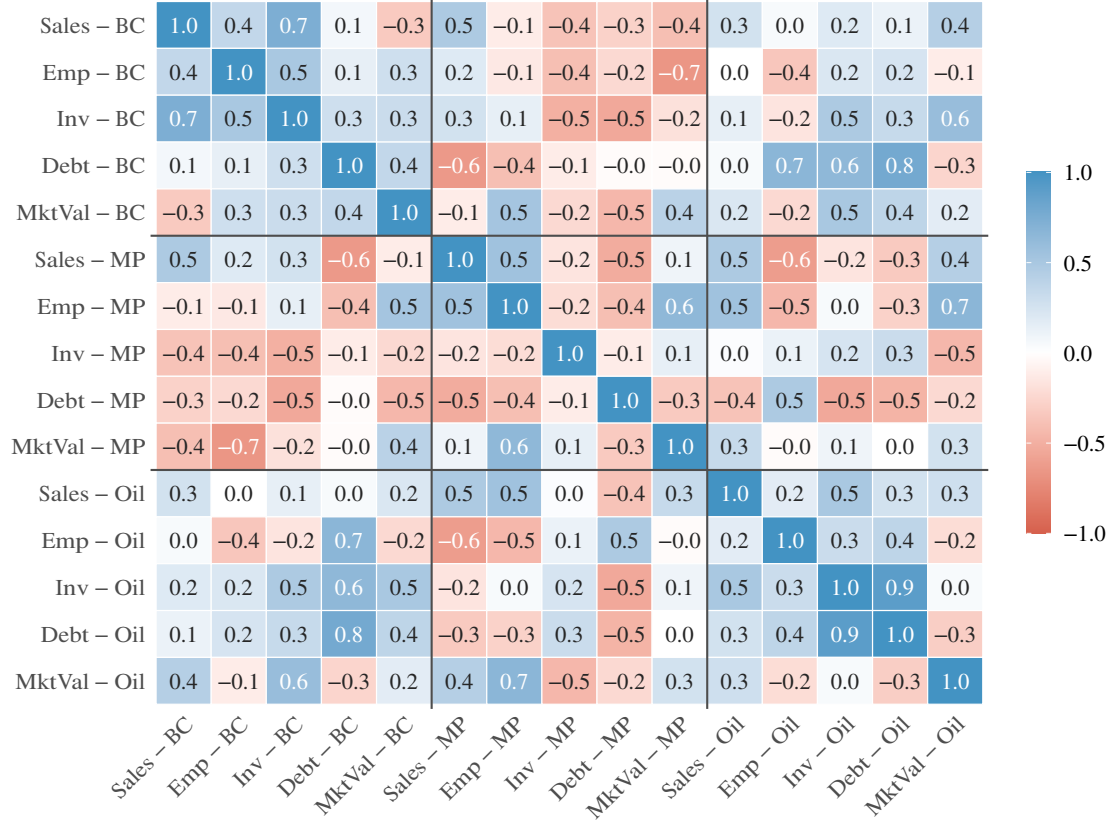
D.3 Comparison across Shocks and Outcome Variables

Figure D.8: Rank Correlation of $\hat{\beta}$



Notes: The figure reports pairwise Spearman rank correlations across firm-level GRF estimates, computed for all outcome-shock combinations at the first horizon. Quarterly treatment effects are first averaged to the firm-year level and then matched with annual employment estimates. A monetary policy shock is a decline in interest rates; an oil price shock is a negative change in oil prices. Darker cells indicate stronger rank association across firms, while the annotated values report the corresponding Spearman correlation coefficients. The block structure groups estimates by shock: business cycle, monetary policy, and oil price.

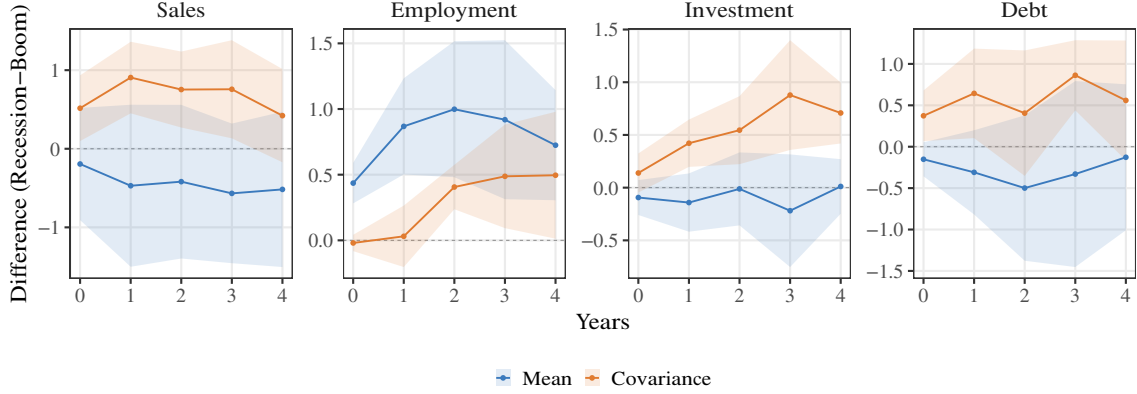
Figure D.9: Rank Correlation of Shares of Importance



Notes: The figure reports pairwise Spearman rank correlations across GRF variable-importance profiles for each outcome-shock combination at the first horizon. For each cell, the correlation is computed across the vector of firm characteristics used by the forest, so higher values indicate that two outcome-shock pairs rely on a similar ranking of covariates. The block structure groups estimates by shock: business cycle, monetary policy, and oil price. Darker cells indicate stronger rank similarity, with annotated values reporting the corresponding Spearman correlation coefficients.

D.4 State Dependency of Monetary Policy

Figure D.10: State Dependent Monetary Policy



Notes: The figure reports regime-dependent responses to a monetary policy shock, decomposing the recession-minus-boom difference into the mean and covariance components of the GRF aggregate response. Shaded areas denote 90 percent confidence intervals based on Newey-West standard errors. Recession periods are defined using NBER recession dates, with boom periods as the omitted reference category.

D.5 Firm Entry and Exit

We extend the accounting identity to allow for entry and exit. Without loss of generality, consider the response of an aggregate outcome between a base period $t-1$ and period t , abstracting from longer horizons. Firms alive at $t-1$ split into *continuing* firms $\mathcal{C}$, still active at t , and *exitors* $\mathcal{X}$, which have disappeared by t ; *entrants* $\mathcal{N}$ are active at t but absent at $t-1$. Let Y_i denote the firm-level quantity that defines the aggregation weight (real sales, employment, real capital, or debt), let $\omega_{i,t-1} = Y_{i,t-1} / \sum_{j \in \mathcal{C} \cup \mathcal{X}} Y_{j,t-1}$ be firm i 's share of the base-period aggregate, and let $g_{i,t} = \beta(\mathbf{X}_{i,t-1}) \cdot W_t$ be its response to the shock.

Two extensive-margin objects summarize turnover: the base-weight share of exiting firms, $s_t^X = \sum_{i \in \mathcal{X}} \omega_{i,t-1}$, and the entrants' contribution, $\phi_t^N = \sum_{i \in \mathcal{N}} Y_{i,t} / \sum_{j \in \mathcal{C} \cup \mathcal{X}} Y_{j,t-1}$. Renormalizing shares *within* survivors, $w_{i,t-1} = \omega_{i,t-1} / (1 - s_t^X)$ so that $\sum_{i \in \mathcal{C}} w_{i,t-1} = 1$, the survivor aggregate $\text{agg}_t^{\mathcal{C}} = \sum_{i \in \mathcal{C}} w_{i,t-1} g_{i,t}$ decomposes exactly as in the fixed-firm case,

$$\text{agg}_t^{\mathcal{C}} = \underbrace{\bar{g}_t}_{\text{mean}} + \underbrace{\widetilde{\text{Cov}}_t(w_{i,t-1}, g_{i,t})}_{\text{covariance}}, \quad (\text{D.14})$$

where $\bar{g}_t$ is the unweighted average response among survivors and $\widetilde{\text{Cov}}_t$ the covariance between

within-survivor weights and responses as in Section 2.1. The economy-wide response is the survivors' contribution, weighted by their base share, plus entry:

$$G_t = (1 - s_t^X) \text{agg}_t^C + \phi_t^N. \quad (\text{D.15})$$

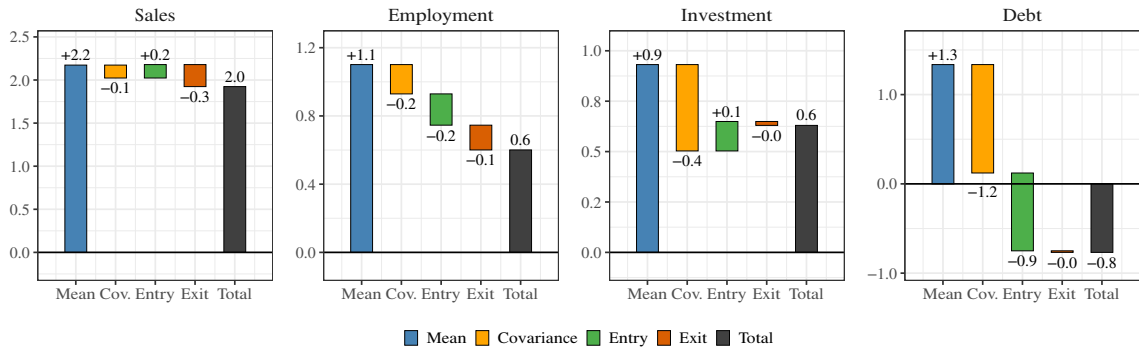
Rearranging (D.15) delivers an additive decomposition into the two intensive-margin terms and a net-entry term:

$$G_t = \underbrace{\bar{g}_t}_{\text{mean}} + \underbrace{\widetilde{\text{Cov}}_t(w_{i,t-1}, g_{i,t})}_{\text{covariance}} + \underbrace{\phi_t^N - s_t^X \text{agg}_t^C}_{\text{net entry}}, \quad (\text{D.16})$$

where net entry adds the entrants' output, ϕ_t^N , and subtracts the response that the exiting firms' base weight would have contributed at the survivor rate, $s_t^X \text{agg}_t^C$. As in the baseline, each term is projected on the shock to obtain its average contribution.

The key observation is that extending the framework to entry and exit does not alter the role of the mean and covariance terms. The decomposition (D.16) is additively separable, and its first two components are defined exactly as in the fixed-firm case—over the set of surviving firms, with weights that sum to one. The mean still measures the average sensitivity of firms to the shock, and the covariance still measures whether firms with larger economic weight are systematically more or less sensitive than average; neither is affected by the arrival or disappearance of firms, whose entire effect is isolated in the separate net-entry margin. Entry and exit thus operate on a distinct, *extensive* margin—changing which firms populate the economy and how base weights are allocated—while the mean and covariance capture the *intensive* margin through which heterogeneity in firm sensitivities shapes the aggregate response. Because the two margins enter additively, the covariance channel at the heart of our results retains the same interpretation once turnover is accounted for.

Figure D.11: Aggregate Effect Accounting for Extensive Margin



Notes: The figure decomposes the aggregate response to business cycle fluctuations into continuing-firm mean and covariance effects, entry, and exit using Equation (D.16).